Sources and attribution

Historical manuscript notes of the I. M. Gelfand Seminar. MCCME digital archive: Notes of the talks at the I. M. Gelfand Seminar.

MCCME identifies M. A. Shubin as the note-taker. The Clay Mathematics Institute funded the scanning. MCCME acknowledges M. A. Shubin, S. I. Gelfand, the Clay Mathematics Institute and Jim Carlson. MCCME archive notice.

Chapter 29. Manuscript frames 1297–1358. Catalogue snapshot: 5 October 2026.

Speakers named in the catalogue associations for the complete edition, using the catalogue's spelling:

I.M. Gelfand; R.L. Dobrushin; Irving Segal; Laurent Schwartz; S.G. Gindikin; A.Ya. Povzner; V.I. Arnold; A.A. Kirillov; I.I. Pyateski-Shapiro; M.I. Vishik; Yu.V. Linnik; E.A. Gorin; D.A. Kazhdan; V.P. Palamodov; A.S. Dynin; S.P. Novikov; Ya.G. Sinai; I.M. Gelfand, A.Ya. Povzner,Berezin; V.N. Latyshev; S.I. Gelfand; I.N. Bernstein; A.M. Stepin; R.A. Minlos; L.D. Faddeev; Paul Halmos; J.J. Kohn; Robert Finn; S.V. Smirnov; V.I. Matsaev; A.G. Kushnirenko; M.S. Agranovich; F.I. Karpelevich; V.A. Borovikov; V.B. Lidskii; G.E. Shilov; Lee Lorch; Louis Nirenberg; O.A. Oleinik; A.N. Parshin; D.B. Fuchs; S.G. Krein; Jean Leray; L.G. Makar-Limanov; A.Yu. Geronimus; Alan Weinstein ; M.L. Gromov; G.A. Margulis; Chandler Davis; V.M. Buchstaber; B.R. Vainberg; Gaetano Fikera; Dusa McDuff; Edwin Hewitt; B.V. Fedosov; A.M. Gabrielov; S. Rolevich; A.M. Perelomov; B.I. Rosenfeld; Vlastimil Ptak; Michael Atiyah; Pierre Deligne; A.M. Vinogradov; L.N. Vasserstein; Jack Morava; M.Sh. Birman; Yu.I. Manin; A.G. Kostyuchenko; V.I. Lomonosov; F.A. Berezin; G.M. Henkin; A.S. Mishchenko; A.N. Rudakov; A.I. Schnirelman; V.A. Ponomarev; Tony Springer; V.E. Zakharov; V.N. Tulovskii; A.B. Shabat; B.M. Gurevich; Yu. Koblenz; Enrico Guisti; Czeslaw Bessaga; B.A. Dubrovin; Graeme Segal; A.M. Polyakov; M.I. Graev; James King; Bert Kostant; E.S. Fradkin; I.M. Krichever; V.G. Drinfeld; M.V. Losik; A.V. Roiter; A.A. Migdal; V.N. Popov ; J.J. Duistermaat; A.M. Vershik; A.S. Tyurin; D.A. Kirzhnits; L.A. Dickey; S.M. Vishik; Bela Sekefalvi-Nagy; Kazuhiko Aomoto; Francesco Calogero; V.S. Kruskal; A.S. Schwarz; Alain Guichardet; A.A. Belavin; A.A. Beilinson; V.I. Ogievetskii; D.R. Lebedev; G.B. Shabat; Renata Kallosh; Raphael Hoegh-Krohn; I.A. Batalin; I.Yu. Kobzarev; A.V. Zelevinsky; Robert MacPherson; G.I. Olshanskii; A.B. Goncharov; Bernard Gaveau; A.G. Khovansky; I.V. Cherednik; B.Ya. Kazarnovskii; P.G. Grinevich; Jean-Luc Brylinski; Eckart Viehweg; Christian Kassel; A.V. Bobylev; B.V. Lidskii; Benedict Seifert; A.G. Elashvili; Werner Mueller; V.L. Ginzburg; John Milnor; Andre Haefliger; A.B. Givental; Mariusz Wodzicki; Al-dr.B. Zamolodchikov; Jurgen Frohlich; B.L. Feigin; Yu.A. Neretin; M.A. Semenov-Tyan-Shanski; George Wilson; Yu.L. Daletskii; V.A. Chulaevskii; P.B. Wiegman; F. Nikiforov; Yu.M. Berezanskii; D.A. Leites; Ivan Penkov; Jean-Pierre Serre; M.M. Kapranov; Marie-France Vigneras; Yu.S. Ryzhikova; V.G. Knizhnik; M.L. Kontsevich; A.Yu. Morozov; A.N. Varchenko; Yan I. Kogan; V.A. Kazakov; V.Yu. Ovsienko; I.S. Zakharevich; Don Zagier; D.V. Alekseevskii; A.P. Veselov; Klyachko; A.O. Radul; N.Yu. Reshetikhin; A.G. Usheridze; Gregory Moore; V.S. Dotsenko.

A catalogue association does not by itself establish an exact talk boundary within a manuscript page or identify the writer of every note.

The modern LaTeX text is AI-transcribed. The English edition includes AI-translated passages and AI-transcribed passages originally written in English. Diagram reconstructions are AI-created and typesetting is AI-produced. Historical lecture and manuscript authorship remains with its human authors.

Rights in the original notes, scans and other third-party material remain with their authors and respective rights holders, including heirs. To the extent that the project holds copyright or related rights in its own contributions, it dedicates those rights under CC0 1.0 Universal. The dedication applies only to rights held by the project.

These editions are freely available and are published solely for non-commercial educational purposes. If a rights holder or their representative objects to publication, please identify the affected material in GitHub Issues so we can take it down.

Exact image URLs, file digests and catalogue associations accompany the editable sources.

Chapter 29

GELFAND-P001297

29/IX-86 Gelfand Sem.

V. Ginzburg [crossed out] Procesi, De Concini, Springer



Uncertain reading. Before ‘Procesi’ is an illegible cancelled entry.

Original source details: GELFAND-P001297

[crossed out] Progress in Math., $N = 60$ (1985), birkhäuser



Uncertain reading. Before ‘Progress in Math.’ the word ‘Springer’ is crossed out.

Original source details: GELFAND-P001297

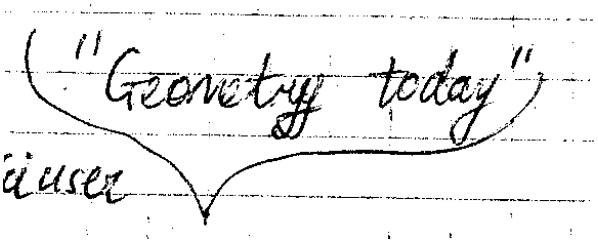


Diagram and typed description.

“Geometry today”

Original source details: GELFAND-P001297

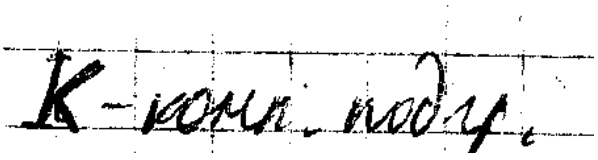
Compactification of quadr. forms

quadr. form [wavy underlining] (Space $\to$ dual space)

— a set of lin. subspaces

$$\underset{\text{compactification (min)}}{\bar{M}} \supset M = G/H$$

G a grp., K a comp. subgrp.



Uncertain reading. K is described using the abbreviation ‘комп.’ (‘comp.’).

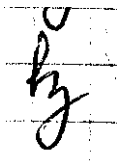
Original source details: GELFAND-P001297

H the normalizer of K

$$H/K = \oplus Z/2Z$$

$\mathfrak{g}$ the Lie alg.

h_g the subalg. corresponding to H



Original source details: GELFAND-P001297

Uncertain reading. The subalgebra symbol has a loop below the baseline. The reading h_g is tentative.

Properties of $\bar{M}$?

1) $\bar{M}$ is smooth

2) $\bar{M}$ consists of finitely many G-orbits

3) $\bar{M} \setminus M$ is a divisor with normal crossings

(a common sol. of k equations, where $k = \text{Codim}(\bar{M} \setminus M)$)

(like a system of coord. planes)

4) every component $D =$ closure of one orbit.

5) There is exactly 1 closed orbit, isom. to G/P, where P is some parab. subgrp. constructed from K.

6) Line bundles on $\bar{M} \xrightarrow{\sim}$ line bundles on G/P.

— · — · —

$$G = GL_n$$

Hecke algebra assoc. with the symm. group [wavy underlining]

H $\quad$ $T_1, \ldots, T_{n-1}$ — gen.

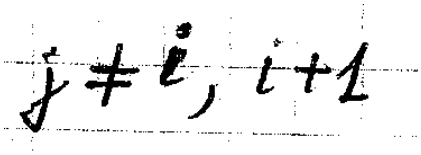
[At left] q-analogue.

$$1) \quad (T_i + 1)(T_i - q) = 0 \Rightarrow T_i^2 = 1 \quad \text{(at } q = 1\text{)} \qquad (q\text{-analogue})$$

$$2) \quad T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$$

(T_i the transpos. of the i-th and $(i+1)$-st elements)

$$3) \quad T_i T_j = T_j T_i, \qquad j \neq i, i+1$$



Original source details: GELFAND-P001297

Uncertain reading. Condition (3) reads $j \neq i, i+1$.

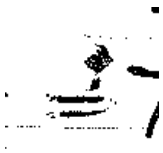
Cherednik devised the embedding $H \subset \widetilde{H}$.

[Under $\widetilde{H}$] gen. $T_1, \ldots, T_{n-1};\ x_1, \ldots, x_n$

and rel.

$$x_i T_i = T_i x_{i+1} - (q-1)x_{i+1} \qquad \Big| \qquad x_i x_j = x_j x_i$$

$$x_{i+1} T_i \overset{?}{=} T_i x_i + (q-1)x_{i+1}$$



Uncertain reading. The mark above the second equals sign resembles a question mark.

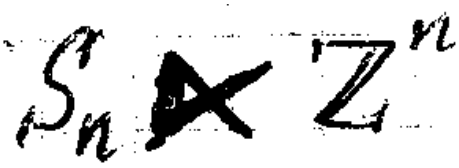
Original source details: GELFAND-P001297

GELFAND-P001298

[source number: 8]

This is related to the crossed prod.

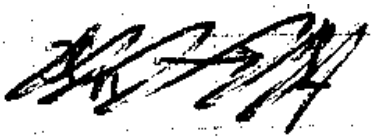
$$S_n \ltimes Z^n$$



Uncertain reading. The crossed-product sign has a vertical stroke at left.

Original source details: GELFAND-P001298

[Crossed out]



Uncertain reading. Below the formula is a crossed-out entry, possibly $x_n \to 1$.

Original source details: GELFAND-P001298

(x_j a basis in Z^n).



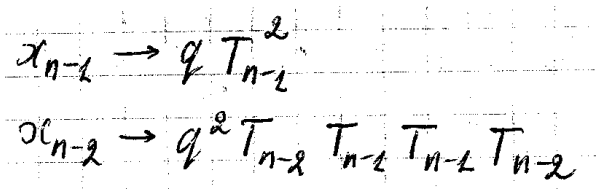
Uncertain reading. The overwritten index after x reads j.

Original source details: GELFAND-P001298

Map of the grp. into the algebra

$$(g, z) \mapsto g$$

$$x_n \to 1$$
$$x_{n-1} \to qT_{n-1}^2$$
$$x_{n-2} \to q^2 T_{n-2}T_{n-1}T_{n-1}T_{n-2}$$
$$\cdots\cdots\cdots$$



Uncertain reading. The first line reads qT_{n-1}^2. The next line has q^2 before the product of four T's.

Original source details: GELFAND-P001298

This is a homomorphism!

At $q = 1$ we obtain 1.

Problem (Cherednik)

(unsolved)

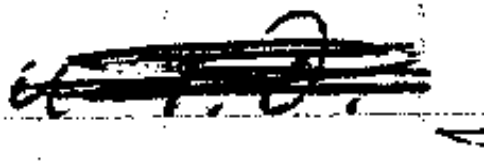
Find a homomorphism starting with $x_n^2 \to 1$

and what is the target of this homomorphism?

Or an arb. polynomial in x_n.

29.9.86 V. Ovsienko

On the connection of the KdV eq. with the Virasoro alg. [Crossed out] and of super KdV with the Schwarz and Ramond algebras.



Uncertain reading. After 'Virasoro' is a crossed-out entry, possibly 'KdV'.

Original source details: GELFAND-P001298

(geom. meaning)

KdV = the Euler eq. corresponding to the Virasoro alg., more precisely, to the Virasoro group.

Super KdV = — " — Schwarz and Ramond

Vect S^1 — alg. of vect. fields on S^1 and its central extension.

[Crossed out] KdV is an eq. on the dual space. It is tangent to the orbits and is Hamiltonian.



Uncertain reading. Before 'KdV' is a crossed-out entry whose reading is unresolved.

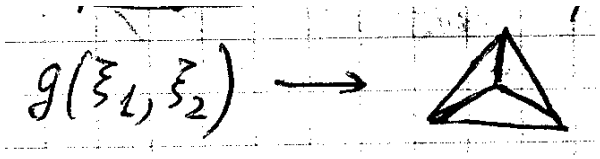
Original source details: GELFAND-P001298

— — — — —

29.9.86 Kazakov

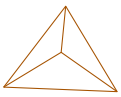
Discretization and exact solution of 2-dimensional gravity

(continued)



Original source details: GELFAND-P001298

Diagram and typed description.

$g(\xi_1, \xi_2) \longrightarrow$ 

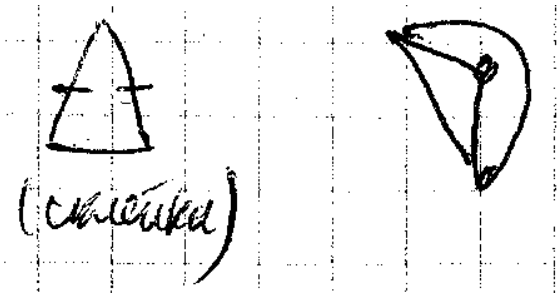
Segments join the interior point to all three vertices. The thickened ends do not show definite arrowheads.

$$Z = \underbrace{\int Dg(\xi) \exp\left\{-\int d^2\xi\sqrt{g}\right\}}_{\text{(surfaces with the topology of a sphere)}} \underset{\text{discretization}}{\Longrightarrow} Z = \sum_n^{\infty} \sum_{\{G^{(n)}\}} e^{-\beta n} \qquad (1)$$

$G^{(n)}$ is a graph with n vertices, with the sum over all graphs.

GELFAND-P001299

$\{G^{(n)}\}$ is the set of combinatorial triangulations of the sphere (simplic. complexes with sphere topology up to permutations).



Original source details: GELFAND-P001299

Diagram and typed description.



(gluings)

The first sketch has two short transverse strokes. In the second, the joins and possible arrowheads at the loops are indistinct. The first caption appears to read ‘склейки’ (‘gluings’).

[To the right of the sketches] this can also be forbidden

(we will leave it for now)

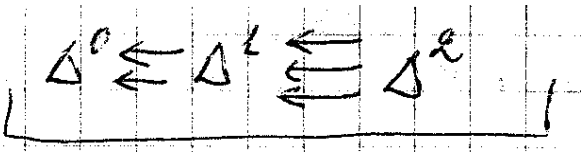
this will not affect the answer



Original source details: GELFAND-P001299

Uncertain reading. The abbreviation reads ‘сказ.’ (‘affect’).

$\underset{\text{sets}}{\Delta^0, \Delta^1, \Delta^2}$ and maps



Original source details: GELFAND-P001299

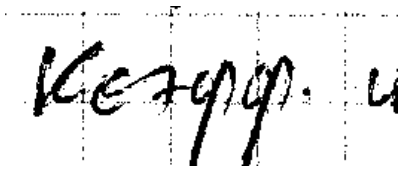
Diagram and typed description.

$\Delta^0 \longleftarrow \Delta^1 \Longleftarrow \Delta^2$

A permutation group acts throughout. We quotient by it.
The singularity of the discr. sum and of the cont. integral is the same (near divergence).

$$Z(\beta) \underset{\beta\to\beta_c}{\approx} (\beta-\beta_c)^{5/2} \tag{2}$$

(β is called the cosmological constant)
Coeffs. and lower terms depend on the type of triangulations.



Original source details: GELFAND-P001299

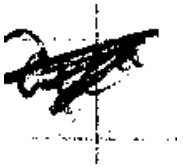
Uncertain reading. The abbreviation is read as 'Коэфф.' ('coeffs.').

$$I(g)=\int d^{N^2}u\,\exp\left[-\operatorname{tr}\left(u^2+\frac{g}{\sqrt{N}}u^3\right)\right] \tag{3}$$

(u an $N\times N$ ['Hermitian' inserted above 'matrix'] matrix)
Claim. [wavy underlining]

$$Z \underset{N\to\infty}{=} \frac{1}{N^2}\ln I(e^{-\beta/2}) \tag{4}$$

Proof [wavy underlining] by expanding in g. The resulting [crossed out] Feynman graphs are triangulations of the sphere.



Original source details: GELFAND-P001299

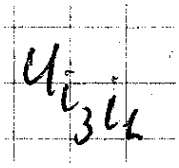
Uncertain reading. Before 'Feynman graphs' is an illegible crossed-out entry.

Integral (3) at $N\to\infty$ is computed exactly. We have

$$I(g)=\sum_n \frac{(-g)^n}{n!N^{n/2}}\int d^{N^2}u\,e^{-\operatorname{tr}u^2}(\operatorname{tr}u^3)^n \tag{5}$$

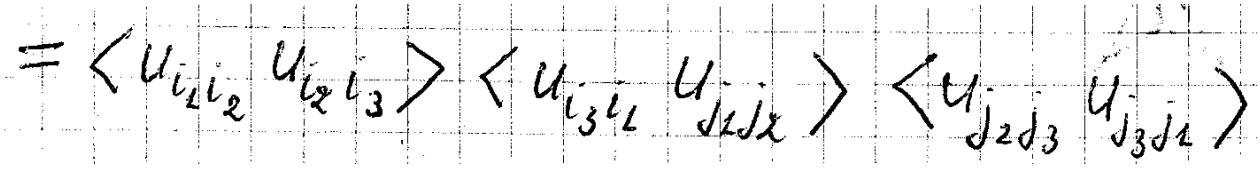
$\langle\ \ \rangle$ denotes av. with Gaussian weight.
Wick's thm.

$$\begin{aligned}\langle\operatorname{tr}(uuu)\operatorname{tr}(uuu)\rangle &= \langle u_{i_1i_2}u_{i_2i_3}u_{i_3i_1}u_{j_1j_2}u_{j_2j_3}u_{j_3j_1}\rangle = \\ &= \langle u_{i_1i_2}u_{i_2i_3}\rangle\langle u_{i_3i_1}u_{j_1j_2}\rangle\langle u_{j_2j_3}u_{j_3j_1}\rangle\end{aligned} \tag{6}$$



Original source details: GELFAND-P001299

Uncertain reading. The third factor in the first average reads $u_{i_3i_1}$.

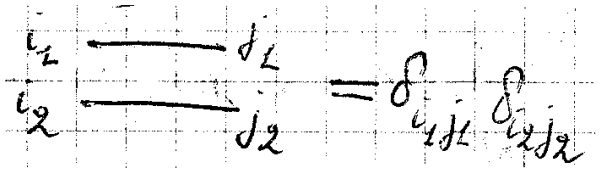


Original source details: GELFAND-P001299

Uncertain reading. The second line of (6) shows three consecutive pair averages. No summation sign is written on this line.

(A Gauss. integral is always expressed in terms of pairs!)

We shall denote the average $\langle u_{i_1 i_2} u_{j_1 j_2} \rangle$ by



Original source details: GELFAND-P001299

Diagram and typed description.

i_1 ——— j_1 $= \delta_{i_1 j_1} \delta_{i_2 j_2}$

i_2 ——— j_2

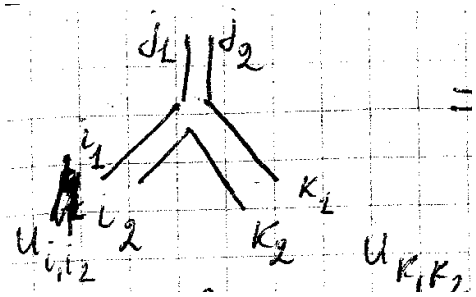
The upper pair is labelled i_1, j_1 and the lower pair i_2, j_2.

[Beneath the average] (normalization is understood)

[Below, with an arrow] in the coupling const.

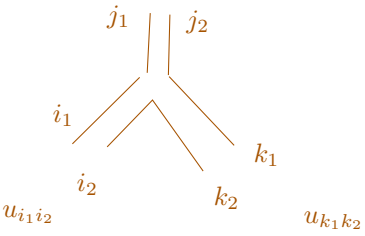
GELFAND-P001300

[source number: 9]



Original source details: GELFAND-P001300

Diagram and typed description.



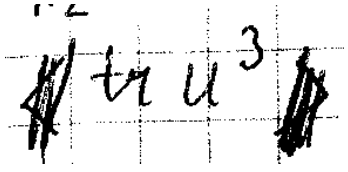
$$= \frac{g}{\sqrt{N}} \delta_{i_1 j_1} \delta_{k_1 j_2} \delta_{i_2 k_2}$$



Original source details: GELFAND-P001300

Uncertain reading. An illegible mark is crossed out to the left of i_1, i_2.

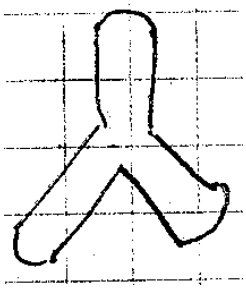
$$\operatorname{tr} u^3$$



Original source details: GELFAND-P001300

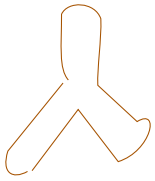
Uncertain reading. Illegible marks are crossed out on either side of $\operatorname{tr} u^3$.

Trace — closing the lines

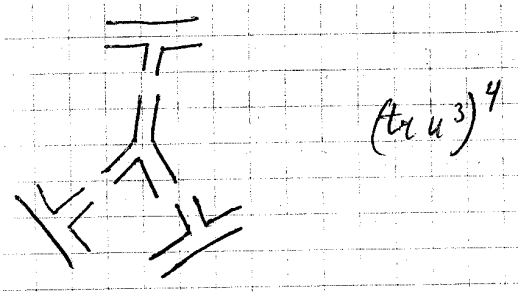


Original source details: GELFAND-P001300

Diagram and typed description.

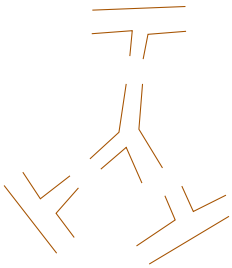


Small gaps are visible at the left of the neck and at the lower left end.



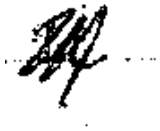
Original source details: GELFAND-P001300

Diagram and typed description.



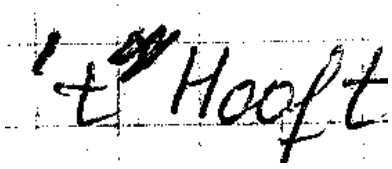
$(\mathrm{tr}\, u^3)^4$

The selection of planar graphs [crossed out] etc. was devised by ’t Hooft.



Original source details: GELFAND-P001300

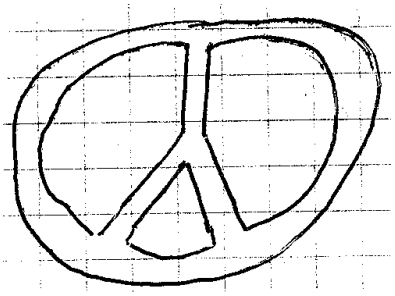
Uncertain reading. The mark after ‘graphs’ is crossed out and undeciphered.



Original source details: GELFAND-P001300

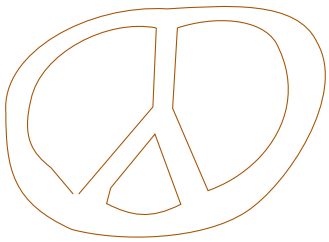
Uncertain reading. The crossed-out superscript mark between ‘’t’ and ‘Hooft’ is illegible.

We join them and sum over all possible ways of joining.



Original source details: GELFAND-P001300

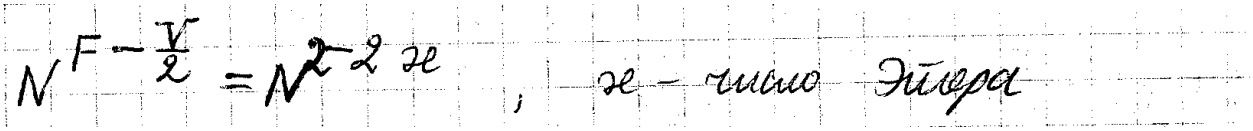
Diagram and typed description.



— tetrahedron. [wavy underlining]

$\forall$ index loop $=$ a factor N. Their number F is the number of faces.

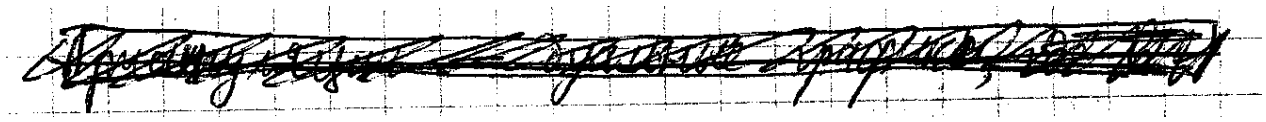
$$N^{F-\frac{V}{2}} = N^{2-2\chi}, \qquad \chi \text{ the Euler number}$$



Original source details: GELFAND-P001300

Uncertain reading. The same glyph appears in the exponent after $2-2$ and before 'Euler number'. The reading χ is tentative.

(to a vertex — $\frac{1}{\sqrt{N}}$)



Original source details: GELFAND-P001300

Undeciphered passage. The line above (7) is heavily crossed out with horizontal and oblique strokes. The original wording is undeciphered.

$$I(g) = \int d^{N^2}u \, \exp\left\{-\operatorname{tr}\left(u^2 + \frac{g}{\sqrt{N}}u^3\right)\right\} \tag{7}$$

(u a Hermitian matrix)

Computed in the work of Parisi, Itzykson, etc.

(in the lim. $N \to \infty$)

Pol. coord. (generalized)

$$u_{ij} = \sum_k \Omega^+_{ik}\lambda_k\Omega_{kj}, \tag{8}$$

Ω a unitary matrix

(separate out the eigenvalues)

$$I(g) = \int \exp\left\{-\sum_{i=1}^{N}\left(\lambda_i^2 + \frac{g}{\sqrt{N}}\lambda_i^3\right)\right\} d(\Omega)\prod d\lambda_j \prod_{i>j}(\lambda_i - \lambda_j)^2 \tag{9}$$



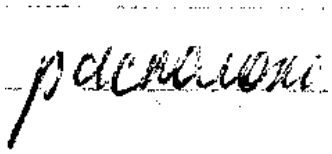
Uncertain reading. An illegible mark is cancelled between the integral sign and exp.

Original source details: GELFAND-P001300

$$\left(du = d(\Omega)\prod_{i>j}(\lambda_i - \lambda_j)^2 \prod_j d\lambda_j\right) \quad \text{— Dyson measure}$$

GELFAND-P001301

Dyson: [wavy underline] the most probable arrangement of the eigenvalues of a unitary matrix of high order is uniform on the circle.



Uncertain reading. The last strokes in the Russian abbreviation for “arrangement” run together.

Original source details: GELFAND-P001301

Rewrite

$$I(g) = \int d(\Omega) \prod d\lambda_j$$

$$\underbrace{\exp\left\{ -\sum_{i=1}^{N} \left(\lambda_i^2 + \frac{g}{\sqrt{N}} \lambda_i^3 \right) + 2 \sum_{i>j} \ln |\lambda_i - \lambda_j| \right\}}_{\sim N^2 \text{ terms}} \tag{10}$$

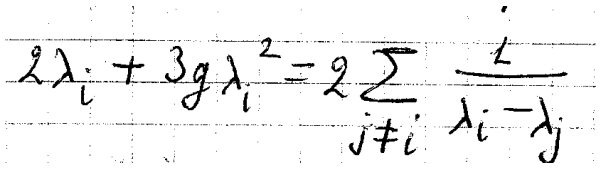
redefine $\lambda_i \rightsquigarrow$

$$I(g) = \int d(\Omega) \prod d\lambda_j$$

$$\exp\left\{ -N \sum_{i=1}^{N} (\lambda_i^2 + g\lambda_i^3) + 2 \sum_{i>j} \ln |\lambda_i - \lambda_j| \right\}$$

and it is calculated by the rules of the saddle-point method. The resulting eq. is

$$2\lambda_i + 3g\lambda_i^2 = 2 \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} \tag{11}$$



Uncertain reading. The manuscript writes (11) without an additional factor of N or $1/N$.

Original source details: GELFAND-P001301

In the limit $N \to \infty$ the eq. reduces to an integral equation!

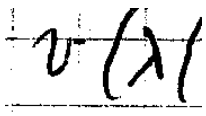
Introduce

$$x = \frac{i}{N}, \qquad \nu(\lambda(x)) = 2 \frac{dx}{d\lambda(x)} \quad \text{— eigenvalue density}$$

(in $\forall$ interval)

$$2\lambda + 3g\lambda^2 = \int \frac{d\mu\, \nu(\mu)}{\lambda - \mu} \quad \text{(principal value)}$$

$$\int d\lambda\, \nu(\lambda) = 2 \quad \text{— normalization.}$$

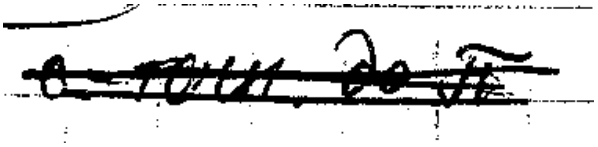


Uncertain reading. The density symbol in the numerator of the integral is overwritten. The reading ν follows the neighboring formulas.

Original source details: GELFAND-P001301

(Dispersion relation; representation of $\Re f(z)$ in terms of $\Im f(z)$)

$$F(\lambda) = 2\lambda + 3g\lambda^2 - \underbrace{[\alpha + \beta\lambda]\sqrt{(\lambda - 2a)(2b - \lambda)}}_{\pi\nu(\lambda)}$$

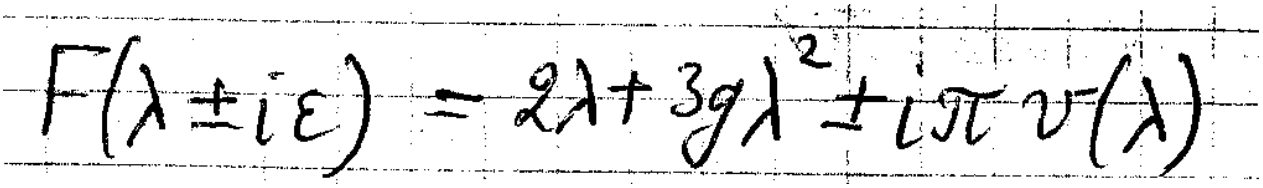


Original source details: GELFAND-P001301

Undeciphered passage. The note to the right of $\pi\nu(\lambda)$ is crossed out. “To” can be read; the remaining signs are uncertain.

a, b, α, β are functions of g; they are chosen.

$$F(\lambda \pm i\varepsilon) = 2\lambda + 3g\lambda^2 \pm i\pi\nu(\lambda)$$

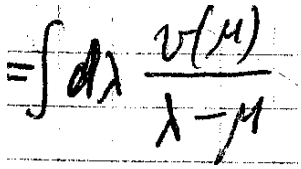


Original source details: GELFAND-P001301

Uncertain reading. The manuscript writes both signs as $\pm$.

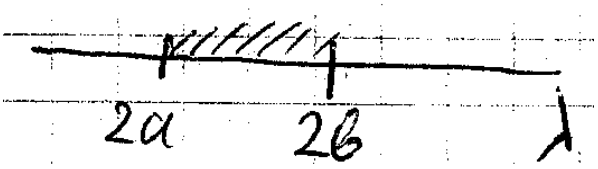
(Continuous spectrum! — analytic in a domain with a cut)

$$F(\lambda) = \int d\lambda \, \frac{\nu(\mu)}{\lambda - \mu}$$



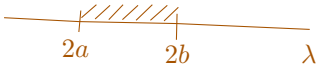
Original source details: GELFAND-P001301

Uncertain reading. The manuscript gives $d\lambda$ as the differential and μ in the fraction.



Original source details: GELFAND-P001301

Diagram and typed description.



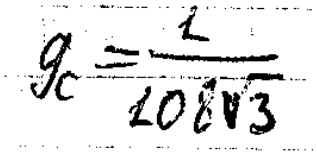
$$F(\lambda) \underset{\lambda\to\infty}{\longrightarrow} \frac{2}{\lambda}$$

α, β, a, b are found so as to cancel the second-degree polynomial.

Answer

$$F(g) = Z = -\frac{1}{2}\sum_{k=1}^{\infty} \frac{(72g^2)^k \Gamma\left(\frac{3k}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)(k+2)!} \tag{12}$$

Critical constant $g_c = \frac{1}{108\sqrt{3}}$, singularity $F(g) \approx (g - g_c)^{5/2}$.



Uncertain reading. The last digit in 108 is read from the lower loop and upper stroke. The manuscript gives g_c without a square.

Original source details: GELFAND-P001301



Undeciphered passage. Beyond the sheet's right edge, three opening parentheses are visible with cropped strokes. Their continuations lie outside the frame.

Original source details: GELFAND-P001301

GELFAND-P001302

[source number: 10]

The number F_n of triangulations with n vertices is then found asymptotically

$$F_n \sim n^{-7/2} g_c^{-2n}$$

Possible generalization: [wavy underline] spin variables at $\forall$ vertex of the triangulation

GELFAND-P001303



Uncertain reading. A short isolated oblique stroke appears above the date and heading. Its meaning is uncertain.

Original source details: GELFAND-P001303

13.10.86 Gelfand Sem.

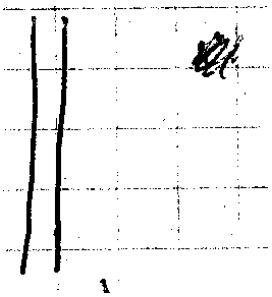
D. Gurevich. Generalized Lie algebras and groups.

Constant solutions of the Yang–Baxter equations

$\Downarrow$

generalization of Lie theory

analogous to the inclusion (supermathematics) $\supset$ (mathematics)



Original source details: GELFAND-P001303

Diagram and typed description.



Two parallel strokes to the right of the text. The small crossed-out mark at upper right is illegible.

$$L, \quad \dim L = n$$

$$L^{\otimes 2} \xrightarrow{S} L^{\otimes 2} \quad \text{(linear maps)}$$

$$L^{\otimes 3} \underset{\mathrm{id}\otimes S = S^{(2)}}{\xrightarrow{S\otimes \mathrm{id} = S^{(1)}}} L^{\otimes 3}$$

(1) Y.–B. equation:

$$1) \quad S^{(1)}S^{(2)}S^{(1)} = S^{(2)}S^{(1)}S^{(2)}$$

$$2) \quad S^2 = \mathrm{id} \quad \overset{\text{def}}{\Longleftrightarrow} \quad S \text{ is a symmetry}$$

Generalization of the notion of a Lie algebra.

L is a finite-dimensional vector space

The structure of a generalized Lie algebra (or S-algebra) is a map [the end of the Russian word for “generalized” is below the line]

$$L^{\otimes 2} \xrightarrow{[\cdot,\cdot]} L \qquad \begin{matrix}\text{(linear)}\\ \text{map}\end{matrix}$$



Original source details: GELFAND-P001303

Uncertain reading. An illegible mark is crossed out before the formula.



Original source details: GELFAND-P001303

Uncertain reading. The superscript at the right of the final L is crossed out. The reading under the strokes is uncertain.

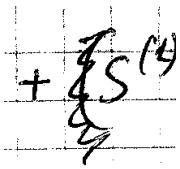
and the axioms hold

$$1) \quad [\cdot,\cdot] = -S[\cdot,\cdot]$$

(S is a symmetry operator satisfying axioms $(1)_{1,2}$ and is fixed)

(operators are written on the right)

$$2) \quad \big(\mathrm{id} + S^{(1)}S^{(2)} + S^{(2)}S^{(1)}\big)\big([\cdot,\cdot]\otimes \mathrm{id}\big)[\cdot,\cdot] = 0$$



Uncertain reading. A mark after the first $+$ and before $S^{(1)}$ is crossed out. Its reading is uncertain.

Original source details: GELFAND-P001303

Here $\mathrm{id} = \mathrm{id}_{L^{\otimes 3}}$

$$3) \quad S^{(1)}S^{(2)}\big([\cdot,\cdot]\otimes \mathrm{id}\big) = \big(\mathrm{id}\otimes[\cdot,\cdot]\big)S$$

Better to write $S^{(1)} = S^{(12)}, \quad S^{(2)} = S^{(23)}$.

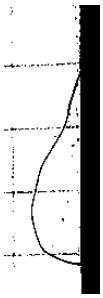


Diagram and typed description.

A thin line at the right edge. Its continuation beyond the visible edge is uncertain.

Original source details: GELFAND-P001303

GELFAND-P001304

S is a permutation of the factors $\Rightarrow$ what do the axioms mean?

Axioms (1) mean that S gives a rep. of the permutation group on 3 elements.

2) becomes the Jacobi identity [wavy underline] for S a permutation and [,] the commutator.

3) a tautology in the ordinary situation.

Supermathematics.

$t_1, \ldots, t_n$ is a homogeneous basis

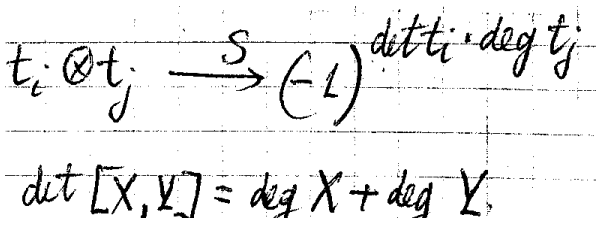


Uncertain reading. A struck-out mark appears before t_1. Its reading is uncertain.

Original source details: GELFAND-P001304

$$t_i \otimes t_j \xrightarrow{S} (-1)^{\mathrm{dt}\, t_i \cdot \deg t_j}\, t_j \otimes t_i$$

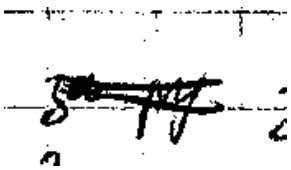
$$\mathrm{dt}[X, Y] = \deg X + \deg Y.$$



Uncertain reading. The two forms before t_i and $[X, Y]$ are rendered dt. The middle stroke is unclear. The other operators read deg.

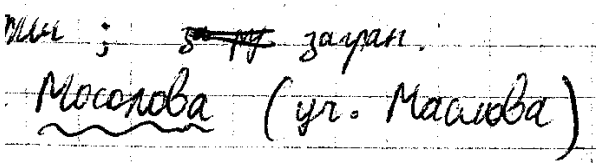
Original source details: GELFAND-P001304

Another example: [wavy underline] color symmetries; specified by Mosolov [wavy underline] (Maslov's student)



Uncertain reading. A word before "specified" is crossed out. It may begin with Russian "з".

Original source details: GELFAND-P001304



Uncertain reading. The surname is provisionally read "Мосолова" (Mosolov) and the abbreviation in parentheses "уч." (student).

Original source details: GELFAND-P001304

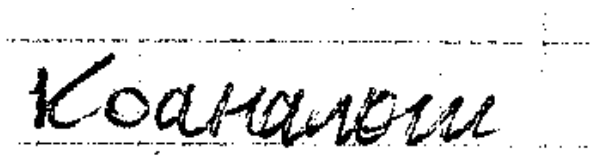
Color mathematics [broken underline]

$$S: \quad t_i \otimes t_j \longmapsto a(i,j) t_j \otimes t_i$$

(Latest work by Trostel, 1984, J. Math. Phys., 25, $N = 11$)

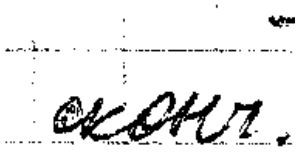
The $a(i, j)$ must satisfy certain conditions (written down from the axioms)

The braids in their final form were devised by Drinfeld.



Uncertain reading. The first word is provisionally read "Косички" (braids). Several letters in the middle run together.

Original source details: GELFAND-P001304



Uncertain reading. The abbreviation is read "оконч." (final). Its last strokes run together.

Original source details: GELFAND-P001304

The associative algebras will be the same.

An S-algebra is constructed from an assoc. algebra as follows

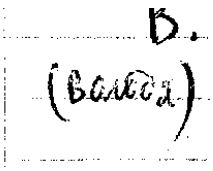
$$[A, B] = AB - (AB)S$$

More precisely, there is a multiplication operator $A^{\otimes 2} \overset{\circ}{\to} A$

$$[f, g] = (f \otimes g) \circ -((f \otimes g)S) \circ$$

It will satisfy all the axioms.

V. Kazakov. Another model of discrete 2-dimensional gravity. (Volodya)



Original source details: GELFAND-P001304

Uncertain reading. The parenthetical name under the initial on the left is provisionally read “Володя” (Volodya).

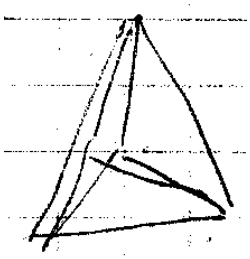
Consider a triangulation of S^2, assign additional variables $\sigma_i = \pm 1$ at the vertices and use Ising statistics



Original source details: GELFAND-P001304

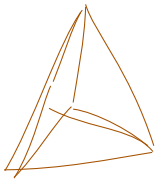
Uncertain reading. A sign before S^2 is crossed out. Its reading is uncertain.

$$Z_n = \sum_{\substack{\text{triang.} \\ \text{of the sphere}}} \sum_{\{\sigma_i\}} \exp[\beta G_{ij} \sigma_i \sigma_j]. \tag{1}$$



Original source details: GELFAND-P001304

Diagram and typed description.



Some slanted lines are retraced. The joins near the middle and at the lower-right corner are unclear.

G_{ij} is the adjacency matrix of the triangulation,

$$G_{ij} = \begin{cases} 1, & \text{if } i, j \text{ are neighbors} \\ 0 & \text{otherwise} \end{cases}$$

GELFAND-P001305

[source number: 14]



Uncertain reading. A short group of letters is crossed out before the first line. Its reading is uncertain.

Original source details: GELFAND-P001305

Why is this a model of QG (quantum grav.)? And of which QG?

Consider the continuous theory

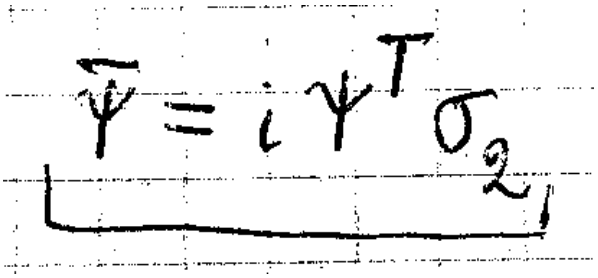
$$Z = \int \mathcal{D}g \int \mathcal{D}\Psi \, \exp\left(-\int d^2\xi \left[\mu\sqrt{g} + \bar{\Psi}\sigma_\mu \partial_\mu \Psi + g^{1/4} m \bar{\Psi}\Psi\right]\right) \tag{2}$$

Ψ is a 2-dimensional Majorana fermion in curved space/time

$(2) \sim (1)$ The proof is difficult, σ_μ are Pauli matrices

Solution of (2): Migdal and Bershadsky [wavy underline]

$\bar{\Psi} = i\Psi^T\sigma_2$ — relation for a Majorana fermion



Uncertain reading. A squared bracket is drawn beneath $\bar{\Psi} = i\Psi^T\sigma_2$.

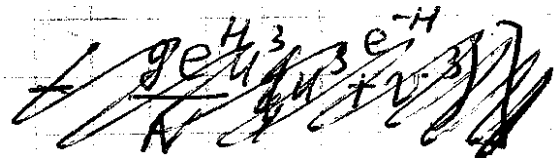
Original source details: GELFAND-P001305

($\Psi, \bar{\Psi}$ are not independent as they are for Dirac fermions)
Instead of (1), a more general one is considered below (with a magnetic field)

$$Z_n = \sum_{\{G^{(n)}\}} \sum_{\{\sigma_i\}} \exp\left[\beta \sum_{i,j=1}^{n} G_{ij}\sigma_i\sigma_j + H\sum_i \sigma_i\right] \tag{3}$$

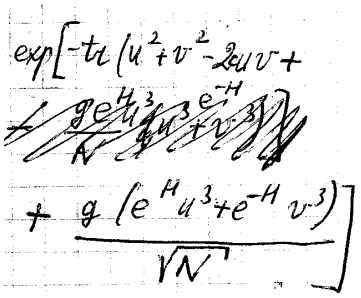
[At right, with a bracket around the note:] $G^{(n)}$ is a graph with n vertices
Introduce the model

$$\begin{gathered} Z(g) = \sum_n g^n Z_n(\beta) \overset{N\to\infty}{=} \frac{1}{N^2}\log \int d^{N^2}u \int d^{N^2}v \\ \exp\big[-\operatorname{tr}\big(u^2+v^2-2cuv+ \\ \text{[crossed-out term]} \\ + \frac{g(e^H u^3 + e^{-H}v^3)}{\sqrt{N}}\big] \end{gathered} \tag{4}$$



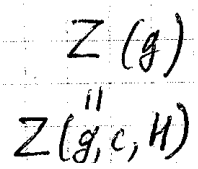
Original source details: GELFAND-P001305

Undeciphered passage. At left the crossed-out line may read $\frac{ge^H}{N}u^3$. Farther right, u^3, a raised e^{-H} and $+v^3$ can be distinguished. The grouping and the end of the entry are uncertain.



Original source details: GELFAND-P001305

Uncertain reading. The manuscript shows $+$ at the end of the first exponent line, $+$ at the start of the replacement line, and a square bracket after the fraction. A separate closing parenthesis for tr(is uncertain.



Original source details: GELFAND-P001305

Uncertain reading. Below the left-hand $Z(g)$ is $Z(g,c,H)$, with two short vertical strokes between the lines.

u, v are $N \times N$ Hermitian matrices

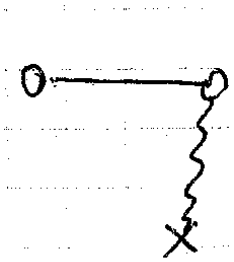
Here $c = e^{-2\beta}$

Solved in work by Zuber, Itzykson and . . .

The coupling constant g may be taken to be complex.

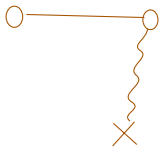
I.M.G. There is a book by Dyson (in Russian): all reasonable problems in symmetric spaces as $N \to \infty$ have an answer.

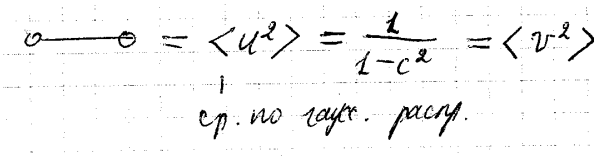
Proof of formula (4): expand exp $\Rightarrow$ diagrammatic technique. Vertices of two types: u-vertices and v-vertices.



Original source details: GELFAND-P001305

Diagram and typed description.



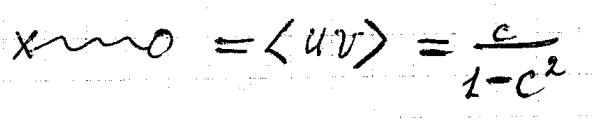


Original source details: GELFAND-P001305

Diagram and typed description.

$\circ\!\!-\!\!\circ = \langle u^2 \rangle = \dfrac{1}{1-c^2} = \langle v^2 \rangle$

[Below $\langle u^2 \rangle$, with a vertical stroke:] average over the Gaussian distribution.



Original source details: GELFAND-P001305

Diagram and typed description.

$\times\!\sim\!\circ = \langle uv \rangle = \dfrac{c}{1-c^2}$

GELFAND-P001306

What is there in model (1)?

Phase transition (spontaneous ordering of the spins) at the value

$$\beta_t^* = \tfrac{1}{2} \ln(2\sqrt{7} + 1).$$



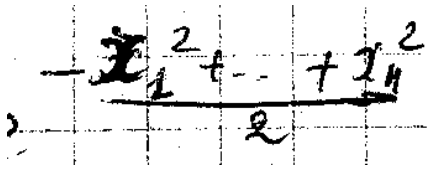
Original source details: GELFAND-P001306

Uncertain reading. The first strokes of the Russian abbreviation for “phase” are overwritten.

Varchenko

Wick's theorem [broken underline]

$$\frac{1}{(2\pi)^{n/2}} \int e^{-\frac{x_1^2+\cdots+x_n^2}{2}} \xi_1 \cdots \xi_N \, dx \stackrel{\text{def}}{=} \langle \xi_1 \cdots \xi_N \rangle$$



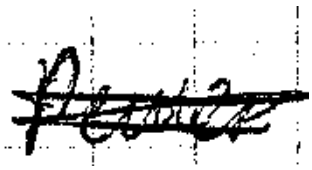
Original source details: GELFAND-P001306

Uncertain reading. The first sign in the exponent numerator is overwritten. The visible entry is rendered x_1^2. The earlier reading is uncertain.

Then

$$\langle \xi_1 \cdots \xi_N \rangle = \sum_{\substack{\text{arrangement} \\ \text{of brackets}}} \langle \xi_{i_1} \xi_{i_2} \rangle \cdots \langle \xi_{i_{N-1}} \xi_{i_N} \rangle$$

ξ_j are linear functions of the x_k

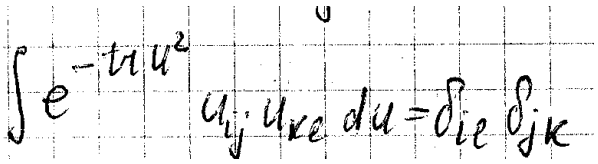


Original source details: GELFAND-P001306

Uncertain reading. The crossed-out word before "Proof" is provisionally read "Решение" (Solution).

Proof. [broken underline] first $\xi_i = x_{k_i} \Rightarrow$ integral = a product, and it remains to use multilinearity

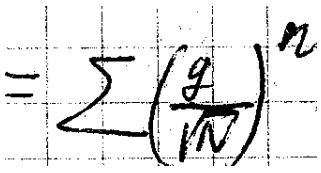
$$\int e^{-\operatorname{tr} u^2} u_{ij} u_{k\ell} \, du = \delta_{i\ell} \delta_{jk}$$



Original source details: GELFAND-P001306

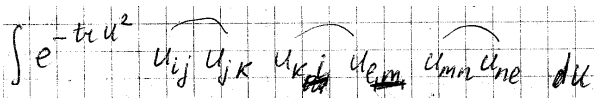
Uncertain reading. The measure is written du; there are no explicit normalization factors in the line.

$$\int du \, e^{-\operatorname{tr}(u^2 + \frac{g}{\sqrt{N}} u^3)} = \sum \left(\frac{g}{\sqrt{N}} \right)^n \int e^{-\operatorname{tr} u^2} (\operatorname{tr} u^3)^n \, du$$



Original source details: GELFAND-P001306

Uncertain reading. The sum has no written limits, and its coefficient has no written minus sign or factorial.



Original source details: GELFAND-P001306

Diagram and typed description.

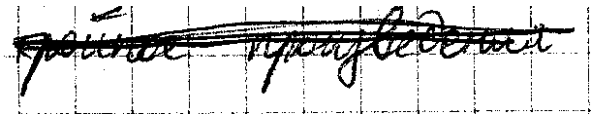
$$\int e^{-\operatorname{tr} u^2} \; \overbrace{u_{ij} u_{jk}} \; \overbrace{u_{ki} u_{\ell m}} \overbrace{u_{mn} u_{n\ell}} \; du$$



Original source details: GELFAND-P001306

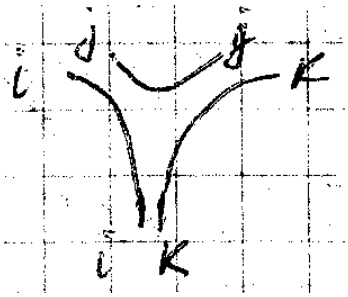
Uncertain reading. Under the third and fourth factors, some indices are crossed out. The uncrossed indices are read ki and ℓm.

Arrangement of brackets ⇒ [crossed out: triple product] graph.



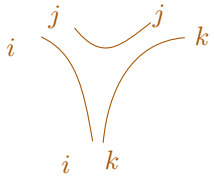
Original source details: GELFAND-P001306

Uncertain reading. The crossed-out words are provisionally read “тройное произведение” (triple product).



Original source details: GELFAND-P001306

Diagram and typed description.

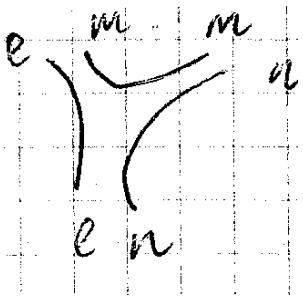


$$\delta_{ik}\delta_{kj}\delta_{km}\delta_{i\ell}\delta_{m\ell}\delta_{nn}$$



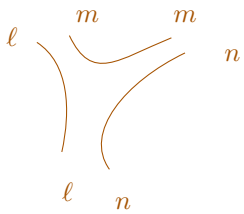
Original source details: GELFAND-P001306

Uncertain reading. The first index of the second factor is retraced. The visible strokes are rendered δ_{kj}.



Original source details: GELFAND-P001306

Diagram and typed description.





Original source details: GELFAND-P001306

Diagram and typed description.



A short, thin wavy stroke in the lower-left margin. It has no label.

GELFAND-P001307

20.10.86 Gelfand Sem.

V. Ginzburg.

On work by A. Givental. (and then on work by P. Deligne — L. Illusie)

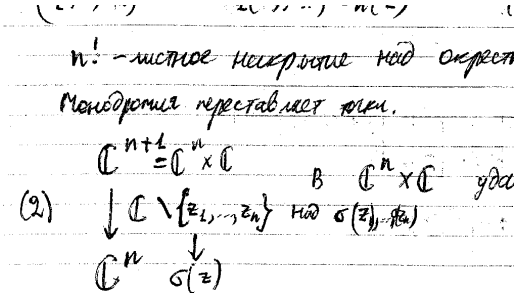
An explicit example of a representation of the Hecke algebra.

(arising from singularity theory)

$$\begin{aligned} &\mathbb{C}^n \longrightarrow \mathbb{C}^n \\ &(z_1, \dots, z_n) \longrightarrow \sigma_1(z), \dots, \sigma_n(z) \quad \text{(elem. symm. polynomials)} \end{aligned} \tag{1}$$

$n!$-sheeted covering over a neighborhood of a general point.

Monodromy permutes the points.



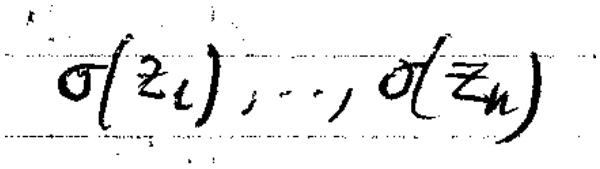
Original source details: GELFAND-P001307

Diagram and typed description.

$$\begin{array}{cll} & \mathbb{C}^{n+1} = \mathbb{C}^n \times \mathbb{C} & \\ (2) & \downarrow & \mathbb{C} \setminus \{z_1, \dots, z_n\} \text{ over } \sigma(z_1, \dots, z_n) \\ & \mathbb{C}^n & \downarrow \sigma(z) \end{array}$$

The last argument in the label of the fiber is overwritten. The reading z_n follows the adjacent notation.

In $\mathbb{C}^n \times \mathbb{C}$, remove $\{z_1, \dots, z_n\}$ over $\sigma(z_1), \dots, \sigma(z_n)$.



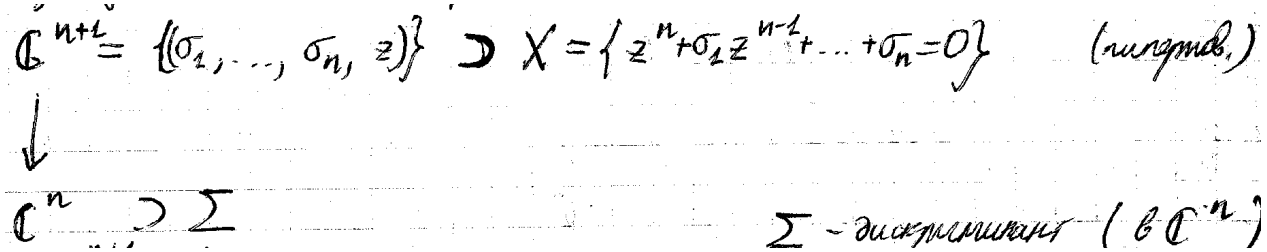
Original source details: GELFAND-P001307

Uncertain reading. The last line writes $\sigma(z_1), \dots, \sigma(z_n)$ separately. In the diagram a single σ takes the list of arguments.

GELFAND-P001308

[source number: 20]

Remove from the base the σ that correspond to roots that have coalesced.



Original source details: GELFAND-P001308

Diagram and typed description.

$$\begin{array}{l} \mathbb{C}^{n+1} = \{(\sigma_1, \dots, \sigma_n, z)\} \supset \\ \qquad X = \{z^n + \sigma_1 z^{n-1} + \dots + \sigma_n = 0\} \\ \downarrow \text{ (hypersurface)} \\ \mathbb{C}^n \supset \Sigma \end{array}$$

Σ is the discriminant (in $\mathbb{C}^n$).

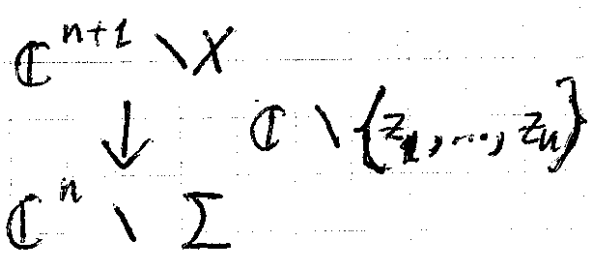


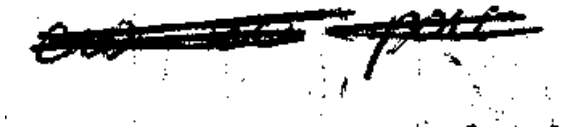
Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^{n+1} \setminus X & & \\ \downarrow & & \mathbb{C} \setminus \{z_1, \ldots, z_n\} \\ \mathbb{C}^n \setminus \Sigma & & \end{array}$$

Original source details: GELFAND-P001308

Pass to 1-dimensional homology with coefficients

f^α — α is a fixed complex number; [crossed out] multivalued function



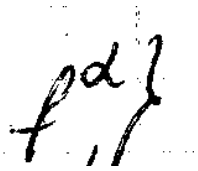
Undeciphered passage. Two groups of letters are crossed out after “number”. Their reading is uncertain.

Original source details: GELFAND-P001308

$$f = z^n + \sigma_1 z^{n-1} + \cdots + \sigma_n$$

f^α defines a covering; consider cohomology with coefficients in it

$$H^1\left(\mathbb{C} \setminus \{*, *, \ldots, *\}, f^\alpha\right\}$$



Uncertain reading. The left delimiter of H^1 is round. The mark after f^α resembles a closing brace.

Original source details: GELFAND-P001308

There is a rep. of the braid group $\Rightarrow$ of the fund. group (of braids) $B(n)$, depending on the parameter α. [Below $B(n)$: of n strands.]

fund. group of $\mathbb{C}^n \setminus \Sigma = B(n)$.

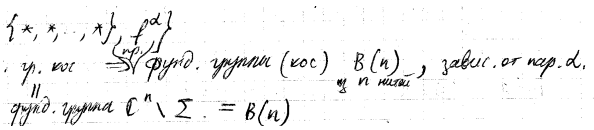


Diagram and typed description.

$$\begin{array}{ccc} & f^\alpha & \\ & \text{repr.} & \\ \text{braid group} & \Rightarrow & \text{fund. group} \\ || & & \\ \mathbb{C}^n \setminus \Sigma = B(n) & & \end{array}$$

The curved strokes below f^α meet in a point beside the arrow. The reading “repr.” and the attachment of the strokes to the words are provisional. Two vertical strokes appear below “braid group”.

Original source details: GELFAND-P001308

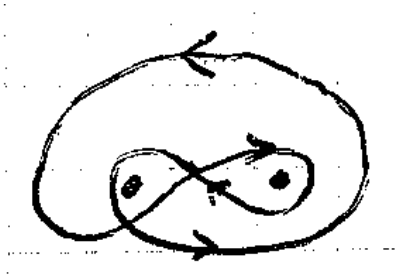
The braid group has generators $T_1, \ldots, T_{n-1}$ with rel.

$$T_1 T_2 T_1 = T_2 T_1 T_2.$$

(T_j corresponds to an elementary circuit)
This rep. has an additional relation.

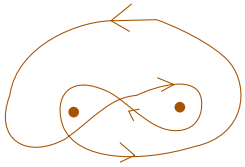
$$(T_i + 1)(T_i - \alpha) = 0 \quad \Rightarrow$$

$\Rightarrow$ the rep. factors through a rep. of the Hecke alg.



Original source details: GELFAND-P001308

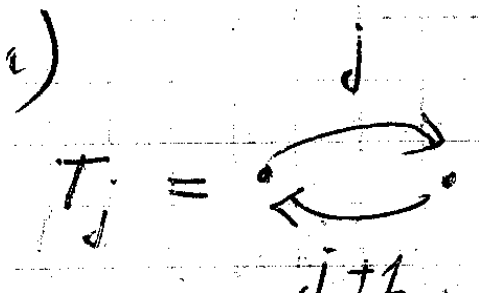
Diagram and typed description.



The points and arrow directions are visible. The exact connections at the two crossings are uncertain.

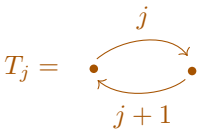
$$\pi_1(\mathbb{C}^n \setminus \Sigma) = B(n)$$

generators



Original source details: GELFAND-P001308

Diagram and typed description.



(lin. transformation corresponding to a transposition of 2 roots)

$$\downarrow$$

connect to Id.

1) $T_i T_j = T_j T_i$ when $|i - j| \geq 2$
2) $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$
3) $(T_i + 1)(T_i - \alpha) = 0$

The aim was to prove faithfulness of the rep. of the fund. group in a group of matrices whose entries are Laurent polynomials.

For [crossed out] $n = 3$ this is known.



Original source details: GELFAND-P001308

Uncertain reading. An entry before $n = 3$ is crossed out. The visible strokes could be n, but the reading is uncertain.



Uncertain reading. A faint curved mark appears to the right of “known”. Its meaning is uncertain.

Original source details: GELFAND-P001308

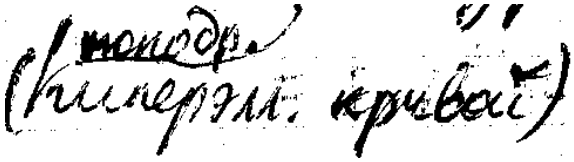
Specialists in braid group theory call this the Burau representation.



Uncertain reading. The manuscript spells the name “Бурао” (Burau).

Original source details: GELFAND-P001308

$\alpha = -1 \Rightarrow$ monodromy group of elliptic integrals ([inserted above: monodr.] of a hyperelliptic curve)



Uncertain reading. The abbreviation “монодр.” (monodr.) is written above “hyperell.” and underlined with a curved connecting stroke.

Original source details: GELFAND-P001308

GELFAND-P001309

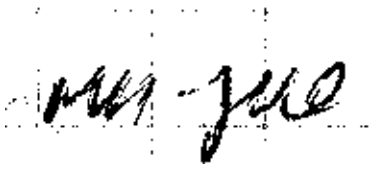
Beilinson

on work by P. Deligne and L. Illusie. Preprint Orsay

Algebraic proofs of facts formerly proved only by transcendental methods

(Laplace operator, etc.)

X is a smooth compact alg. variety over $\mathbb{C}$



Uncertain reading. The initial strokes in the Russian abbreviation “мн-ие” for “variety” run together. The reading follows the context.

Original source details: GELFAND-P001309

De Rham complex: [wavy underline]

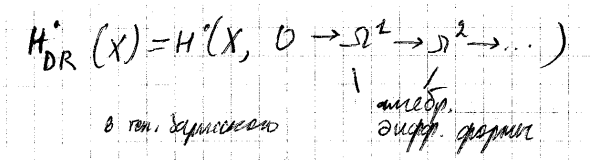
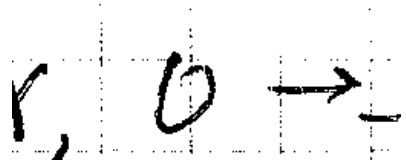


Diagram and typed description.

$H^{\bullet}_{\mathrm{DR}}(X) = H^{\bullet}(X, O \to \Omega^1 \to \Omega^2 \to \dots)$

in the Zariski top. — algebraic diff. forms

Original source details: GELFAND-P001309



Uncertain reading. The initial term of the complex is written with a sign resembling O.

Original source details: GELFAND-P001309

Thm.

$$E_1^{p,q} = H^p(X, \Omega^q)$$

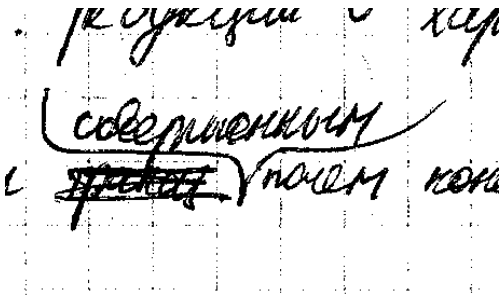
(the spectral sequence degenerates already at the first term)

1st [inserted above: alg.] proof — Faltings — nonelem.

Deligne and Illusie [wavy underline] — elementary algebraic

Idea of the proof: in place of Δ use reduction to char. p!!

X/k over a finite field k or a [crossed out] [inserted above: perfect] field of finite char.



Original source details: GELFAND-P001309

Undeciphered passage. The word before "field" is crossed out and its reading is uncertain. The insertion "perfect" is legible.

There is cohomology.

Kazakov V. (continued).

Formula on which the evaluation of the two-matrix integral is based

$$\int d(\Omega)_{U(N)} e^{\mathrm{tr}(\lambda\Omega^+\mu\Omega)} = \left(\prod_{k=1}^{N-1} k!\right) \frac{\det_{i,j} e^{\lambda_i\mu_j}}{\Delta(\lambda)\Delta(\mu)} \tag{1}$$

λ, μ are diagonal matrices

$$\Delta(\lambda) = \prod_{i>j}(\lambda_i - \lambda_j)$$

$$\lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_N \end{pmatrix}, \qquad \mu = \begin{pmatrix} \mu_1 & & 0 \\ & \ddots & \\ 0 & & \mu_N \end{pmatrix}$$

First in work by Itzykson, Zuber. Proof using the heat eq. on the matrix group.

I.M.G. There is a similar formula in the book by Gelfand, Naimark [wavy underline], "Unit. rep. of class. groups" (1951)

$$\Omega^{-1}\lambda\Omega = H \qquad \Delta_1^{\mu_1}\Delta_2^{\mu_2}\dots\Delta_n^{\mu_n}$$



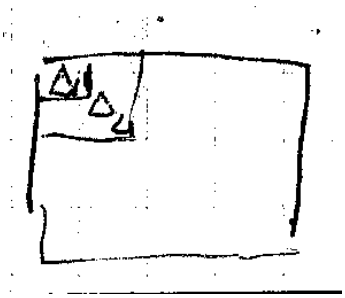
Original source details: GELFAND-P001309

Uncertain reading. The first glyph in $\Omega^{-1}\lambda\Omega$ is retraced. The visible outline reads Ω.

Ω is unitary

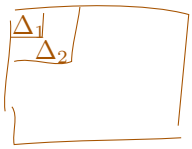
λ is diag. ($\lambda_i > 0$)

(principal minors)



Original source details: GELFAND-P001309

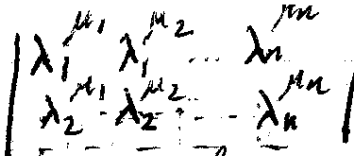
Diagram and typed description.



GELFAND-P001310

[source number: 21]

$$\int \Delta_1^{\mu_1} \Delta_2^{\mu_2} \dots \Delta_n^{\mu_n} \, d\Omega$$
$$= \begin{vmatrix} \lambda_1^{\mu_1} & \lambda_1^{\mu_2} & \dots & \lambda_n^{\mu_n} \\ \lambda_2^{\mu_1} & \lambda_2^{\mu_2} & \dots & \lambda_n^{\mu_n} \\ - & - & - & - \end{vmatrix} \left[\prod (\lambda_i - \lambda_k) \prod (\mu_i - \mu_k) \right]^{-1} \tag{2}$$



Original source details: GELFAND-P001310

Uncertain reading. In the right-hand column of both rows the visible indices read $\lambda_n^{\mu_n}$. Short horizontal strokes appear below the two rows.



Original source details: GELFAND-P001310

Uncertain reading. The lower-left part of Ω in $d\Omega$ is retraced. The earlier sign is indistinct.

(written in the paper of G. and N.)

From this we obtain Lidskii's [wavy underline] thm. on the eigenvalues of a product of (Hermitian) matrices [below the parentheses: $H_j > 0$].

What can they be?

The eigenvalues of $H_1 H_2$ are always > 0. The region is described.

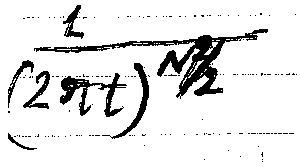
Proof using elliptic coordinates (generalized)

K. Proof of formula (1).

Write the heat equation on Hermitian matrices

$$\left(\frac{\partial}{\partial t} - \frac{1}{2}\Delta_u\right) f(t,u,v) = 0, \quad t > 0; \qquad f|_{t=0} = \delta(u-v) \tag{3}$$

$$f(t,u,v) = \frac{1}{(2\pi t)^{N^2/2}} e^{-\frac{1}{2t}\operatorname{tr}(u-v)^2} \tag{4}$$



Uncertain reading. The exponent of $(2\pi t)$ is retraced. A thick mark appears beside N above the slash. The dimension supports $N^2/2$, while $N/2$ remains possible from the strokes.

Original source details: GELFAND-P001310

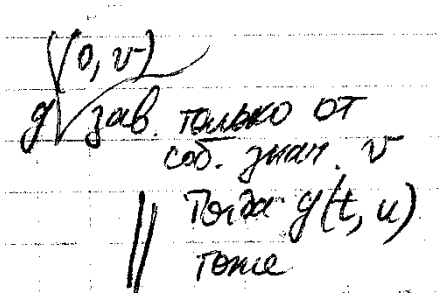
$\Delta_u = \sum$ of 2nd derivatives with respect to Re and Im of the entries (incl. diagonal)

$$g(t,u) = \int d^{N^2} v \, f(t,u,v) g(0,v) \tag{5}$$

Pass to Dyson's variables (variables in terms of eigenvalues), assuming g[inserted above: $(0,v)$] depends only on the eigenvalues of v

$$g(t,\mu) = \int d(\Omega) \int \prod d\lambda \, \Delta^2(\lambda) f(t,\mu,\Omega^+ \lambda \Omega) g(0,\lambda) \tag{6}$$

[In the right margin, after a double vertical stroke:] then $g(t,u)$ does too.



Uncertain reading. The insertion $(0,v)$ is linked to g by a curved stroke. The separate remark on the right is marked by two vertical strokes.

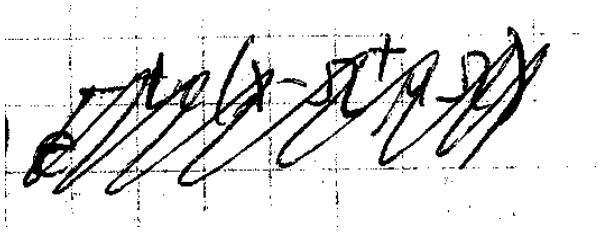
Original source details: GELFAND-P001310

$$u = \Omega_1^+ \mu \Omega_1, \qquad v = \Omega_2^+ \lambda \Omega_2; \quad \text{then} \quad \Omega = \Omega_2 \Omega_1^+$$

$$\underbrace{\Delta(\mu) g(t,\mu)}_{\xi(t,\mu)} = \int \prod d\lambda \, K(t;\mu,\lambda) \Delta(\lambda) g(0,\lambda) \tag{7}$$

Here

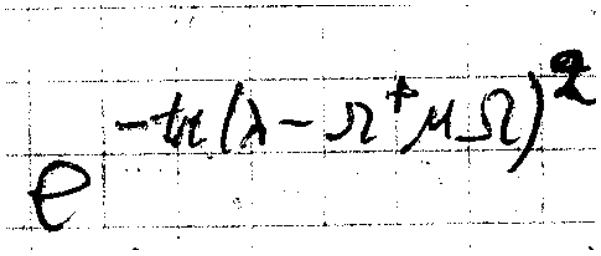
$$K(t,\mu,\lambda) = \Delta(\lambda)\Delta(\mu) \int d(\Omega)\, f(t,\mu,\Omega^+ \lambda \Omega) \tag{8}$$



Uncertain reading. The expression crossed out after (8) provisionally reads $e^{-\operatorname{tr}(\lambda-\Omega^+\mu\Omega)}$. Its final strokes are unclear.

Original source details: GELFAND-P001310

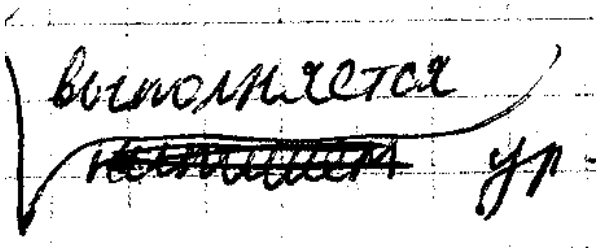
$$e^{-\operatorname{tr}(\lambda-\Omega^+\mu\Omega)^2}$$



Uncertain reading. Below (8) is a separate exponential with a square after the closing parenthesis. No factor of $1/(2t)$ is written in this line.

Original source details: GELFAND-P001310

For $\xi(t,\mu)$ [crossed out] the eq. [inserted above: holds] (!)



Uncertain reading. The crossed-out word may read “we will write”. The word “holds” is inserted above it with a pointer.

Original source details: GELFAND-P001310

$$\left(\frac{\partial}{\partial t} - \frac{1}{2} \sum_{i=1}^{N} \frac{\partial^2}{\partial \lambda_i^2} \right) \xi(t,\lambda) = 0, \quad t > 0. \tag{9}$$



Uncertain reading. A mark resembling f is crossed out before ξ. Its reading is uncertain.

Original source details: GELFAND-P001310

Although

$$\left(\frac{\partial}{\partial t} - \frac{1}{2} \Delta_\lambda \right) g(t,\lambda) = 0,$$
$$\text{where} \quad \Delta_\lambda = \frac{1}{\Delta(\lambda)^2} \sum_i \frac{\partial}{\partial \lambda_i} \left(\Delta^2(\lambda) \frac{\partial}{\partial \lambda_i} \right) \tag{10}$$

I.M.G. says this is in F. A. Berezin's candidate dissertation, Proc. Moscow Math. Soc. (≈ 1956)

GELFAND-P001311

Substituting $\dfrac{\xi(t,\lambda)}{\Delta(\lambda)} = g(t,\lambda)$ into (10), we arrive at (9).



Original source details: GELFAND-P001311

Uncertain reading. The letter before (9) is heavily retraced. The visible strokes are read "к" (at).

I.M.G. Berezin did it, and Harish-Chandra used it repeatedly.

K. Now write $\xi(t,\lambda)$ and so on.

Question: how to evaluate the integral

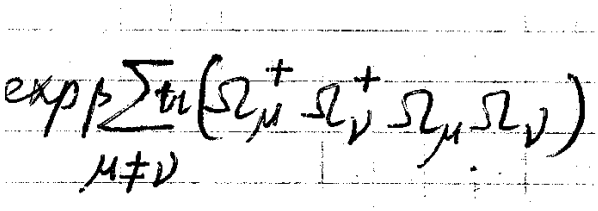
$$\int d\Omega\, e^{\mathrm{tr}(\lambda_1\Omega\mu_1\Omega^+)+\mathrm{tr}(\lambda_2\Omega\mu_2\Omega^+)} \tag{11}$$

— solution of the chiral field on $U(\infty)$

Evaluated in principle (Wiegmann and Polyakov solved the chiral field —
— Eguchi–Kawai reduction)

The most interesting case is connected with chromodynamics

$$\int \prod_{\mu=1}^{d} d\Omega_\mu \,\exp\beta \sum_{\mu\neq\nu} \mathrm{tr}(\Omega_\mu^+\Omega_\nu^+\Omega_\mu\Omega_\nu) \tag{12}$$



Original source details: GELFAND-P001311

Uncertain reading. The sign after exp and before the sum reads β. The manuscript has no outer delimiters around the exponent.

not evaluated, and gauge theory is unsolved [wavy underline]

All the d-dimensional physics is contained here ($d \leq 4$ suffices).

(12) is the Y.–M. partition function on the lattice.

GELFAND-P001312

27/X̄-86 Gelfand Sem.

Zakharevich

V is the space of $N \times N$ Hermitian matrices

Consider a function $g(t,x)$, $t \in \mathbb{R}$, $x \in V$

and the heat eq.

$$g_t = \Delta g \tag{1}$$

Claim.

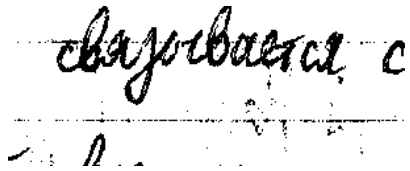
$$\int_{U(N)} d\Omega\, e^{\mathrm{tr}(\lambda\Omega^+\mu\Omega)} = \left(\prod_{k=1}^{N-1} k!\right) \frac{\det \|e^{\lambda_i\mu_j}\|}{\Delta(\lambda)\Delta(\mu)}, \tag{2}$$

$$\lambda = \mathrm{diag}(\lambda_i), \qquad \mu = \mathrm{diag}(\mu_i),$$

$$\Delta(\lambda) = \prod_{i>j}(\lambda_i - \lambda_j)$$

I.M.G. I.M. Gelfand, Naimark, Gindikin and Karpelevich, and Fateev also evaluated such integrals,

Important: this is connected with the combinatorial problem of triangulations of the sphere.



Original source details: GELFAND-P001312

Uncertain reading. The verb reads "связывается" (is connected).

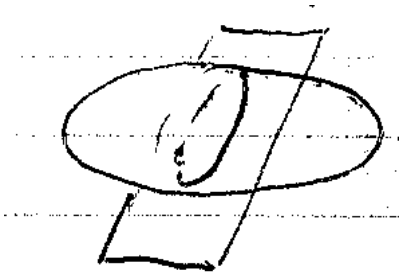
Problem from which representation theory arose

$$H_1 H_2 \qquad H_j^* = H_j > 0.$$

Given the eigenvalues of H_j, how are the eigenvalues of $H_1 H_2$ distributed?

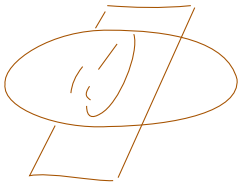
Elliptic coordinates on the set of unitary matrices

Consider an ellipsoid with axes a, b, c



Original source details: GELFAND-P001312

Diagram and typed description.



The inner line is broken at left. Two thickened ends of the quadrilateral are indistinct.

a_1, b_1 are the axes of the ellipse obtained in the section

$$a \le a_1 \le b \le b_1 \le c$$

$$\begin{array}{ccccccc} a & & b & & c & & d \\ & a_1 & & b_1 & & c_1 & \\ & & a_2 & & b_2 & & \\ & & & a_3 & & & \end{array}$$

We need to evaluate $d\Omega$ for matrices of order 2.

$$\int a_{11}^{\lambda_1} \left| \begin{array}{cc} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right|^{\lambda_2} \dots \, d\Omega,$$

λ_j are arbitrary parameters.

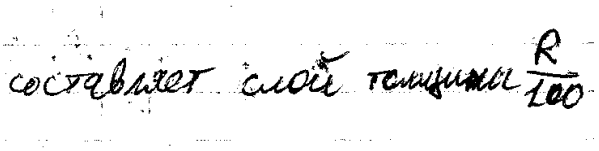
μ is a fixed diag. matrix.

$$A = \Omega^+ \mu \Omega$$

μ is given

Spectral analysis [broken underline]: analysis in $\mathbb{R}^N$ (or $\mathbb{C}^N$) as $N \to \infty$.

Question for schoolchildren [wavy underline]: what fraction of a 1000-dimensional watermelon is the layer of thickness $\frac{R}{100}$ around the equatorial plane? Answer [wavy underline]: almost all of it.



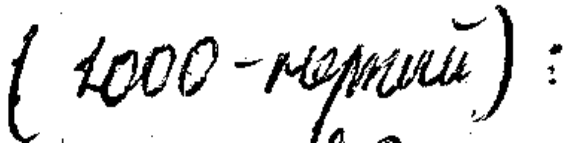
Uncertain reading. The manuscript gives the thickness as $R/100$.

Original source details: GELFAND-P001312

GELFAND-P001313

[source number: 24]

Given a simplex [broken underlining] (1000-dimensional), a particle reflects off the faces. [Crossed out] what fraction of the time will it spend in a $\frac{1}{100}$-neighborhood of the faces?



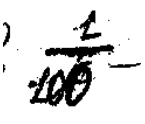
Uncertain reading. The digits read 1000.

Original source details: GELFAND-P001313



Undeciphered passage. A short word is crossed out before “what”. The opening loop allows the reading “она” (it).

Original source details: GELFAND-P001313



Uncertain reading. The fraction reads $\frac{1}{100}$. Its bar is thickened and strokes in the denominator meet in places.

Original source details: GELFAND-P001313

Answer [broken underlining]. Almost all the time.

B. Lidskii. Moment map [wavy underlining].



Original source details: GELFAND-P001313

Uncertain reading. The initial is provisionally read “Б.” (B.) from the lower stroke and catalogue record 690. The manuscript form can also be read “Е.” (E.).

Consider Hermitian matrices with fixed eigenvalues and the map on them

$$H \longrightarrow \text{diagonal of } H$$

The Fourier transform of the moment map is written down.



Original source details: GELFAND-P001313

Uncertain reading. The Russian abbreviation reads “преобр.” (transform). The second stroke is thickened.

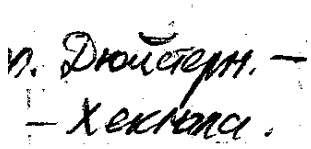
I.M.G. What is the superanalogue of the symmetric group?



Original source details: GELFAND-P001313

Uncertain reading. In the opening of “супераналог” (superanalogue), several strokes meet.

V. Ginzburg. Idea: find the critical points of the map. Take the first term of the asymptotics. It will be an exact formula by the Duisterm.-Heckman theorem.

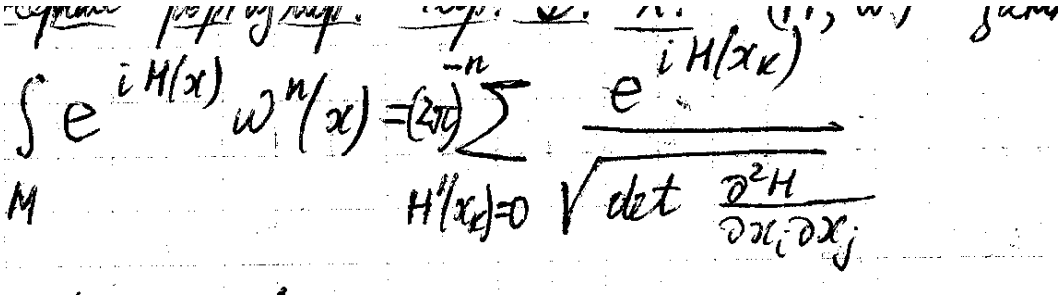


Original source details: GELFAND-P001313

Uncertain reading. The first surname is abbreviated, provisionally read “Дюйстерм.” (Duisterm.). “Хекмана” (Heckman) appears on the line below at right.

Approximate statement of the D.–H. theorem. (M, ω) is a closed symplectic manifold.

$$\int_M e^{iH(x)}\omega^n(x) = (2\pi)^{-n} \sum_{H'(x_k)=0} \frac{e^{iH(x_k)}}{\sqrt{\det \frac{\partial^2 H}{\partial x_i \partial x_j}}}$$

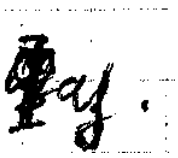


Original source details: GELFAND-P001313

Uncertain reading. The summation condition reads $H'(x_k) = 0$. The tall stroke to the right of the prime forms the opening parenthesis. The prefactor is written $(2\pi)^{-n}$.

$$dH(\cdot) = \omega(\xi, \cdot)$$

Condition. The phase flow ξ must be periodic.



Original source details: GELFAND-P001313

Uncertain reading. The first character of the Russian abbreviation “фаз.” (phase) is retraced several times.

Semenov-Tian-Shansky [broken underlining] showed that this agrees with Kirillov’s [inserted above the line: formula] in the case of an orbit.

Idea: $i\Omega^+\mu\Omega$ is an orbit with a symplectic structure

$$M = \text{quotient of unitary matrices by the diagonal}$$

$$X = \Omega^+\mu\Omega;$$

$$\omega(X)(\xi, \eta) = \operatorname{tr}(X \cdot [\xi, \eta]), \qquad \xi, \eta \in T_X$$

Diagram and typed description.

(!)



Original source details: GELFAND-P001313

I.M.G.'s question [wavy underlining]. When is the second term of the asymptotics also needed? Analyze the topological reasons why the second term is 0, etc. Connection with Morse inequalities.

For which integrals does the first term of the asymptotics give the exact answer?

Method [wavy underlining]. Look for cases where the second term of the asymptotics is $= 0$ at the current level.

Subproblem [wavy underlining]. So that characteristic numbers arise.

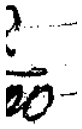
Hypothesis (mine). In place of [crossed out] an S^1-action, specify an action of a nilpotent Lie grp. (e.g., 2-dimensional $\Rightarrow$ two terms will arise).



Original source details: GELFAND-P001313

Undeciphered passage. The cancelled mark before S^1 is obscured by slanted strokes.

[At the cropped left edge, near the bottom.]



Original source details: GELFAND-P001313

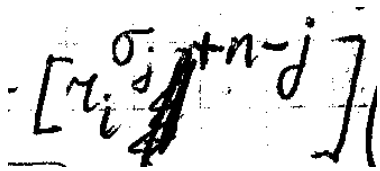
Undeciphered passage. The end of the fraction bar and two loops, provisionally "00", are visible in the denominator. The upper sign is clipped.

GELFAND-P001314

3/XI-86 Gelfand Sem.

G. Olshanskii.

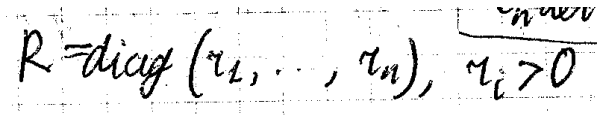
$$\int_{U(n)} \Delta_1(uRu^{-1})^{\sigma_1-\sigma_2}\cdots\Delta_{n-1}(uRu^{-1})^{\sigma_{n-1}-\sigma_n}\Delta_n(uRu^{-1})^{\sigma_n}\,du = \\ = c_n \det\left[r_i^{\sigma_j+n-j}\right]\left(\prod_{i<j}(r_i-r_j)\prod_{i<j}(\sigma_i-\sigma_j+j-i)\right)^{-1}, \tag{1}$$



Original source details: GELFAND-P001314

Uncertain reading. The exponent reads $\sigma_j + n - j$. Between the subscript j and the $+$, overlapping strokes are crossed out.

$$R = \operatorname{diag}(r_1,\dots,r_n), \quad r_i > 0$$

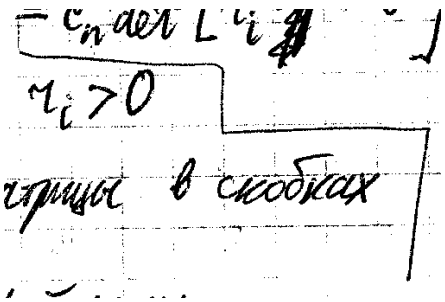


Original source details: GELFAND-P001314

Uncertain reading. The lowercase letters in the list of diagonal entries and before > 0 read r.

Δ_m — principal minors of the matrices in parentheses.

$$c_n = 1!2!\ldots(n-1)!$$



Original source details: GELFAND-P001314

Diagram and typed description.

The callout starts beneath the second equals sign. Its final stroke passes to the right of the explanation of Δ_m.

— an integral from the book by Gelfand and Naimark.

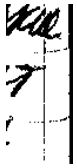


Original source details: GELFAND-P001314

Uncertain reading. The preposition before “книги” reads “из” (from).

Formula (2) on p. 24 is derived from (1).

[At the cropped left edge.]



Original source details: GELFAND-P001314

Uncertain reading. The cropped ending resembles “ми” or “ши”. Below is an arrow ↗, then a clipped hook and a separate mark, possibly part of a question mark.

GELFAND-P001315

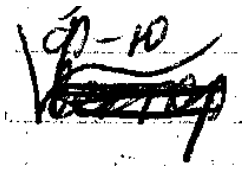
Proof of (1). [Wavy underlining.] (1) = H. Weyl’s character formula.

$e_1, \ldots, e_n$ is the canonical basis.

$$ue_i = f_i$$

Consider GL_n and its unitary representation of class I (i.e., ∃ a vector invariant under the compact subgroup). Let ξ_0 be a vector with $T_u\xi_0 = \xi_0$.

Consider [crossed out: the vector] [above the line: the fn.] $(T_g\xi_0, \xi_0)$.



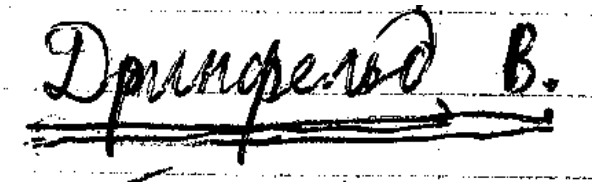
Original source details: GELFAND-P001315

Uncertain reading. The crossed-out entry reads “вектор” (vector). Above it is “ф-ю” (fn.). A curved line passes between the two entries.

If the representation is realized on the space of flags, this function is written as an integral of the form (1). The integral is a spherical fn. of GL_n relative to the unitary subgroup, etc.

I.M.G. General question. Which integrals can be evaluated in this way?

Drinfeld V. Quantization of KdV according to Belavin



Original source details: GELFAND-P001315

Diagram and typed description.
Drinfeld V.



Original source details: GELFAND-P001315

Uncertain reading. A short slanting stroke appears to the right of the heading. Its purpose is unclear.

On an idea of Belavin's.

KdV had previously been quantized incorrectly.

KdV is Hamiltonian with respect to two Gelfand–Dikii brackets.

Which one is needed? One must quantize with respect to both brackets.

Any linear combination of them is also a bracket.

One must introduce 2 Planck constants!

$$u_t = u_{xxx} + uu_x \tag{1}$$

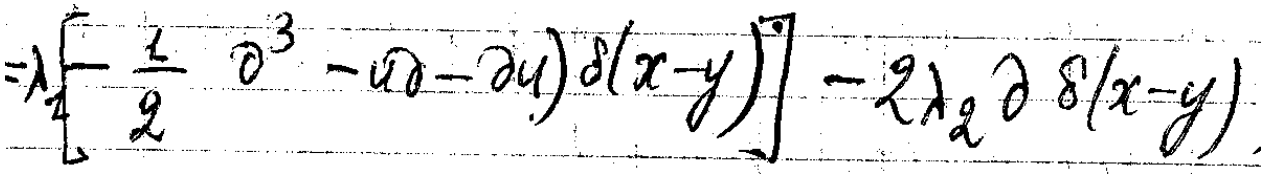
Brackets.

$$\text{II}: \ -\tfrac{1}{2}\partial^3 - u\partial - \partial u; \qquad \text{I}: \ -2\partial \tag{2}$$

— Hamiltonian operators for these brackets.

This means that

$$\{u(x), u(y)\} = \lambda_1\big[-\tfrac{1}{2}\partial^3 - u\partial - \partial u\big)\delta(x-y)\big] - 2\lambda_2\partial\delta(x-y), \tag{3}$$



Original source details: GELFAND-P001315

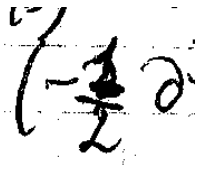
Uncertain reading. There is a square bracket after λ_1 and a closing parenthesis after $-\partial u$. The square bracket closes after $\delta(x-y)$.

where $\partial = \frac{\partial}{\partial x}$

The idea of the Hamiltonian operator goes back to Berezin (he proposed the matrix inverse to the symplectic one).

After quantizing (3) we obtain

$$[u(x), u(y)] = h_1 \left(-\tfrac{1}{2}\partial^3 - u\partial - \partial u\right) \delta(x-y) - 2h_2 \partial \delta(x-y). \tag{4}$$



Original source details: GELFAND-P001315

Uncertain reading. The numerator is heavily retraced. The fraction reads $\frac{1}{2}$, as in (2) and (3).

I.M.G. [Wavy underlining.] There is no nonlinear equation that has been properly quantized.

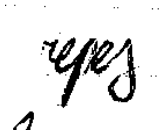


Original source details: GELFAND-P001315

Uncertain reading. The joined strokes before “следует” read “как”, forming the expression “properly”.

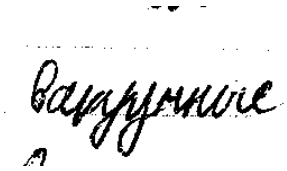
The first attempts are quantum groups.

Proposal. Do it by perturbing the measure. $P(\varphi)_2$ is done in this way,



Original source details: GELFAND-P001315

Uncertain reading. The full form “через” (through) is visible.



Original source details: GELFAND-P001315

Uncertain reading. The letters run together. The word reads “возмущение” (perturbation).

Dr. No one has compared quantization according to Faddeev and according to Albeverio–Høegh-Krohn.



Original source details: GELFAND-P001315

Uncertain reading. After “Kro” the scan appears to have “n” and a Russian case ending. The English line uses the published spelling Høegh-Krohn.

GELFAND-P001316

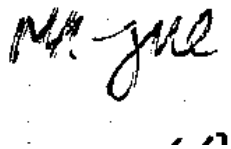
[source number: 28]

Dr. We will deal only with the 2nd bracket.

$$[u(x), u(y)] = h\left(-\frac{1}{2}\partial^3 - u\partial - \partial u\right)\delta(x-y) \tag{5}$$

This gives the Virasoro algebra (h is the central extension parameter).

The Poisson manifold is an orbit of the coadjoint rep. of the Virasoro alg. Quantization is according to Kirillov.



Uncertain reading. The Russian abbreviation after “Пуасс.” (Poisson) is provisionally read “мн-ие” (manifold). A long descender follows the dash.

Original source details: GELFAND-P001316

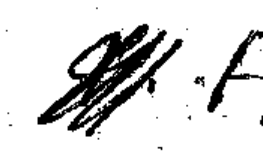
$\exists$ a countable number of integral functionals in involution.

$$F_n(u) = \int \varphi_n(u, u', \ldots, u^{(n)})\, dx \tag{6}$$

A quantum analogue?

u and u' do not commute.

Zamolodchikov. There is a quantum analogue [crossed out] of F_n.



Undeciphered passage. In the crossed-out entry before F_n, a loop, a long stroke and a separate mark at right are visible. Possible readings include φ and φ_n.

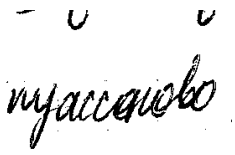
Original source details: GELFAND-P001316

We will leave the Hamiltonian aside.

Miura transform.

$$u = v' - v^2 \tag{7}$$

It is Poisson if on $\{u\}$ we take the 2nd G.–D. bracket,

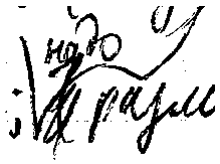


Uncertain reading. The Russian spelling is provisionally read “пуассаново” (Poisson). The vowel after “сс” merges with the next stroke.

Original source details: GELFAND-P001316

$$\text{on } \{v\} \quad — \quad \{v(x), v(y)\} = \delta'(x-y)$$
$$[v(x), v(y)] = h\delta'(x-y) \qquad (x \in S^1) \tag{8}$$

— the Heisenberg alg. [Crossed out] [above the line: one must] expand in a Fourier series [crossed out], and we obtain



Original source details: GELFAND-P001316

Uncertain reading. Before “разложить” (expand), a mark resembling “И” (and) is crossed out. Above it are “надо” (one must) and a curved callout stroke.



Original source details: GELFAND-P001316

Undeciphered passage. After “Fourier” a sign is crossed out. The visible loop permits v or ξ.

$$[v_m, v_n] = hm\delta_{m,-n}I \tag{9}$$

[To the right of (9)] (1-dimensional extension of a commutative algebra).

2 infinite-dimensional Lie algs. (enveloping algs.).

a) Heisenberg alg. b) Virasoro alg. [Both entries have wavy underlining.]

Claim. There is a simple homomorphism from the enveloping alg. in b) to the enveloping alg. in a).

Virasoro generators $\to$ quadratic expressions in Heisenberg generators.

Virasoro alg.

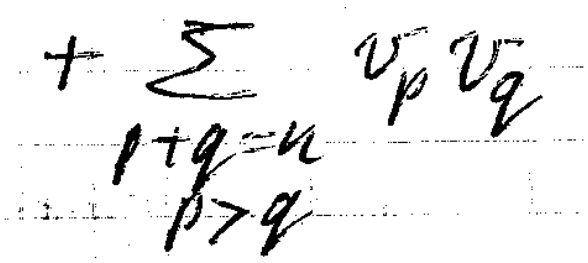
$$L_n = e^{in\varphi}\frac{d}{d\varphi} \qquad (\text{on } S^1) \tag{10}$$

and a 1-dimensional extension by means of the Gelfand–Fuchs cycle.

Fuchs. The corresponding transformation from (9) to (10).

$$v_i,\ i \neq 0, \qquad L_n = (\alpha n + \beta)v_n + \sum_{\substack{p+q=n \\ p>q}} v_p v_q \tag{11}$$

Heisenberg alg. $\quad n \neq 0$

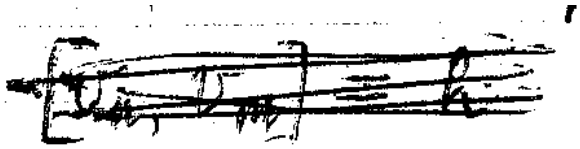


Original source details: GELFAND-P001316

Uncertain reading. The lower summation condition reads $p > q$. There is no second stroke below $>$.

This is the Miura transform.

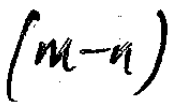
[An equation is crossed out.]



Undeciphered passage. Square brackets, two v signs, an equals sign and h are visible. The indices provisionally read $[v_n, v_m] = h$. The reverse order is also possible.

Original source details: GELFAND-P001316

$$[L_n, L_m] = (m-n)L_{n+m} + \delta_{n,-m}\frac{n^3-n}{12}c, \qquad c \text{ commutes with all } L_n$$



Uncertain reading. The factor before L_{n+m} reads $(m-n)$.

Original source details: GELFAND-P001316

Quantization = embedding in the enveloping alg. of the Heisenberg algebra [the ending of the Russian word “algebra” is below the line].



Uncertain reading. The ending “ры” is written below the line before “Гейзенберга” (Heisenberg).

Original source details: GELFAND-P001316

Problem (interesting). Write down all the Hamiltonians.

GELFAND-P001317

KdV is connected with the Sturm–Liouville operator.

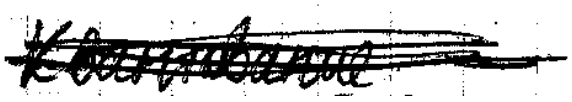


Uncertain reading. The first surname is abbreviated “Шт.” (St.).

Original source details: GELFAND-P001317

$$L = \partial^2 + u \tag{12}$$

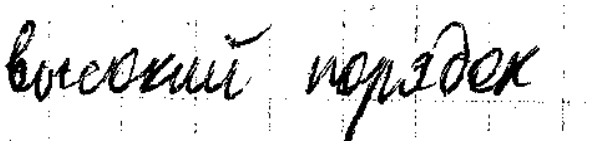
[Crossed out: Quantization.]



Uncertain reading. The lower curves and projecting strokes of the cancellation allow the reading “Квантование” (Quantization).

Original source details: GELFAND-P001317

how to generalize to high order



Uncertain reading. The word before “порядок” (order) reads “высокий” (high).

Original source details: GELFAND-P001317

$$L = \partial^n + u_{n-2}\partial^{n-2} + \ldots + u_0 \tag{13}$$

and there is an analogue of the Miura transf. ([above the line: Wilson,] Drinfeld, Sokolov).



Uncertain reading. “Wilson,” is inserted above “Drinfeld” with a curved callout stroke.

Original source details: GELFAND-P001317

Operator (13) is connected with $sl(n)$.

Instead of

$$u = v' - v^2$$



Undeciphered passage. In the left margin is a looped entry with overlapping diagonal strokes. Its reading is unresolved.

Original source details: GELFAND-P001317

it is more correct to write

$$L = \partial^2 + u = (\partial - v)(\partial + v) \tag{14}$$

Factor.

$$L = \partial^n + u_{n-2}\partial^{n-2} + \ldots + u_0 = (\partial + v_1)\ldots(\partial + v_n), \quad v_1 + \ldots + v_n = 0 \tag{15}$$

This gives the Miura transf.

This is connected with Picard–Vessiot.

$$\{u(x), u(y)\} = \left(-\tfrac{1}{2}\partial^3 - u\partial - \partial u\right)\delta(x-y) \tag{16}$$

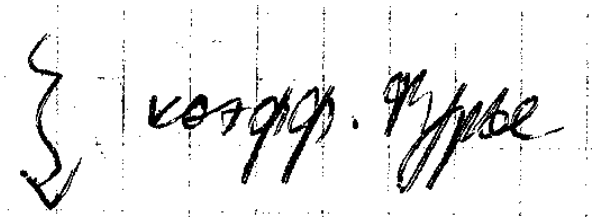
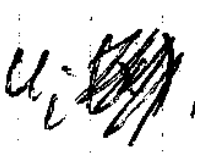


Diagram and typed description.

Fourier coeffs.

Original source details: GELFAND-P001317

$$\{u_i, u_j\} = \text{quadr. rels.} \tag{17}$$



Undeciphered passage. After u_i a group of signs is crossed out. A curve is visible on the left of the group. Its interior signs overlap.

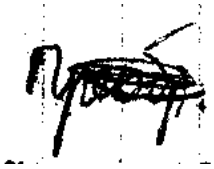
Original source details: GELFAND-P001317



Original source details: GELFAND-P001317

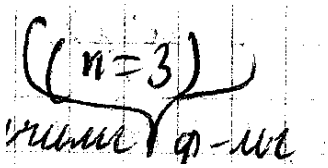
Undeciphered passage. After u_j, before the right brace, a second group is crossed out. A curved stroke is visible at its left edge.

[A crossed-out entry above “Fateev”.] Fateev and Zamolodchikov obtained [inserted above: $(n = 3)$] formulas that solve the problem.



Original source details: GELFAND-P001317

Undeciphered passage. An opening sign resembling “П” and long descenders are visible. The word is unresolved.

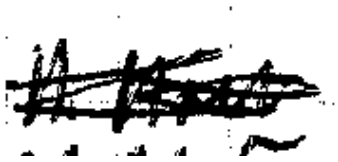


Original source details: GELFAND-P001317

Uncertain reading. The entry $(n = 3)$ is above the start of the abbreviation for “formulas”, with a curved callout stroke.

B. Feigin [wavy underlining] did something (1986, April [below the date: conf. at VDNKh]). Generalization to $sl(3)$.

Sklyanin [wavy underlining]. Quantization of KdV on a finite interval, with boundary conditions.

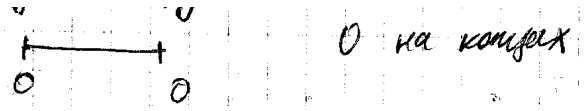


Original source details: GELFAND-P001317

Undeciphered passage. A short entry above “I.M.G.” is crossed out. The first sign resembles “И” (I). The other strokes overlap.

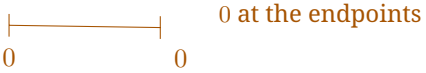
I.M.G. [Wavy underlining.] The sweep method (Gelf., Lokutsievskii) is analogous to the Miura transf.

$$y'' - a(x)y = 0$$



Original source details: GELFAND-P001317

Diagram and typed description.

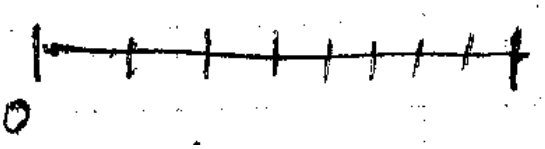


One cannot take an exp growing solution (accuracy is lost).

($\Rightarrow$ one cannot take 2 lin. indep. solutions)

GELFAND-P001318

[source number: 29]



Original source details: GELFAND-P001318

Diagram and typed description.



Move the boundary condition to the neighboring point.

$$y'(0) + \alpha y(0) = 0.$$

At the neighboring point α changes.

$$\alpha' - \alpha^2 = 0 \quad (!) \quad \text{— Miura.}$$



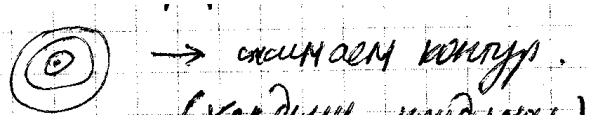
Original source details: GELFAND-P001318

Uncertain reading. The manuscript gives $\alpha' - \alpha^2 = 0$ followed by "(!)".

There is a sweep method for n-th order too!!

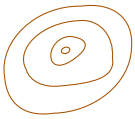
It fixes the correct factorization.

$$\Delta u = 0$$



Original source details: GELFAND-P001318

Diagram and typed description.



⟶ shrink the contour.

Inside the three contours is a small closed or nearly closed stroke.

(Keldysh devised this.)

B. Feigin. [Wavy underlining.]

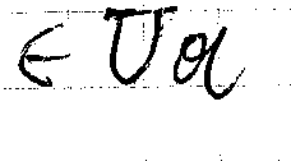
Finite-dimensional Lie alg. a and an inner product. [Wavy underlining.]



Original source details: GELFAND-P001318

Uncertain reading. The looped letter before "and" reads a.

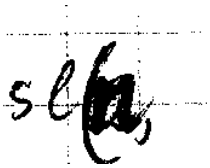
$\sum e_i e^i$ — Casimir elem. $\in Ua$ (from the invariant inner prod.)



Original source details: GELFAND-P001318

Uncertain reading. A looped a follows U, matching the heading.

$$sl(n, \mathbb{C}[\lambda, \lambda^{-1}])$$



Original source details: GELFAND-P001318

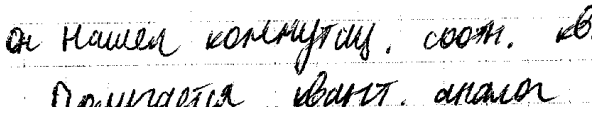
Uncertain reading. The first argument is retraced. The following line with $sl(n, \mathbb{C}[\lambda, \lambda^{-1}])$ supports the reading n.

There are many inner products (invariant).

$$u, v \in sl(n, \mathbb{C}[\lambda, \lambda^{-1}])$$

$$\text{coeff.}_k \operatorname{Tr}(uv)$$

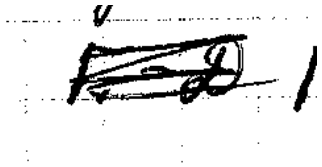
He found the commutation rels. of the quadr. and cubic Casimirs.



Original source details: GELFAND-P001318

Uncertain reading. The line begins “он нашёл коммутац. соотн.” (he found the commutation rels.). The first sign and the middle of the Russian abbreviation for commutation are joined.

This gives a quantum analogue of the formulas of [crossed out: G.–D.] Gelfand–Dikii.



Original source details: GELFAND-P001318

Uncertain reading. The crossed-out abbreviation has an angular first sign and a loop at the end. It is provisionally read “Г.-Д.” (G.–D.).

What problems are there?

1° a) Quantize the KdV integrals.

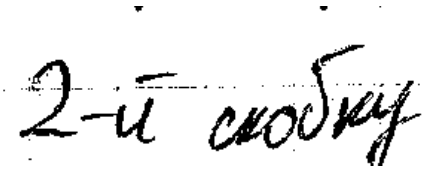
[Crossed out: 2°] b) Compute the spectrum of irreducible reps. of the Virasoro alg.



Original source details: GELFAND-P001318

Uncertain reading. The mark before “b)” is crossed out and provisionally reads 2°.

2°. Quantize the 2nd G.–D. bracket in the general case.



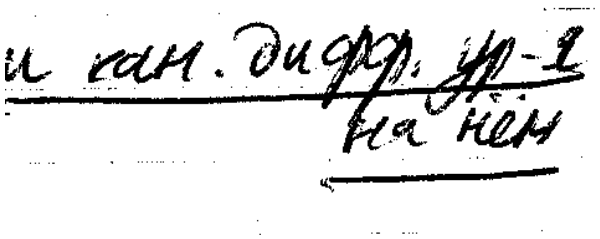
Original source details: GELFAND-P001318

Uncertain reading. Before “скобку” (bracket) the manuscript reads “2-й” (2nd), with a short stroke over its final letter.

GELFAND-P001319

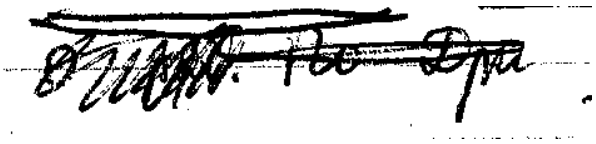
10.11.86 Gelfand Sem.

Beilinson. Action of the Virasoro alg. on the moduli space of algebraic curves and canonical differential eqs. on it.



Original source details: GELFAND-P001319

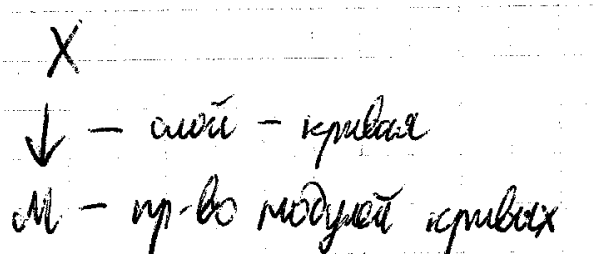
Uncertain reading. The abbreviation before “дифф.” (differential) reads “кан.” (canonical). The strokes under “and canonical differential eqs.” and “on it” have left-pointing tips.



Original source details: GELFAND-P001319

Undeciphered passage. An opening loop and a final group resembling “Дра” are visible through the cancellation. The full line remains unresolved.

On Belavin’s work according to Drinfeld.



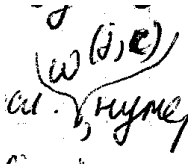
Original source details: GELFAND-P001319

Diagram and typed description.

X
$\downarrow$ — the fiber is a curve
M — moduli space of curves

$$\text{genus } g \quad \Rightarrow \quad \text{dimension } 6g-6$$

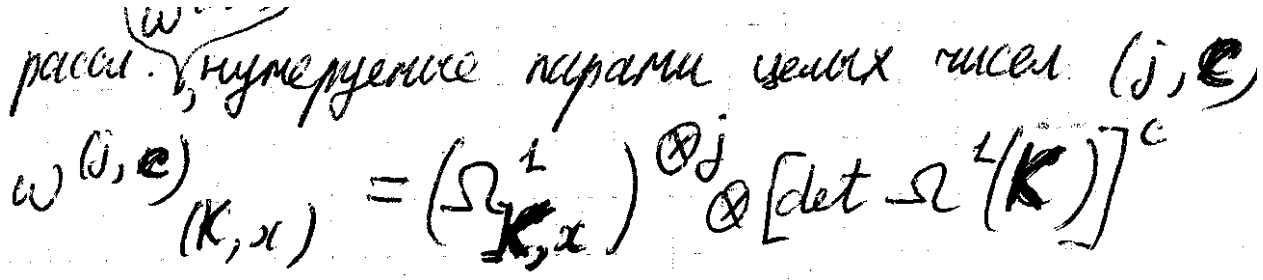
On X are line bundles [inserted above: $\omega^{(j,c)}$] indexed by pairs of integers (j,c).



Original source details: GELFAND-P001319

Uncertain reading. The notation $\omega^{(j,c)}$ is inserted above the line with a curved callout stroke. The c is thickened.

$$X \ni (K,x), \qquad \text{fiber } \omega^{(j,c)}_{(K,x)} = (\Omega^1_{K,x})^{\otimes j} \otimes [\det \Omega^1(K)]^c$$



Original source details: GELFAND-P001319

Uncertain reading. The c in (j,c) is retraced. In the equation its reading follows the exponent on $[\det \Omega^1(K)]^c$. The K signs on the right are also thickened.

[Below K: curve corresponding to a point of M, $x \in K$.]



Original source details: GELFAND-P001319

Uncertain reading. The first letter of (K,x) is heavily retraced. The remaining strokes and the note below support K.

[Below the right-hand side:] j-differentials of the curve K at x.

Differential eqs. are written for sections [wavy underlining] of this bundle.

Extend X.

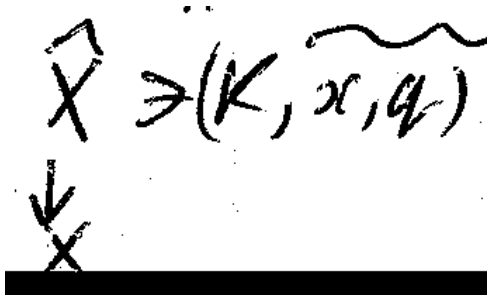


Diagram and typed description.

$\widehat{X} \quad \ni (K, x, q)$

$\downarrow$

X

Original source details: GELFAND-P001319

q is a formal parameter at x.

(choice of a formal coordinate system at x)

GELFAND-P001320

The Virasoro algebra acts on $\widehat{X}$.

(Kontsevich [crossed out] and Beilinson–Schechtman independently.)



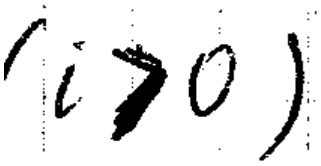
Undeciphered passage. A short entry between "Kontsevich" and "and" is crossed out. Oblique strokes and a projecting vertical line are visible.

Original source details: GELFAND-P001320

For now, the alg. of formal vector fields (without central charge).

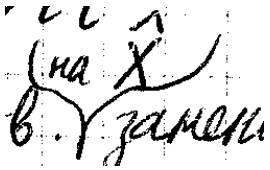
$$Vir = \left\{ \sum a_i t^i \frac{d}{dt} \right\} \supset Vir^+$$

Half of this alg. ($i > 0$) acts [inserted above: on $\widehat{X}$] by changing the parameter q.



Uncertain reading. The sign between i and 0 is heavily retraced. Its upper and lower strokes meet on the right, giving $>$.

Original source details: GELFAND-P001320



Uncertain reading. "on $\widehat{X}$" is inserted above the line with a curved callout stroke.

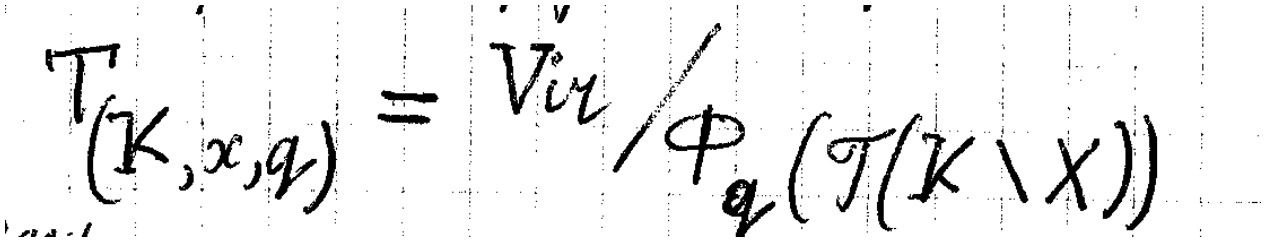
Original source details: GELFAND-P001320

(Taylor series in q.) This action extends to an action of the full alg. Vir on $\widehat{X}$.

The conformal class of the curve will then change.

According to Kodaira–Spencer [wavy underlining], the tangent space to $\widehat{X}$ has the form

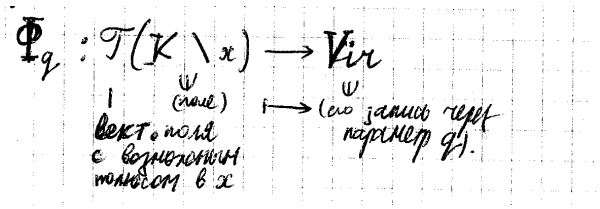
$$T_{(K,x,q)} = Vir/\Phi_q(\mathcal{T}(K\backslash X)) \tag{1}$$



Original source details: GELFAND-P001320

Uncertain reading. In (1), the letter after the backslash is capital X. The diagram below and (2) have lowercase x.

Here



Original source details: GELFAND-P001320

Diagram and typed description.

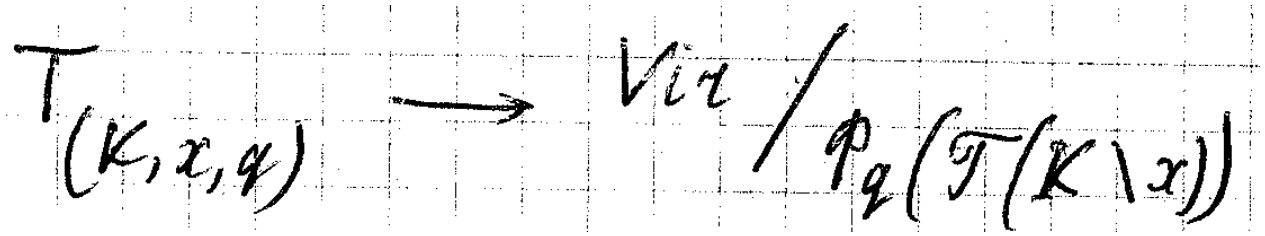
$\Phi_q : \mathcal{T}(K\backslash x) \longrightarrow Vir$

$\downarrow$ (field) $\longmapsto$ $\downarrow$ (its expression in the parameter q)

vect. fields with a possible pole at x

Explanation. [Wavy underlining.] Construct a map

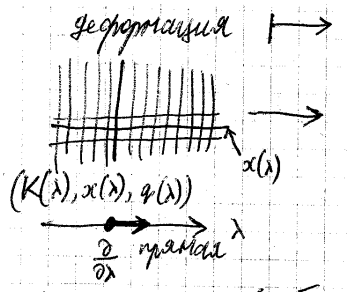
$$T_{(K,x,q)} \longrightarrow Vir/\varphi_q(\mathcal{T}(K\backslash x)) \tag{2}$$



Original source details: GELFAND-P001320

Uncertain reading. In (2) the map is written with lowercase φ_q. In (1) and in the diagram it has uppercase Φ_q.

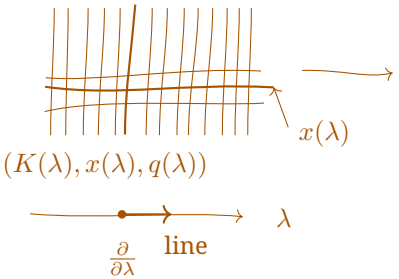
(which will be the isomorphism).



Original source details: GELFAND-P001320

Diagram and typed description.

deformation $\longmapsto$



$x(\lambda)$

$(K(\lambda), x(\lambda), q(\lambda))$

λ

$\frac{\partial}{\partial\lambda}$ line

The formal neighborhood (tubular) of the section is trivialized —
— by the coordinates λ and q.

a) Lift $\dfrac{\partial}{\partial\lambda}$ horizontally to this tubular neighborhood.

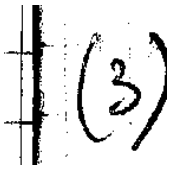
b) $\dfrac{\partial}{\partial\lambda}$ can also be lifted to the complement of $x(\lambda)$ (another lift).



Original source details: GELFAND-P001320

Uncertain reading. Near the right edge of the main leaf, level with (2), there is a faint winding stroke. Its meaning is uncertain.

[On the strip of the facing leaf beyond the gutter: (3).]



Original source details: GELFAND-P001320

Uncertain reading. The number (3) on the facing leaf is visible in this frame.

GELFAND-P001321

Consider a Verma module with highest weight (c, g).

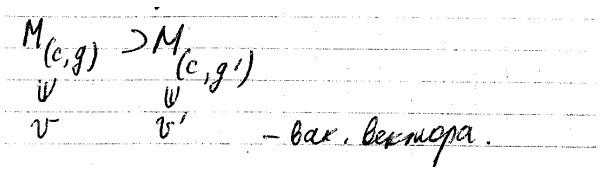
$$M_{(c,g)}$$



Original source details: GELFAND-P001321

Uncertain reading. The second sign in the pair reads g, as in the module notation below.

Suppose it is reducible.



Original source details: GELFAND-P001321

Diagram and typed description.

$$\begin{array}{ccc} M_{(c,g)} & \supset & M_{(c,g')} \\ \downarrow & & \downarrow \\ v & & v' \end{array}$$

— vacuum vectors.

$$v' = L(v), \qquad L \in U(\mathit{Vir}).$$

The eqs. are canonical overdetermined differential eqs. on M.



Original source details: GELFAND-P001321

Uncertain reading. The first sign in the Russian abbreviation “Ур-я” (eqs.) is heavily retraced.

(Associated with singular vectors in Verma modules over *Vir*.)

$$\psi \in \Gamma(\omega_{X/M}^{\otimes j} \otimes \lambda^c)$$

Lift ψ to $\widehat{X} \Rightarrow$ obtain a singular vector $\Rightarrow$ apply L to it and obtain a new singular vector, i.e., a new section of $\omega_{X/M}^{\otimes j'} \otimes \lambda^c$.

Solutions [wavy underlining] form a local system of finite rank.

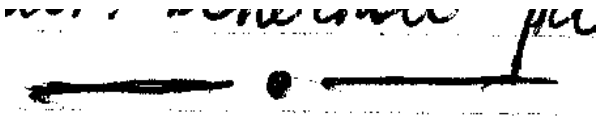


Diagram and typed description.

A dot separates the two lines. At the left end of the first line are two short slanted strokes.

Original source details: GELFAND-P001321

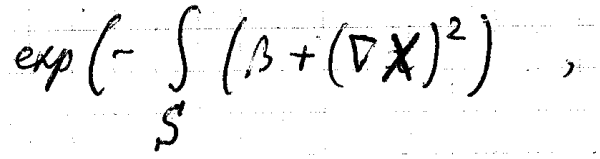
GELFAND-P001322

17.11.86 Gelfand Sem.

Kazakov

Zakharevich

$$Z = \int Dg \int DX \ \exp(-\int_S (\beta + (\nabla X)^2), \tag{1}$$

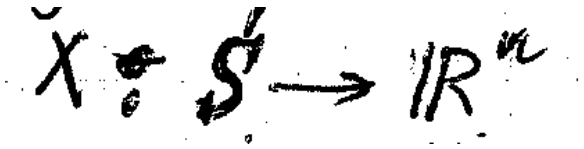


Uncertain reading. One closing parenthesis is visible after $(\nabla X)^2$, before the comma. The parenthesis opened after exp remains unclosed.

Original source details: GELFAND-P001322

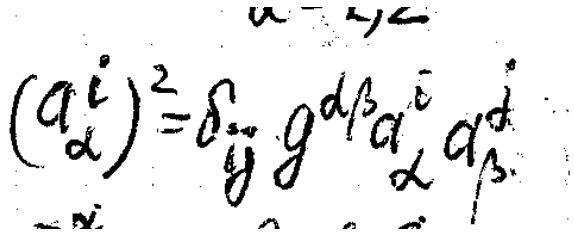
[To the right of the vertical line.] g is the metric on S.

$$X : S \longrightarrow \mathbb{R}^n$$
$$(\nabla X)^i_\alpha = \frac{\partial X^i}{\partial \xi^\alpha}$$
$$i = 1, \ldots, n, \qquad \alpha = 1, 2$$
$$(a^i_\alpha)^2 = \delta_{ij} g^{\alpha\beta} a^i_\alpha a^j_\beta$$



Uncertain reading. Two dots appear between X and S. The upper dot is retraced and is read as part of the colon.

Original source details: GELFAND-P001322



Uncertain reading. The lower signs on δ read i, j. On the left the square is outside the parentheses enclosing a^i_α.

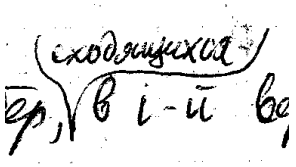
Original source details: GELFAND-P001322

ξ^α are coord. on S.

[Left column continued.]

$$Z = \sum_{\substack{T \\ \text{triang. of the sphere}}} \int \prod dx_i \exp\left(-\sum_i \beta\sigma_i - \sum_{\substack{i,j \\ \text{neighbors}}} (x_i - x_j)^2 \right) \tag{2}$$

$$\sigma_i = \frac{n_i}{3}, \qquad n_i = \#\text{ edges (meeting) at the } i\text{-th vertex} \tag{2'}$$

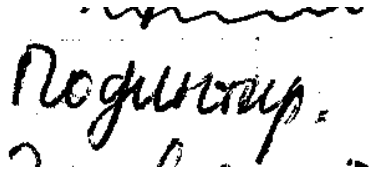


Uncertain reading. “(meeting)” is inserted above the line after “edges”, with a curved callout stroke.

Original source details: GELFAND-P001322

(this is called the defect [wavy underlining]).

(3) Claim. The integrand expressions agree for regular embeddings (when all $\triangle$s are regular).



Uncertain reading. The opening abbreviation in (3) reads “Подынтгр.” (integrand). Some middle strokes are joined.

Original source details: GELFAND-P001322

GELFAND-P001323

(1) is a bosonic string.

K. Wilson and Polyakov wrote the Yang–Mills Lagrangian in this way, but there the space is flat and one can take a cubic lattice.

I.M.G.’s question [wavy underlining]. How should one write the Y.–M. model? Characteristic classes arise there and one cannot restrict to a cubic lattice.

Z. Consider an arb. triangulated surface of the sphere. Then on average (for a random embedding X) all triangles of the embedding will be regular (the entire [crossed out] mass of a watermelon is in the rind!!).



Undeciphered passage. An entry between “entire” and “mass” is crossed out. Projecting strokes are visible, including a long descender.

Original source details: GELFAND-P001323

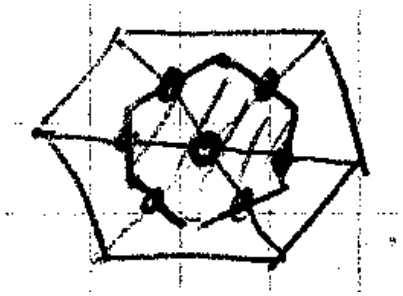
Thus it suffices to consider any metric g (fixed) and a linear map on each Δ.

K. [above at right: 4?] Why $\frac{n_i}{3}$?



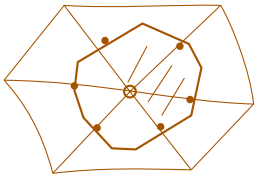
Uncertain reading. The small sign to the right of “K.” resembles 4 or “a”. Its meaning is uncertain.

Original source details: GELFAND-P001323



Original source details: GELFAND-P001323

Diagram and typed description.



Inside the hexagonal outline, six marks on the segments and a central mark are visible. Some slanted strokes inside are hard to distinguish from the notebook ruling.

Cut off [crossed out] $\frac{1}{2}$ of the edge and obtain area $= \frac{1}{3}$ (?).



Original source details: GELFAND-P001323

Undeciphered passage. A short sign is crossed out before $1/2$. A loop is visible at its top.

(all Δs are regular).

Z. In place of (2), now consider

$$Z_2 = \sum_m e^{-\beta m} \sum_{|T|=m} \left(\int \prod d\tilde{x}_i \cdot \exp - \sum_{i,j} (\tilde{x}_i - \tilde{x}_j)^2 \right)^n \tag{4}$$

K. It can only be evaluated at $n = 0$ and [crossed out] $n = -2$ (!!).



Original source details: GELFAND-P001323

Undeciphered passage. An entry between “and” and $n = -2$ is crossed out by diagonal strokes.

Z. Here the $\tilde{x}_i$ are already one-dimensional!

This follows because different components are integrated independently.



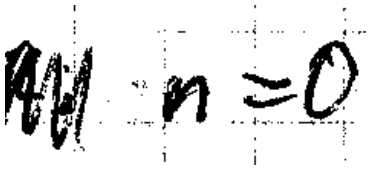
Original source details: GELFAND-P001323

Uncertain reading. A faint open loop appears in the right margin. Its purpose is uncertain.

The parameter n enters analytically!

At $n = 0$ and $n = -2$ the sum can be evaluated.

At $n = 0 \quad \Rightarrow$



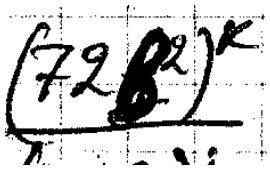
Original source details: GELFAND-P001323

Uncertain reading. The word before $n = 0$ is heavily retraced and provisionally reads “При” (at).

$$Z_2 = \sum_m \sum_{|T|=m} e^{-\beta m} = \sum_m e^{-\beta m} \#\{T : |T| = m\} \tag{5}$$

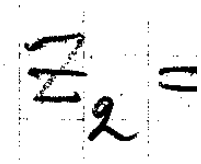
Answer.

$$Z_2 = -\frac{1}{2} \sum_{k=1}^{\infty} \frac{(72b^2)^k}{(k+2)!} \frac{\Gamma\left(\frac{3k}{2}\right)}{\Gamma\left(\frac{k+2}{2}\right)}, \quad b = b(\beta) \tag{6}$$



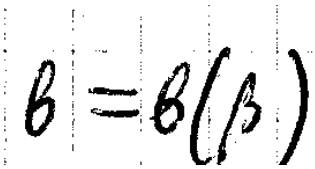
Original source details: GELFAND-P001323

Uncertain reading. The sign before the superscript 2 is retraced. The loops and the entry $b = b(\beta)$ at right support the reading b^2.



Original source details: GELFAND-P001323

Uncertain reading. The Z on the left of (6) has an additional horizontal stroke through its middle.



Original source details: GELFAND-P001323

Uncertain reading. To the right of the series is $b = b(\beta)$. The b and β have distinct forms.

$b(\beta)$ has not been evaluated, but physical quantities do not depend on this.

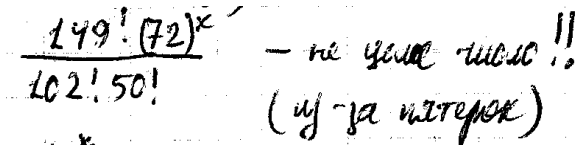
GELFAND-P001324

[source number: 37]

K.

$$\#\{T, |T| = 2k\} = \frac{(72)^k}{(k+2)!} \frac{\Gamma\left(\frac{3k}{2}\right)}{\Gamma\left(\frac{k+2}{2}\right)} \tag{7}$$

$k = 100 \quad \Rightarrow \quad \frac{149!\,(72)^k}{102!\,50!}$ is not an integer!! (because of the fives)



Original source details: GELFAND-P001324

Uncertain reading. The numerator reads 149!, and the denominator reads 102! 50!.

K. Radius of conv. of the series is b^* [below the line: (in (6))].



Original source details: GELFAND-P001324

Uncertain reading. Above the line about the radius is a thin winding stroke with a loop. Its purpose is uncertain.

$$Z_2 \underset{b\to b^*}{\cong} a + c(b-b^*)^{5/2} \tag{8}$$

Then

$$Z(\beta) = A + B(\beta-\beta^*)^{5/2} \tag{9}$$

The critical exponent matters.

This is a typical situation in field theory. For example, the mass spectrum

$$m_n = m_0(g) f(n)$$

Z. Combinatorial problem [wavy underlining]. Find

$$\frac{1}{n!}\#\{\tilde{T}, |\tilde{T}| = n\} = \text{[crossed out] } W(n)$$



Undeciphered passage. A group of signs between the equals sign and $W(n)$ is crossed out. Its interior strokes overlap.

Original source details: GELFAND-P001324

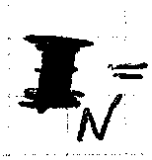
$\tilde{T}$ is a labeled triangulation of the sphere (with numbered Δs).

$\Rightarrow$ the answer need not be an integer.

K. Asymptotically $W(m) \sim m^\gamma e^{\beta^* m}, \quad \gamma = -7/2$. How does one compute the asymptotics?

Consider the integral

$$I_N = \int d^{N^2}u\, e^{-N\,\mathrm{tr}(u^2+gu^3)}, \quad u \text{ is Hermitian, } u^+ = u \tag{10}$$

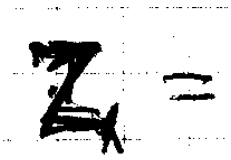


Uncertain reading. The main letter in (10) is retraced. Its shape and (11) support the reading I_N.

Original source details: GELFAND-P001324

Expanding in g and evaluating the Gaussian integrals, we obtain

$$Z_{[?]} = \sum_{n=0}^{\infty} \Pi_n g^n$$



Uncertain reading. Crossing strokes are visible at the lower right of Z. The subscript remains unresolved.

Original source details: GELFAND-P001324

$$Z = \lim_{N\to\infty} \log I_N \tag{11}$$

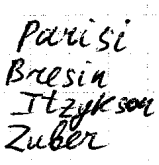
— the generating fn. for the number of triangulations of the sphere.

Idea [wavy underlining]. Interpret the integrals obtained by series expansion graphically. This interpretation gives triangulations of the sphere.

One can replace $u^2 + gu^3$ by $u^2 + gu^3 + \lambda u^4$.

This was evaluated by Feynman, t' Hooft, Parisi

Bresin
Itzykson
Zuber



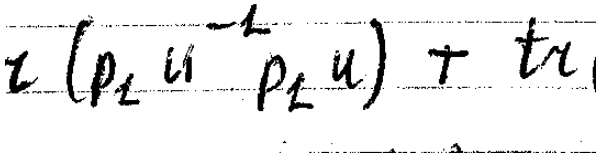
Original source details: GELFAND-P001324

Uncertain reading. The first surname in the lower-right list is written "Bresin".

GELFAND-P001325

Other integrals

$$\int du\, e^{\mathrm{tr}(p_1 u^{-1} p_1 u) + \mathrm{tr}(p_2 u^{-1} p_2 u)}, \qquad u^+ = u^{-1} \tag{12}$$



Original source details: GELFAND-P001325

Uncertain reading. The letter in (12) and the diagonal matrices reads as p, though its shape is close to ρ.

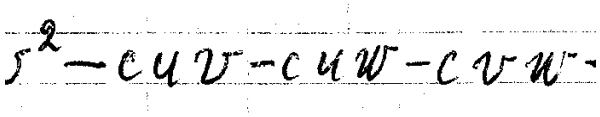
equiv. to the principal chiral field (2-dimensional) field.

$$S = \int d^2x\ \mathrm{tr}(\partial_\mu u^{-1}\ \partial_\mu u)$$

$$u^+ u = I, \qquad N \longrightarrow \infty$$

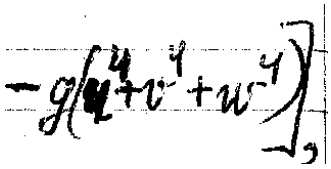
In principle solved by Polyakov and Wiegmann ⇒ such an integral ought to be calculable

$$\int d^{N^2}u\, d^{N^2}v\, d^{N^2}w\ \exp\big[-\mathrm{tr}(u^2 + v^2 + w^2 - cuv - cuw - cvw - g(u^4 + v^4 + w^4)\big], \tag{13}$$



Original source details: GELFAND-P001325

Uncertain reading. The final mixed term reads as $-cvw$. The final stroke of w matches the preceding term.



Original source details: GELFAND-P001325

Uncertain reading. The first letter after $g($ is heavily overwritten. It reads as u^4.



Original source details: GELFAND-P001325

Uncertain reading. After w^4 there is one closing parenthesis before the square bracket.

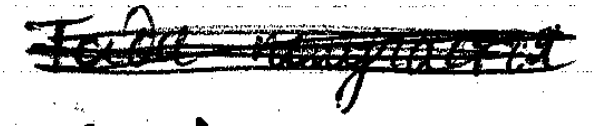
u, v, w are Hermitian

(problem of summing over triangulations of the sphere, but with a Z_3 spin at each vertex instead of Z_2)

In (12)

$$p_1 = \mathrm{diag}\big((p_1)_1, \ldots\big)$$
$$p_2 = \mathrm{diag}\big((p_2)_1, \ldots\big).$$

[Crossed out: Then the integral.]

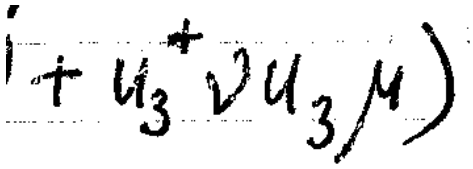


Original source details: GELFAND-P001325

Uncertain reading. The cancellation partly covers the final strokes of “integral”.

(13) reduces to an integral over the group (pass to eigenvalues)

$$\int d(u_1)d(u_2)d(u_3)\, e^{\mathrm{tr}(u_1^+ \lambda u_1 \mu + u_2^+ \mu u_2 \nu + u_3^+ \nu u_3 \mu)} \tag{14}$$



Original source details: GELFAND-P001325

Uncertain reading. The final factor in (14) is written as μ.

For Yang–Mills

$$Z = \int \prod_{\mu=1}^{d} du_\mu \, \exp\Big[\beta N \sum_{\mu>\nu} \mathrm{tr}(u_\mu u_\nu u_\mu^+ u_\nu^+)\Big] \tag{15}$$

$d = 2$ solved

Need $d = 3, 4$

GELFAND-P001326

[source number: 38]

[Strip of the facing leaf at the left.]



Original source details: GELFAND-P001326

Uncertain reading. Only the tips of the brackets and the comma are visible,)], or],.

I. M. Gelfand,

(+ V. B. Lidskii and Ponomarev)

In work that took 10 years.

There are 2 methods of working, lattices and categories.

Lattices are more convenient (but have been discredited by specialists in [crossed out: ...] [above the line: general] algebra)



Original source details: GELFAND-P001326

Undeciphered passage. The cancellation covers the word under the insertion “general”.

Nontrivial work on lattices [wavy underline]

1) Dedekind [wavy underline]

2) Weyl [wavy underline] on the axiomatics of projective geometry

dim. = max. length of an ordered chain.

$n \geq 3$ $\Rightarrow$ Desargues's axiom holds, etc.

3) von Neumann [wavy underline] removed the finiteness requirement and connected this with factors

(recall in connection with the work of [crossed out: ...] Dison [?], which [crossed out: ...] Connes promotes)



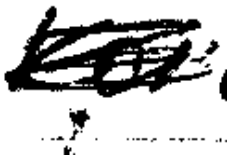
Original source details: GELFAND-P001326

Undeciphered passage. Several slanted strokes cover the short entry before the surname.



Original source details: GELFAND-P001326

Uncertain reading. The surname after the cancellation reads tentatively as “Dison”.



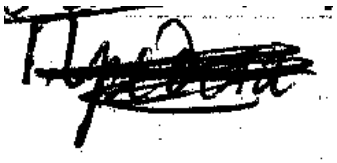
Undeciphered passage. The short entry before “Connes” is crossed out.

Original source details: GELFAND-P001326

4) Gelfand and Ponomarev [wavy underline]

Problem. There are k linear subspaces. How are they to be classified?

[Crossed out: Pred...]



Undeciphered passage. The initial fragment “Pred...” is visible. The cancellation covers the rest.

Original source details: GELFAND-P001326

$k = 2$ trivial

$k = 3$ a (good) problem in linear algebra

Decomp. into a direct sum of subspaces on a line and three

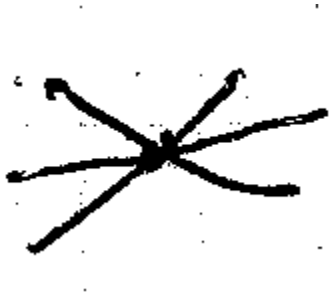
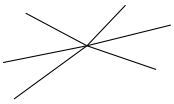


Diagram and typed description. 

Original source details: GELFAND-P001326

The corresponding lattice contains 28 elements

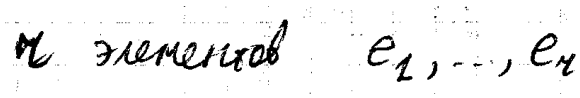
$k = 4$ took a long time (Gelf. and Ponomarev). See Manin’s book for the formulation [below the line: on logic]

Consider r elements $e_1, \ldots, e_r$ and the free modular lattice generated by [crossed out: ...] these letters.



Undeciphered passage. The cancellation covers the short entry before “these letters”.

Original source details: GELFAND-P001326



Uncertain reading. The number of generators and the last subscript read as r.

Original source details: GELFAND-P001326

(see Birkhoff, Lattice Theory)

operations, sum $+$ and intersection $\cdot$

equal if they can be transformed using the axioms

$r \geq 5$, the word identity problem is undecidable

Rep. $x \rightsquigarrow \underset{\text{subspace}}{\varrho(x)} \subset V$

$$\varrho(x_1 x_2) = \varrho(x_1) \cap \varrho(x_2), \qquad \varrho(x_1 + x_2) = \varrho(x_1) + \varrho(x_2)$$

Ryzhikova [wavy underline] (Semenov's diploma student). Can linear equivalence be formulated with finitely many axioms?



Original source details: GELFAND-P001326

Uncertain reading. The surname reads as "Ryzhikova". Compare Yu. S. Ryzhikova, "Decidability of the identity problem in lattices of linear subspaces" (1987).

Let there be 2 reps. ϱ_j in V_j, $j = 1, 2$.

A morphism φ is a linear map $\varphi : V_1 \to V_2$ if $\varphi(\varrho_1(x)) \subset \varrho_2(x)$.

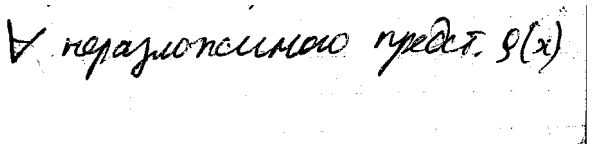
Indecomposable rep. if V_ϱ does not decompose into a direct sum ...



Original source details: GELFAND-P001326

Uncertain reading. The heavily overwritten subscript on V reads tentatively as ϱ.

A rep. ϱ is called preprojective [wavy underline] if $\forall$ indecomposable rep. $\varrho(x)$



Original source details: GELFAND-P001326

Undeciphered passage. The writing at the foot of the leaf breaks off after $\varrho(x)$.

GELFAND-P001327

$\exists$ only finitely many morphisms $\tau : \varrho \longrightarrow \varrho_0$



Original source details: GELFAND-P001327

Uncertain reading. The word in the first line reads as "morphisms".

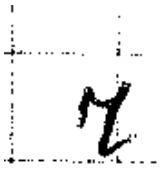
According to Arnold, these are reps. without parameters.

Lattice elts. are called preprojectively equiv. [wavy underline] if no preprojective rep. distinguishes them.

Identify them and we get the reduction.

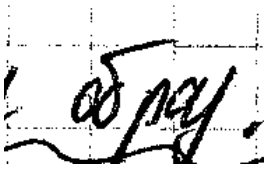
Problem (will be done) [below the line: actually already done]. Find the red. of the free lattice.

Thm. 1. All preprojective reps. of the [above the line: free] lattice on r gens. are described. [Wavy underline under the statement.]



Original source details: GELFAND-P001327

Uncertain reading. The letter reads as r, matching the notation for the number of generators on the preceding leaf.



Original source details: GELFAND-P001327

Uncertain reading. The abbreviation after r reads “образ.” (generators).

Thm. 2. The preprojective reps. of the lattice and of its reduction coincide. [Wavy underline under the statement.]

Def. A lattice elt. a is called perfect [wavy underline] if for every rep. [above the line: (in the space V)] either

$$\varrho(a) = V \quad \text{or} \quad \varrho(a) = 0$$

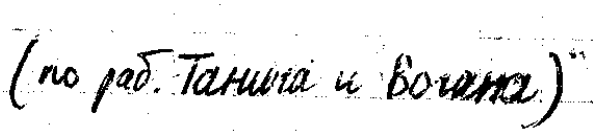
(this is an analogue of the centre of the group ring)

Perfect elements can be described.

GELFAND-P001328

24/XI-86 Gelfand Sem.

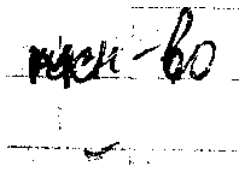
Zelevinsky A. Irreducible representations of $GL(n, \mathbb{C})$ (following the work of Tadić and Vogan)



Original source details: GELFAND-P001328

Uncertain reading. The first surname reads as “Tadić”. Two short marks follow the closing parenthesis.

Consider reps. of the group $GL(n, \mathbb{C})$. Denote the set of irreduc. unit. reps. by $GL(n, \mathbb{C})\hat{}$.



Original source details: GELFAND-P001328

Uncertain reading. The first strokes of the abbreviation for “set” are overwritten.



Original source details: GELFAND-P001328

Uncertain reading. A mark before the superscript $\wedge$ is crossed out. Its reading as u is tentative.

[Inserted above the line at left: Now] All unitary representations of this group are known. (it is proved that the ones found

$$D^u = (\text{multipl. chars. of the field } \mathbb{C}), \quad \text{i.e. } D^u \ni \delta :$$

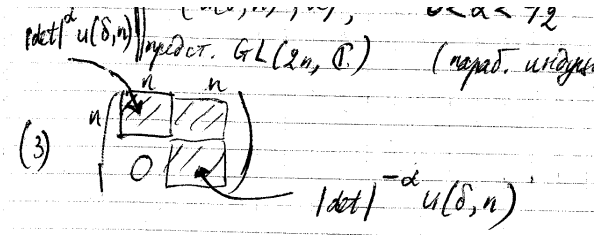
$$\delta(z) = \left(\frac{z}{|z|}\right)^k |z|^{i\rho}, \qquad k \in \mathbb{Z},\ \rho \in \mathbb{R} \tag{1}$$

$$u(\delta, n)(g) = \delta(\det g) \quad \text{is a unitary rep. of } GL(n, \mathbb{C}). \tag{2}$$

Additional family of representations [Double vertical stroke to the left of the following two lines.]

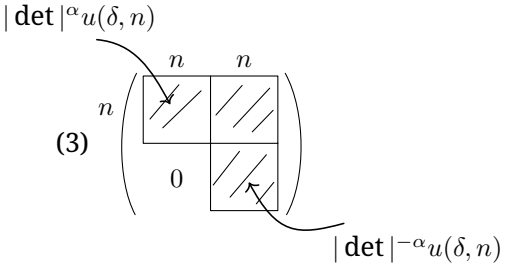
$$\pi(u(\delta, n), \alpha), \qquad 0 < \alpha < \frac{1}{2}$$

a rep. of $GL(2n, \mathbb{C})$ (parab. induction)



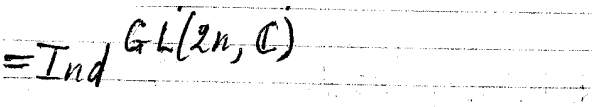
Original source details: GELFAND-P001328

Diagram and typed description.



and π is induced by this 1-dimensional representation.

$$\pi(u(\delta, n), \alpha) = \text{Ind}^{GL(2n,\mathbb{C})} \tag{4}$$



Original source details: GELFAND-P001328

Undeciphered passage. The line ends at $\text{Ind}^{GL(2n,\mathbb{C})}$.

I.M.G. Def. of an induced rep. [wavy underline] $G \supset H$, $\pi(h)$ is given Consider fns. $f(g)$ on the group G with values in the space of the rep. π such that $f(hg) = \pi(h)f(g)$

GELFAND-P001329

[source number: 42]

The space of such fns. carries an action of G and this rep. is called induced.

The restriction functor is dual to the induction functor.

3. Any representation [inserted below the line: (irreduc. unit.)] is obtained from these as follows

Consider a tuple $(n_1, \dots, n_k)$, $n_1 + n_2 + \dots + n_k = n$, and a collection $\sigma_i \in GL(n_i, \mathbb{C})^\vee$ of one of the 2 types described

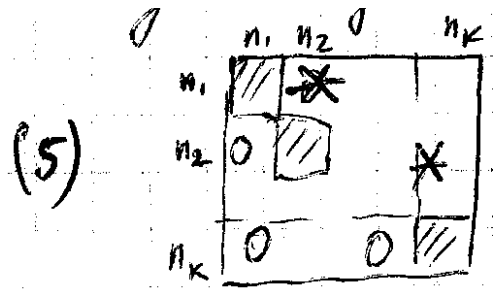


Diagram and typed description.

(5)

	n_1	n_2	n_k
n_1	//////	*	
n_2	0	//////	*
n_k	0	0	//////

Original source details: GELFAND-P001329

Consider the rep. $\bigotimes \sigma_i$ for such a group and induce.

Theorem

a) The result is irreduc. and unitary.

b) Every irreduc. unit. rep. has this form.

c) equiv. $\iff$ the σ_i differ by a permutation

— . —

$D^u =$ square-integrable representations of the groups $GL(k, F)$ for some k

(with square-integrable matrix coefficients).

Then the statement of thm. (5) remains valid for every field.

(Previously $k = 1$.)

$$u(\delta, n) \in GL(nk, F)^\wedge$$

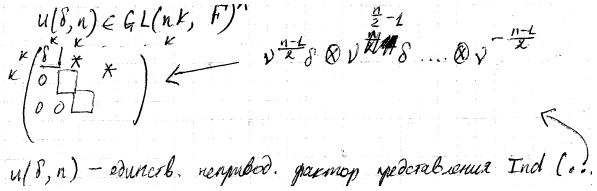


Diagram and typed description.

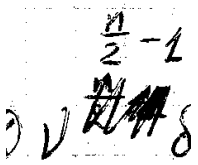
(6)

$$\nu^{\frac{n-1}{2}}\delta \otimes \nu^{\frac{n}{2}-1}\delta \ \dots \ \otimes \nu^{-\frac{n-1}{2}}$$

	k	k	
k	δ	*	
k	0	//////	*
	0	0	//////

$\mathrm{Ind}(\dots)$

Original source details: GELFAND-P001329



Undeciphered passage. The exponent $\frac{n}{2} - 1$ is written above the cancellation at the second ν. Traces of a fraction remain beneath it. There is no δ after the final power of ν on the line.

Original source details: GELFAND-P001329

$u(\delta, n)$ is the unique irreduc. quotient of the representation Ind$(\dots)$

I.M.G.'s question [wavy underline]. What about other groups?

Kirillov [wavy underline] [crossed out nearby: Kir. (?)]



Uncertain reading. Tentatively "Kir." under the cancellation.

Original source details: GELFAND-P001329

In addition to $GL(n, \mathbb{C})$, all unit. irreduc. reps. are also known for the groups [wavy underline]

$$B_2 = C_2\ ; \qquad G_2\ ; \qquad SU(2,2)\ ; \qquad SU(2,n)$$

$$SO(5) \sim Sp(4) \qquad \text{Knapp}$$

(Duflo)

Vogan did the quaternionic ones

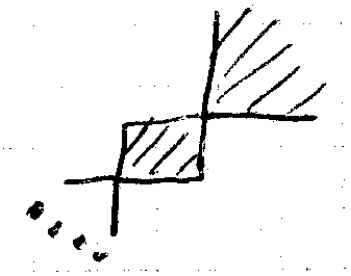
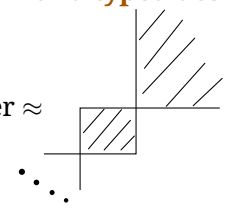


Diagram and typed description.

answer ≈



Original source details: GELFAND-P001329

GELFAND-P001330

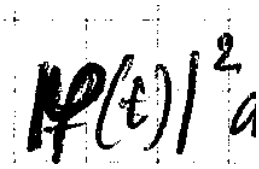
I.M.G. Introduce the inner prod.

$$\int |x_1 - x_2|^{-\varrho} f(x_1)\overline{f(x_2)}\, dx_1 dx_2$$

For $\varrho < 2$ the integral converges

$\varrho > 2 \Rightarrow$ need to continue analytically and the form will fail to be positive def.

$$\int t^{\tilde{\varrho}} |\varphi(t)|^2\ dt$$



Uncertain reading. The function is written φ. The sign under the overwriting may be f.

Original source details: GELFAND-P001330

Positive when $\varphi(0) = 0$, and otherwise (regularization) we must also impose $\varphi'(0) = 0$, etc.
Degree of diverg. $\to$ number of minus signs
Kazakov.
Integrals that need to be evaluated

$$\int d(\Omega)_{U(N)} \exp[\mathrm{tr}\, \Omega^{+} p^{\mu} \Omega p^{\mu}], \qquad \Omega^{+} = \Omega^{-1} \tag{1}$$



Original source details: GELFAND-P001330

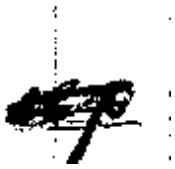
Uncertain reading. A superscript above the second Ω is crossed out, possibly $+$.

$$p^{\mu} = \mathrm{diag}(p_1^{\mu}, p_2^{\mu}, \ldots, p_N^{\mu}), \qquad \mu = 1, 2$$

The generalization is also of interest

$$\int d(\Omega)_{U(N)} \exp[\mathrm{tr}(\Omega^{-1} p^{\mu} \Omega q^{\mu})]. \tag{2}$$

Without μ, a 1-dimensional [crossed out: “кр.” (?)] chiral field (triv. problem)



Original source details: GELFAND-P001330

Uncertain reading. The short cancellation before the abbreviation for “chiral” tentatively reads “кр.”.

$\mu = 1, 2$, a 2-dimensional chiral field (not evaluated)

$$\downarrow$$

done by another method.

Perturbation theory shows that all [above the line: planar] diagrams of the 2-dimensional chiral field are reproduced. Namely, write

$$\Omega = e^{iA} \tag{3}$$

and expand in powers of A.

Approximation with $\Omega = 1$, then account for fluctuations.

$$(2) = Z(p, q) \tag{4}$$

Write the Haar measure and substitute $\Omega = e^{iA}$ in $d\Omega$ as well.

At leading order, a Gaussian term.



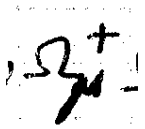
Uncertain reading. Beyond the right edge of the leaf, three clipped marks from the facing leaf are visible, including “(6”.

Original source details: GELFAND-P001330

GELFAND-P001331

[source number: 43]

$$Z = \int \prod_{\mu=1}^{d} d(\Omega_\mu) \exp\left[\beta \sum_{\mu,\nu} \operatorname{tr} \Omega_\mu \Omega_\nu \Omega_\mu^+ \Omega_\nu^+\right] \tag{5}$$



Uncertain reading. The index on the first adjointed factor in (5) is heavily retraced. It reads as μ.

Original source details: GELFAND-P001331

This is equiv. to ordinary gauge theory (as fns. of β) as $N \to \infty$.

— · —

Evaluation of integrals



Original source details: GELFAND-P001331

Uncertain reading. The ending of the word "integrals" is heavily retraced.

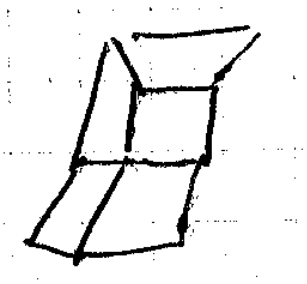
$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \int d^{N^2}u\, d^{N^2}v \exp \operatorname{tr}\left(-u^2 - v^2 + 2cuv - \frac{g}{N}(u^4+v^4)\right), \quad u^+ = u, \quad v^+ = v \tag{6}$$

$F(c,g)$ is the generating fn. of graphs, coverings of the sphere by quadrilaterals



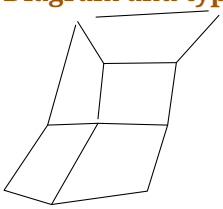
Original source details: GELFAND-P001331

Uncertain reading. Two horizontal strokes are visible above the word "graphs".



Original source details: GELFAND-P001331

Diagram and typed description.



All the quadrilaterals are coloured in 2 colours (white and black)

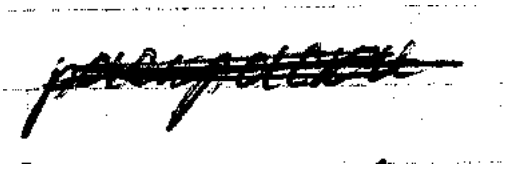
$$F(c,g) = \sum T_n^m g^n \quad \text{[crossed out: exp]} \quad e^{2cm}$$



Original source details: GELFAND-P001331

Uncertain reading. The cancelled fragment after g^n reads tentatively as exp.

T_n^m is the number of graphs for which the boundary length of the [crossed out: coloured (?)] coloured patches is m. n is the number of little squares

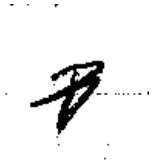


Original source details: GELFAND-P001331

Uncertain reading. The crossed-out word before “coloured patches” tentatively repeats “coloured”.

(edge length $\stackrel{\text{def}}{=} 1$)

This is also equiv. to the Ising model. ↗



Original source details: GELFAND-P001331

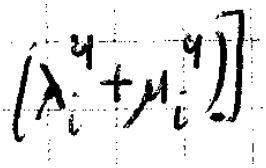
Uncertain reading. The mark after “Ising” resembles a retraced slanting arrow.

Let

$$u = \Omega_1^+ \lambda \Omega_1, \quad v = \Omega_2^+ \mu \Omega_2, \qquad \lambda, \mu \text{ are diagonal matrices}$$

Then

$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \iint \left(\prod_{i=1}^{N} d\lambda_i \, d\mu_i \right) \Delta^2(\lambda)\Delta^2(\mu)$$
$$\exp\Big[- \sum_{i=1}^{N} \big[\lambda_i^2 + \mu_i^2 + \frac{g}{N}(\lambda_i^4 + \mu_i^4) \big] \Big] \tag{7}$$
$$\int d\Omega \, \exp 2c \, \mathrm{tr}(\Omega^+ \lambda \Omega \mu),$$



Original source details: GELFAND-P001331

Uncertain reading. After $\lambda_i^4 + \mu_i^4$, one closing parenthesis and one closing square bracket are visible.

$$\underbrace{\Omega = \Omega_1 \Omega_2^+}_{\substack{\text{relative matrix} \\ \text{angle}}}$$



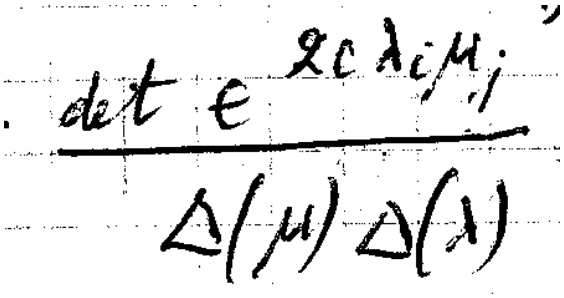
Original source details: GELFAND-P001331

Uncertain reading. The mark to the left of the formula resembles a crossed-out 4 or a slanting arrow.

Here $\Delta(\lambda) = \prod_{i<j}(\lambda_i - \lambda_j)$

But

$$\int d\Omega \, \exp 2c \operatorname{tr}(\Omega^{+}\lambda\Omega\mu) = \frac{\det e^{2c\lambda_i\mu_j}}{\Delta(\mu)\Delta(\lambda)} \tag{8}$$

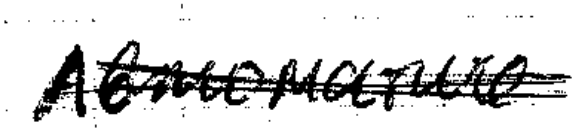


Uncertain reading. The right-hand side is written as a fraction with the determinant above two Δ factors.

Original source details: GELFAND-P001331

There are now $2N$ variables and the saddle-point method applies!

[Crossed out: Automatic...] We obtain



Uncertain reading. The beginning "Automatic..." is visible under the cancellation.

Original source details: GELFAND-P001331

$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \int \prod_{1}^{N} d\lambda_i \, d\mu_i \, \Delta(\lambda)\Delta(\mu) \times \underbrace{\times \exp\left[-\sum_{1}^{N}\left[\lambda_i^2 + \mu_i^2 + \frac{g}{N}(\lambda_i^4 + \mu_i^4) - 2c\lambda_i\mu_i\right]\right]}_{w(\lambda_i,\mu_i)} \tag{9}$$



Uncertain reading. The mark under the second line of (9) is labelled $w(\lambda_i, \mu_i)$.

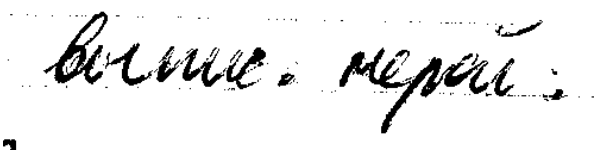
Original source details: GELFAND-P001331

$\Delta(\lambda)$ antisymm., $\Rightarrow$ antisymmetrization takes place automatically

GELFAND-P001332

$$F(c,g) = \lim_{N\to\infty} \frac{1}{N^2} \log \int\int \prod_{i=1}^{N} d\lambda_i \, d\mu_i \, w(\lambda_i,\mu_i)\Delta(\lambda)\Delta(\mu). \tag{10}$$

There are polynomials, orthog. with weight W and with a highest measure.



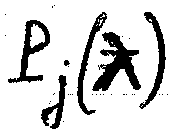
Uncertain reading. The word after "высше." (highest) reads tentatively as "мерой" (measure).

Original source details: GELFAND-P001332

$$\Delta(\lambda) = \det(\lambda_i^{j-1})_{i,j=1}^N = \det[P_{j-1}(\lambda_i)], \tag{11}$$

where

$$P_j(\lambda) = \lambda^j + \sum_{k=0}^{j-1} a_k \lambda^k$$



Uncertain reading. The argument of P_j reads as λ. There may be a z under the overwriting.

Original source details: GELFAND-P001332

Construct a system of orthog. polynomials P_j such that



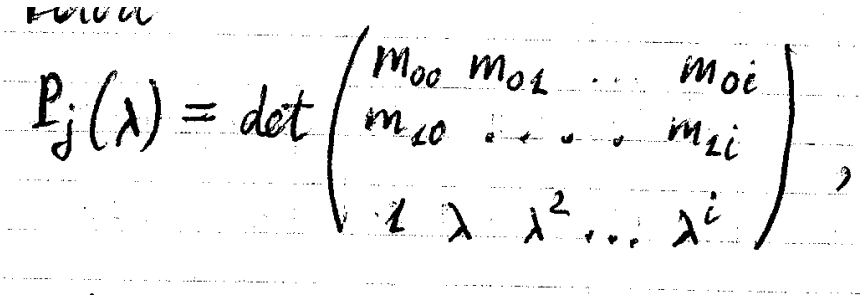
Uncertain reading. A separate mark before "Construct" is heavily overwritten, possibly Russian «И» ("And").

Original source details: GELFAND-P001332

$$\int_{-\infty}^{\infty} d\lambda\, d\mu\, W(\lambda,\mu) P_i(\lambda) P_j(\mu) = h_i \delta_{ij} \tag{12}$$

Then

$$P_j(\lambda) = \det \begin{pmatrix} m_{00} & m_{01} & \cdot & \cdot & m_{0i} \\ m_{10} & \cdot & \cdot & \cdot\cdot & m_{1i} \\ 1 & \lambda & \lambda^2 & \cdots & \lambda^i \end{pmatrix}, \tag{13}$$



Original source details: GELFAND-P001332

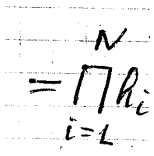
Uncertain reading. The left-hand side is $P_j(\lambda)$. The last column reads m_{0i}, m_{1i}, λ^i.

where

$$m_{ij} = \int d\lambda\, d\mu\, w(\lambda,\mu) \lambda^i \mu^j. \tag{14}$$

For h_j we get a system of recurrence eqs. In the limit of infinite N it becomes a single eq. Next,

$$\iint \prod d\lambda_i\, d\mu_i\, w(\lambda_i,\mu_i) \Delta(\lambda)\Delta(\mu) = \prod_{i=1}^N h_i \tag{15}$$



Uncertain reading. The right-hand side reads $\prod_{i=1}^N h_i$.

Original source details: GELFAND-P001332

Answer

$$\begin{cases} F(c,z) = \frac{1}{2}\ln\frac{z}{g(z)} + \frac{1}{2g^2(z)}\int_0^z \frac{dt}{t}g^2(t) - \frac{1}{g(z)}\int_0^z \frac{dt}{t}g(t) \\ g\ [\text{crossed out}] = \frac{z}{(1-3z)^2} - c^2z + 3c^2z^3 \end{cases} \tag{16}$$



Undeciphered passage. A short entry after g is crossed out, possibly (z).

Original source details: GELFAND-P001332

Eliminating z from this gives $F(c,g)$.

(this gives the answer parametrically)

This makes it possible to study the asymptotics of the number of graphs

([crossed out: ...] use $g'_z(z) = 0$ for the singularities)



Undeciphered passage. Slanted strokes cover the short fragment before “use”.

Original source details: GELFAND-P001332

GELFAND-P001333

9.2.87 Gelfand Sem.

Don Tsagir (Don Zagier)

I.M.G. [wavy underline] Faltings’s idea is to make p-adic strings.

One piece of work (problem for the next seminar) Don Zagier.

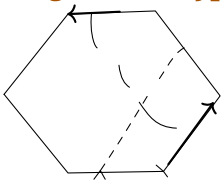
Given an oriented [wavy underline] $2N$-gon, its sides are divided into pairs (each glued to some other side) [Inserted above the line: it cannot be glued to itself]

Identify in reverse (with a change of orientation)

We obtain a surface of genus g



Diagram and typed description.



Some short strokes on the boundary and joins to the interior lines are uncertain.

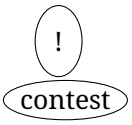
Original source details: GELFAND-P001333

[In the left margin.]



Original source details: GELFAND-P001333

Diagram and typed description.



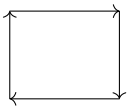
How many gluings yield a surface of genus g? [Wavy underline under the question.]

Write a formula. (2 gluings that differ by a rotation are counted as different) (a fn. of N and g).



Original source details: GELFAND-P001333

Diagram and typed description.



we can obtain a torus and a sphere.

This is the basis for computing the Euler characteristic of Teichmüller space!

D.Z. 2 things.

Modular forms and special values of zeta-functions

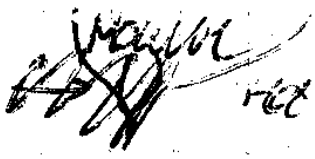
1. Classical Modular forms

2. Special values of zeta-functions

Modular forms. (parabolic) [wavy underline]

$$\Gamma = SL(2,\mathbb{Z})/\{\pm 1\}, \qquad H = \{z = x + iy;\ y > 0\}$$

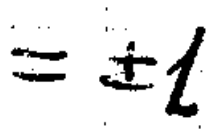
$f(z)$ holom. [above the line: on H], [crossed out: ...] [above the cancellation: zero] at ∞, and



Original source details: GELFAND-P001333

Undeciphered passage. The crossed-out fragment before "at ∞" is unclear. "Zero" is written above it.

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \tag{1}$$
$$a,b,c,d \in \mathbb{Z}, \qquad ad-bc = \pm 1$$



Uncertain reading. The right-hand side is written as ± 1.

Original source details: GELFAND-P001333

[At right:] Zeros at ∞ — cusp.

$$S_k = \{f\}, \qquad \dim S_k < \infty$$

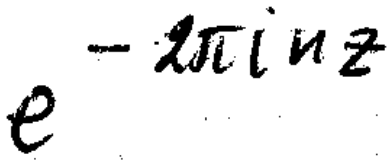
GELFAND-P001334

[source number: 69]

Hecke: $\exists$ many

$$f(z) = \sum_{n=1}^{\infty} a(n) e^{-2\pi i n z} \tag{2}$$

(since it follows from (1) that $f(z+1) = f(z)$)



Uncertain reading. In (2) there is a minus sign before $2\pi inz$.

Original source details: GELFAND-P001334

$\exists$ a basis in which $a(n) \in \mathbb{Z}$

(3) Example. [broken underline]

For $k = 2,4,6,8,10,14$, $S_k = \{0\}$ (no cusp forms)

Modular form [wavy underline] $O(1)$ at ∞ (as opposed to zero)

$$M_k \ni f(z) = \sum_{n=0}^{\infty} a(n) q^n, \qquad q = e^{2\pi i z} \tag{4}$$

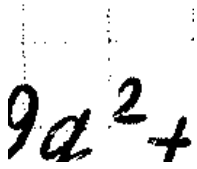
Then $\exists$ an Eisenstein series [deletion between «ряд» and «Eisenstein'a»]



Undeciphered passage. Before «Eisenstein'a», $E\ldots$ is crossed out.

Original source details: GELFAND-P001334

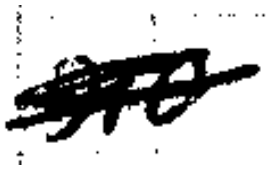
$$E_k(z) = \frac{1}{2} \sum_{(c,d)=1} \frac{1}{(cz+d)^2} \tag{5}$$



Original source details: GELFAND-P001334

Uncertain reading. In (5) the exponent in the denominator is written as 2.

Ex. Check [crossed out] automorphy.



Original source details: GELFAND-P001334

Undeciphered passage. The crossed-out word before “automorphy” may read «ero» (“its”).

$$\begin{aligned} E_4(z) &= 1 + 240(q + 9q^2 + 28q^3 + 73q^4 + \ldots) = \\ &= 1 + 240 \sum_{n=1}^{\infty} \underbrace{\sigma_3(n)}_{\| \\ \sum_{d|n} d^3} q^n \end{aligned} \tag{6}$$

(7) k odd $\Rightarrow$ no cusp forms (obvious from the signs)

$$\begin{aligned} S_k \ni \Delta(z) &= q - 24q^2 + 252q^3 + \ldots - 6048q^6 + \ldots \\ &= \sum \tau(n) q^n \end{aligned} \tag{8}$$

(no constant term $\Rightarrow$ cusp form

$$24 \cdot 252 = 6048$$

$$\tau(nm) = \tau(n)\tau(m) \quad \text{when } (n,m) = 1. \tag{9}$$

(10) Claim. $\exists$ a basis of cusp forms whose coefficients have property (9)

$$\begin{aligned} \frac{E_4}{240} &= \frac{1}{240} + \sum_{n=1}^{\infty} \sigma_3(n) q^n, \\ \sigma_3(6) &= \sigma_3(2)\sigma_3(3) \end{aligned} \tag{11}$$

(12) Def. $f(z)$ is called a Hecke form [«Гекке» with “(Hecke)” underneath] if $f = \sum_0^\infty a(n)q^n$

$$a(nm) = a(n)a(m) \quad \text{when } (n,m) = 1$$

(13) Claim. The number of Hecke forms = the dimension of the subspace, and they form a basis.

GELFAND-P001335

Proof: Hecke op. T_n:



Diagram and typed description. S_k

Original source details: GELFAND-P001335

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad f = f(z)$$

(14) notation:

$$(f \mid \gamma)(z) = \left(\frac{cz+d}{\sqrt{ad-bc}}\right)^{-k} f\left(\frac{az+b}{cz+d}\right)$$

(action of $SL_2(\mathbb{R})$) [added below] and even $GL_2^+(\mathbb{R})$

[The denominator $\sqrt{ad-bc}$ is inserted and the parenthesis extended.]



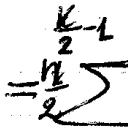
Uncertain reading. A stroke crosses the last letter of the numerator, which reads as b.

Original source details: GELFAND-P001335

$$M_k \ni f \Longleftrightarrow f \mid \gamma = f \qquad (\gamma \in \Gamma).$$

$$T_1 = 1.$$

$$T_n\big(\overset{S_k}{\downarrow} f\big)(z) = \frac{n^{\frac{k}{2}-1}}{2} \sum_{\gamma \in \Gamma \backslash M_n} f \mid \gamma. \tag{15}$$



Uncertain reading. In (15), a denominator 2 is visible under $n^{k/2-1}$.

Original source details: GELFAND-P001335

$$M_n \subset M(2,\mathbb{Z}) \quad \underset{2\times 2}{\text{— matrices with det} = n}.$$

$n = 2 \Rightarrow$ there are 3 terms

$$\begin{aligned} T_2(f) &= 2^{\frac{k}{2}-1}\left[f \mid \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} + f \mid \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} + f \mid \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\right] = \\ &= 2^{k-1} f(2z) + \frac{1}{2} f\left(\frac{z}{2}\right) + \frac{1}{2} f\left(\frac{z+1}{2}\right) \end{aligned} \tag{16}$$



Undeciphered passage. Strokes by the first f on the second line of (16) are crossed out. They may be a denominator 2.

Original source details: GELFAND-P001335

Example: $k = 12$. Then for Δ, given by its Fourier series,

$$T_2(\Delta) = 2^{11} \sum_{n=1}^{\infty} \tau(n) q^{2n} + \sum_{n=1}^{\infty} \tau(2n) q^n \tag{17}$$

From the condition $T_2(\Delta) = \lambda_2 \Delta$ we find $\lambda_2 = -24$, etc.

(18) The eigenvalue λ_n is always a Fourier coeff. (follows directly)



Undeciphered passage. Fragmentary strokes are visible in the left margin by (18), without a secure reading.

Original source details: GELFAND-P001335

(19) Def. 2. f is a Hecke form $\Longleftrightarrow T_n(f) = \lambda_n f, \forall n$.

The existence of a basis follows from 2 facts.

$$\text{a)} \quad T_n T_m = T_m T_n \tag{20}$$

b) S_k has an inner product (Petersson) in which the T_n are self-adjoint!

Then $\exists$ a basis of common eigenvectors.

Always:

$$\begin{aligned} a(1) &= 1 \\ a(n) &= \lambda_n \end{aligned} \tag{21}$$

$$T_n T_m = T_{nm} \quad \underset{\text{when } (n,m)=1}{\Longrightarrow} \quad a(n)a(m) = a(nm) \tag{22}$$

(Mordell [wavy underline] introduced the Hecke operators in 1917)

Hecke proved implication (22).

GELFAND-P001336

[source number: 70]

Question [wavy underline]: compute $\mathrm{Tr}(T_n)$



Diagram and typed description. $T_n : S_k$ 

Original source details: GELFAND-P001336

$\underline{k = 12}$ $\quad \Delta = \sum_1^\infty \tau(n) q^n.$

$$T_n(\Delta) = \tau(n)\Delta, \quad T_n = (\tau(n)), \quad \mathrm{Tr}\, T_n = \tau_n. \tag{23}$$

The trace formula will give, in particular, a formula for $\tau(n)$

Inner product:

$$(f, g) = \int_{\Gamma\backslash H} f(z)\overline{g(z)} y^k \underbrace{dV}_{\frac{dx\,dy}{y^2}}, \qquad f, g \in S_k. \tag{24}$$

(inv. measure)

([Inserted above a deletion] $f(z)\overline{g(z)}y^k$ is Γ-invariant ($\Rightarrow$ the integral makes sense))



Undeciphered passage. The word below the insertion, possibly «оно» (“it”), is crossed out.

Original source details: GELFAND-P001336

(25) Lemma. Let $F(z)$ be holomorphic on H and sufficiently small at ∞, $k \geq 2$, $z_0 \in H$. Then

$$\int_H \frac{F(z)}{(\bar{z} - z_0)^k} y^k \, dV = C_k F(z_0), \tag{26}$$

where

$$C_k = \frac{(-1)^{k/2}\pi}{2^{k-2}(k-1)} \qquad \text{for } k \text{ even}.$$

Proof [broken underline]:

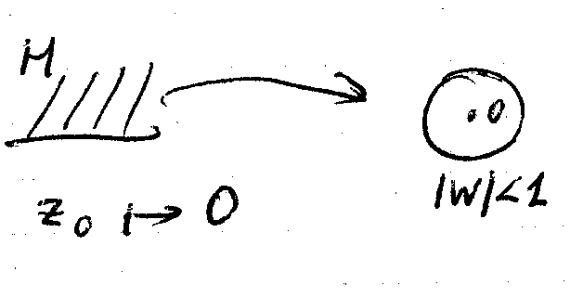
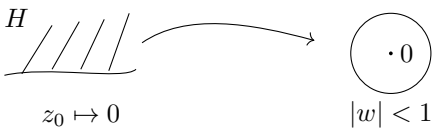


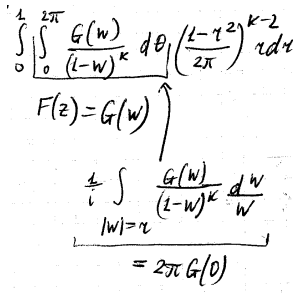
Diagram and typed description.



Original source details: GELFAND-P001336

$$w = \frac{z - z_0}{z - \bar{z}_0} = re^{i\theta}, \quad 0 \le r < 1,$$
$$z \in \mathbb{R} \Rightarrow |w| = 1$$

We obtain



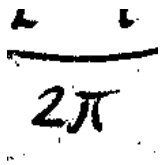
Original source details: GELFAND-P001336

Diagram and typed description.

$$\int_0^1 \left[\int_0^{2\pi} \frac{G(w)}{(1-w)^k}\, d\theta \right] \left(\frac{1-r^2}{2\pi} \right)^{k-2} r\, dr$$

$$F(z) = G(w)$$

$$\underbrace{\frac{1}{i} \int_{|w|=r} \frac{G(w)}{(1-w)^k} \frac{dw}{w}}_{= 2\pi G(0)}$$



Original source details: GELFAND-P001336

Uncertain reading. The denominator of the radial factor reads 2π.

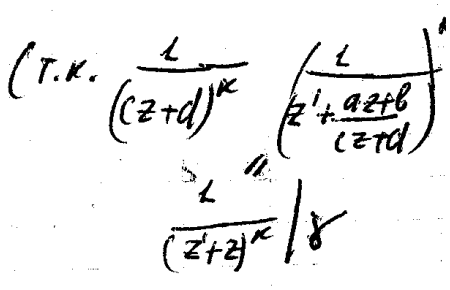
▷

$$K_n(\underset{\uparrow H}{z}, \underset{\uparrow H}{z'}) = \frac{1}{2} \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in M_n} \frac{1}{[(cz+d)z' + (az+b)]^k} \tag{27}$$
$$\in S_k \otimes S'_k.$$

(Bergman kernel) converges for $k \ge 4$.

[At lower right:]

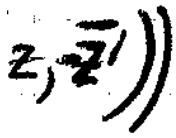
$$(\text{since } \frac{1}{(cz+d)^k} \left(\frac{1}{z' + \frac{az+b}{cz+d}} \right)^k$$
$$= \frac{1}{(z'+z)^k} \mid \gamma$$



Original source details: GELFAND-P001336

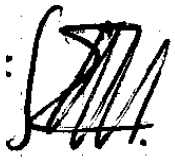
Uncertain reading. The two slanting strokes between the expressions have been read as an equals sign.

$$\big(f(z), K_n(z, -\bar{z}')\big) = \int_{\Gamma\backslash H} \sum_{\gamma\in M_n} \frac{f(z)}{(\bar{z}-z')^k} \mid \gamma \Big] y^k \, dV. \tag{28}$$



Uncertain reading. The short stroke before $\bar{z}'$ has been read as a minus sign.

Original source details: GELFAND-P001337

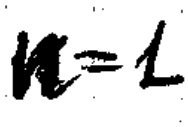


Undeciphered passage. An entry, possibly a $\sum$ sign, is crossed out after the integral sign in (28).

Original source details: GELFAND-P001337

$$n = 1 \quad \Rightarrow \quad \int_H \frac{f(z)}{(\bar{z}-z')^k} \, dz$$

and one has to use the lemma.



Uncertain reading. The first letter of the condition is overwritten. Tentative reading n.

Original source details: GELFAND-P001337

Any $n \Rightarrow$ we obtain $n^{1-k} C_k T_n(f)(z)$, i.e. K_n is the kernel of the Hecke op.

$f_1, \ldots, f_d$ is a basis of Hecke forms in S_k

$$f(z) = \sum_{n=1}^{\infty} a_j(n) q^n, \qquad a_j(1) = 1, \qquad T_n(f_j) = a_j(n) f_j$$

$$K_n(z, z') = \frac{C_k}{n^{k-1}} \sum_{j=1}^{d} \frac{a_j(n)}{(f_j, f_j)} f_j(z) f_j(z') \tag{29}$$

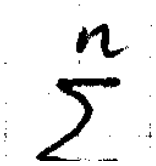
(obvious because the matrix is diagonal)



Undeciphered passage. A symbol, possibly $\sum$, is crossed out between the equals sign and the fraction in (29).

Original source details: GELFAND-P001337

$$\int_{\Gamma\backslash H} K_n(z,-\bar{z})y^k\,dV = \frac{C_k}{n^{k-1}} \underbrace{\sum_{j=1}^{n} \lambda_j(n)}_{\| \operatorname{Tr} T_n} \tag{30}$$

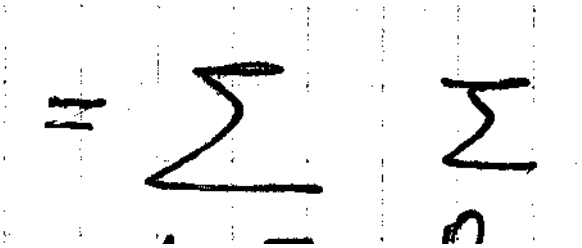


Original source details: GELFAND-P001337

Uncertain reading. The upper limit of the sum in (30) is written as n.

But:

$$\begin{aligned} K_n(z,-\bar{z}) &= \frac{1}{2} \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in M_n} \frac{1}{[c|z|^2 + d\bar{z} - az - b]^k} = \\ &= \sum_{\substack{t\in\mathbb{Z} \\ | \\ \text{trace}}} \sum_{\substack{Q \\ | \\ \operatorname{disc} Q = t^2-4n}} \frac{1}{(A|z|^2 + Bx + C - ity)^k} \end{aligned} \tag{31}$$



Original source details: GELFAND-P001337

Uncertain reading. In the second line of (31), the summation sign immediately follows the equals sign.



Original source details: GELFAND-P001337

Undeciphered passage. An isolated curved stroke is visible in the left margin next to (31).

[To the left of the second line of (31)] Put

$$\begin{aligned} d + a &= t \\ c &= A \\ d - a &= B \\ -b &= C \end{aligned}$$

$$\begin{aligned} Q = [A,B,C] &= Au^2 + Buv + Cv^2 \\ \operatorname{disc} Q = t^2 - 4n &= B^2 - 4AC = \\ &= (d-a)^2 + 4bc = (d+a)^2 - 4(ad-bc) \end{aligned}$$



Original source details: GELFAND-P001337

Undeciphered passage. A short isolated vertical stroke is visible in the lower left margin.

$K_n(z,-\bar{z})y^k$ must be Γ-invariant!

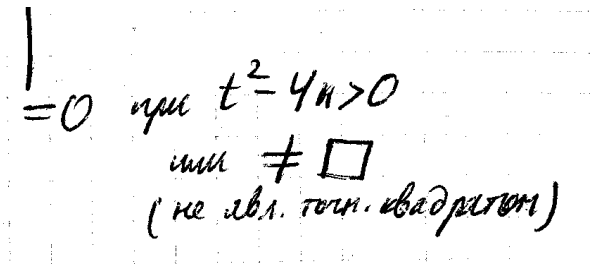
GELFAND-P001338

[source number: 71]

We obtain

$$K_n(z,-\bar{z})y^k = \sum_{t\in\mathbb{Z}} H(t^2-4n)(\cdot\ \cdot\ \cdot), \tag{32}$$

[Arrow to H] number of quadratic forms with the given discr. (class number)
[Arrow to the parentheses with dots] $=0$ when $t^2-4n>0$ or $\neq \square$ (is not a perfect square)



Original source details: GELFAND-P001338

Uncertain reading. The condition reads «или» (“or”).

and this is already a finite sum!

$$Q=[A,B,C], \qquad A,B,C\in\mathbb{Z}$$

$$Au^2+Buv+Cv^2$$

Γ acts [added below] on the set of such Q:

$$(Q\circ\gamma)(u,v) = Q\big((u,v)\gamma\big)$$

For $t\in\mathbb{R}$ put

$$R_{Q,t}(z) = \frac{y^k}{(A|z|^2+Bx+C-ity)^k} \tag{33}$$

$$K_n(z,-\bar{z})y^k = \sum_{t\in\mathbb{Z}} \sum_{\substack{Q \\ \operatorname{disc} Q=t^2-4n}} R_{Q,t}(z) \tag{34}$$



Original source details: GELFAND-P001338

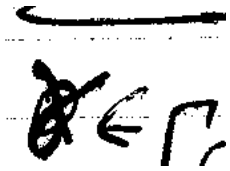
Undeciphered passage. A symbol at the end of (34) is crossed out. Two horizontal strokes are visible under the crossing-out.

Since

$$R_{Q,t}(\gamma z) = R_{Q\circ\gamma,t}(z), \tag{35}$$

the sum in (34) is automatically invariant

$$(34) = \sum_{t\in\mathbb{Z}} \sum_{\substack{[Q] \bmod \Gamma \\ \operatorname{disc} Q = t^2-4n}} \underbrace{\sum_{\gamma\in\Gamma_Q\backslash\Gamma} R_{Q\circ\gamma,t}(z)}_{\substack{\text{the integral of this is computed} \\ \text{— this is } (\cdots) \text{ in (32)}}} \tag{36}$$

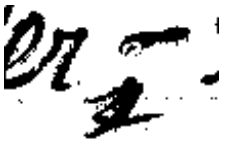


Uncertain reading. The index of the last sum is overwritten. It has been read as γ, as in the subscript of R in the same line.

Original source details: GELFAND-P001338

The important point is that the factor $H(t^2-4n)$ appears

(37) Theorem (Eichler-Selberg trace formula, 1956)



Undeciphered passage. A small sign resembling 2 or z is crossed out below the hyphen.

Original source details: GELFAND-P001338

Let $k \geq 4$ be even

$$\begin{aligned} \mathbb{Z} \ni \operatorname{Tr}(T_n : S_k \circlearrowleft) &= -\frac{1}{2} \sum_{\substack{t\in\mathbb{Z} \\ |t|\leq 2\sqrt{n}}} p_k(t,n) H(t^2-4n) \\ &\quad - \frac{1}{2} \sum_{d|n} \min\left(d, \frac{n}{d}\right)^{k-1} \end{aligned} \tag{38}$$

$$\begin{pmatrix} \tau(n) & k=12 \\ 0 & k=8 \end{pmatrix} \qquad \text{[below the trace]}$$

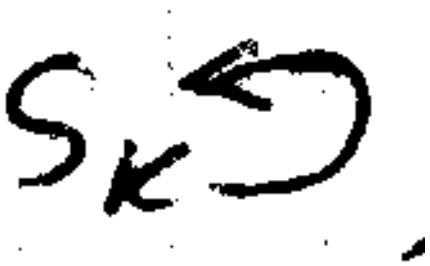


Diagram and typed description. $S_k \circlearrowleft$

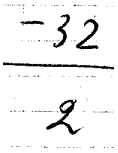
Original source details: GELFAND-P001338

GELFAND-P001339

Δ	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
$H(\Delta)$	$-\frac{1}{12}$	0	0	$\frac{1}{3}$	$\frac{1}{2}$	0	0	1	1	0	0	1	$\frac{4}{3}$	0

-14	-15	-16	-17	-18	-19	-20	-21	-22	-23	-24	-25
0	2	3/2	0	0	1	2	0	0	3	2	0

-26	-27	-28	-29	-30	-31	-32	-33
0	4/3	2	0	0	3	2	0



Uncertain reading. The table entry under -32 reads 2.

Original source details: GELFAND-P001339

Number of classes accounting for automorphisms ⇒ denominators appear



Uncertain reading. The words after the arrow read «появл. знаменателей» (“denominators appear”).

Original source details: GELFAND-P001339

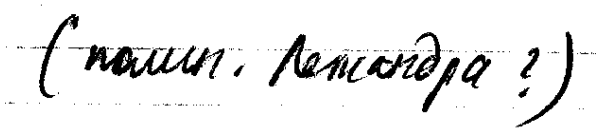
[At right, with a bracket] only the following occur

$$\Delta = B^2 - 4AC \equiv 0, 1 \pmod 4$$
otherwise $H(\Delta) = 0$.

$$\begin{aligned} p_2(t,n) &= 1 \\ p_3 &= t \\ p_4 &= t^2 - n \\ p_5 &= t^3 - 2tn \\ p_6 &= t^4 - 3t^2 n + n^2 \end{aligned}$$

$$p_k(t,n) = \frac{\alpha^{k-1} - \bar{\alpha}^{k-1}}{\alpha - \bar{\alpha}}$$

(Legendre polyn.?)



Uncertain reading. The manuscript has a question mark after “Legendre”.

Original source details: GELFAND-P001339

where α is such that

$$\alpha + \bar{\alpha} = t, \qquad \alpha\bar{\alpha} = n$$

(39) Corollary. $k = 4$, n prime. Then

$$-2 = \sum_{t^2 \le 4n} (t^2 - n) H(t^2 - 4n)$$



Uncertain reading. At the beginning of $(t^2 - n)$, the letter t is overwritten.

Original source details: GELFAND-P001339

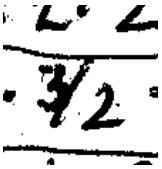


Undeciphered passage. An isolated stroke under H has no secure reading as a symbol.

Original source details: GELFAND-P001339

$n = 5$ check

t	
0	$-5 \cdot 2$
± 1	$-4 \cdot 1 \cdot 2$
± 2	$-1 \cdot 3/2 \cdot 2$
± 3	$+4 \cdot 1 \cdot 2$
± 4	$+11 \cdot 4/2 \cdot 2$



Uncertain reading. In the row $t = \pm 2$ the fraction reads $3/2$. A stroke intersects the numerator.

Original source details: GELFAND-P001339



Uncertain reading. In the row $t = \pm 4$ the fraction reads $4/2$. The numerator may be 1 joined to the slash.

Original source details: GELFAND-P001339

In the general Selberg formula the sum is infinite.

GELFAND-P001340

[source number: 72]

Eisenstein series (— Maass) are nonholomorphic.

$$E(z, \underset{\uparrow C}{s}) = \frac{1}{2} \sum_{(c,d)=1} \frac{y^s}{|cz+d|^{2s}} \qquad \operatorname{Re} s > 1 \tag{40}$$



Uncertain reading. An upward arrow from the letter C appears under s.

Original source details: GELFAND-P001340



Undeciphered passage. A short fragment at the right of (40) is crossed out. The first stroke may belong to a c.

Original source details: GELFAND-P001340

$$E(\gamma z, s) = E(z, s), \quad \gamma \in \Gamma. \qquad \text{(easy)} \tag{41}$$

$$\Gamma_\infty = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \; b \in \mathbb{Z} \right\}$$

Then

$$E(z, s) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} \operatorname{Im}(\gamma z)^s \tag{42}$$

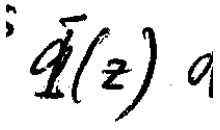
Let Φ be a Γ-invariant function on H (small at ∞). Consider the integral

$$\int_{\Gamma \backslash H} E(z, s)\varphi(z) \underset{\underset{\frac{dx\,dy}{y^2}}{\|}}{dV} = \tag{43}$$

[At right, after a double vertical line:]

$$\varphi(z+1) = \varphi(z)$$
$$\Downarrow$$
$$\varphi(z) = \sum_{n \in \mathbb{Z}} c_n(y) e^{2\pi i n x} \qquad z = x + iy$$

$$= \int_{\Gamma_\infty \backslash H} \operatorname{Im}(z)^s \bar{\varphi}(z)\, dV =$$
$$= \int_0^\infty \int_0^1 \Phi(x+iy) y^{s-2}\, dx\, dy =$$



Uncertain reading. In the first integral continuing (43), a bar appears over φ.

Original source details: GELFAND-P001340



Original source details: GELFAND-P001340

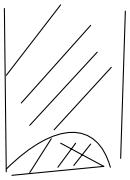


Diagram and typed description.

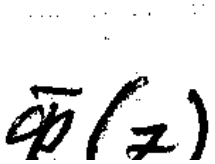
$$= \int_0^\infty c_0(y) y^{s-2}\, dy$$

(Rankin method)

Prop. [Arrow from above to $\varphi(z)$] Γ-invariant with decay

$$\int_{\Gamma\backslash H} \varphi(z) E(z,s)\, dV = \int_0^\infty c_0(y) y^{s-2}\, dy \tag{44}$$

[Arrow from below to $c_0(y)$] const. term in the Fourier exp. of $\bar{\varphi}(z)$.



Original source details: GELFAND-P001340

Uncertain reading. In the annotation to $c_0(y)$, φ also reads as barred.

Applications. [wavy underline]

(45) $E(z,s)$ properties

① merom. continuation in the s-plane (z fixed)

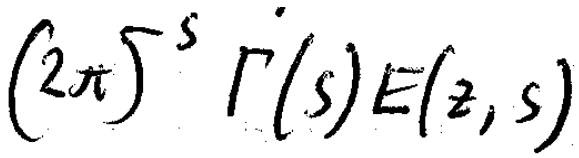
poles at $s = 1$ and $s = \frac{1}{2}\rho$, ρ a root of $\zeta(\rho) = 0$

(there are no other poles)

GELFAND-P001341

② Functional eq.

$$(2\pi)^s \Gamma(s) E(z,s) \quad \text{invariant under } s \mapsto 1-s$$

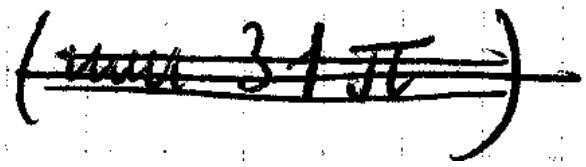


Original source details: GELFAND-P001341

Uncertain reading. The factor before $\Gamma(s)$ reads $(2\pi)^s$.

③

$$E(z,s) = \frac{3/\pi}{s-1} + O(1) \qquad \text{(the residue is independent of } z\text{)}$$

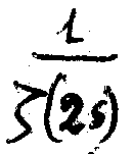


Original source details: GELFAND-P001341

Undeciphered passage. A parenthesized note below the formula is crossed out, possibly "(residue $3/\pi$)".

Reason for meromorphy. Consider

$$\frac{1}{\zeta(2s)} \sum_{\substack{(m,n)\in\mathbb{Z}^2 \\ \neq(0,0)}} \frac{y^s}{|mz+n|^{2s}} \tag{46}$$

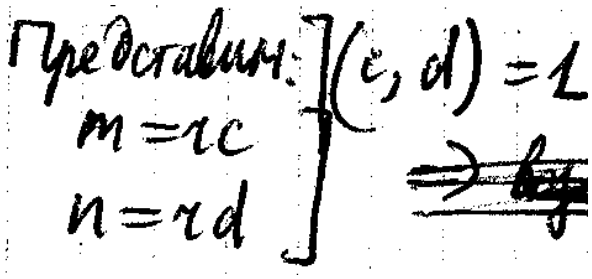


Uncertain reading. The factor in (46) is written $1/\zeta(2s)$.

Original source details: GELFAND-P001341

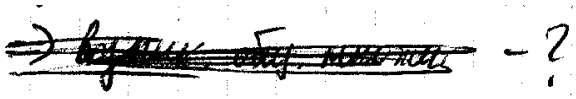
[At right] Write

$$\left.\begin{aligned} m &= rc \\ n &= rd \end{aligned}\right] \quad (c,d)=1 \quad \Rightarrow$$



Uncertain reading. The multiplier in both lines has been read as r.

Original source details: GELFAND-P001341



Undeciphered passage. A line after the arrow is crossed out. "— ?" remains at right.

Original source details: GELFAND-P001341

study using the Poisson summation formula.

$c_0(y)$ has surprising properties! (For every Γ-invariant function $\Phi(z)$!)

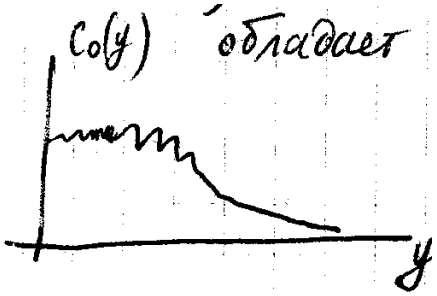
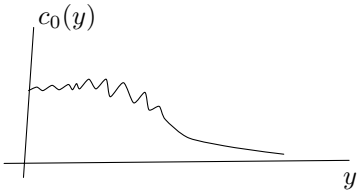


Diagram and typed description.



Original source details: GELFAND-P001341

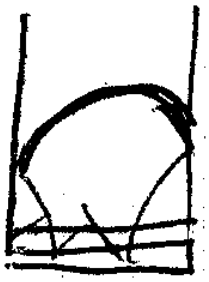
the frequency of oscillations ["as $y \to 0$" inserted above] is given by ρ, a zero, $\rho \mapsto \zeta(2\rho)$



Uncertain reading. The sign in $\zeta(2\rho)$ and before the arrow reads ρ.

Original source details: GELFAND-P001341

(see the behavior of the Mellin transform)



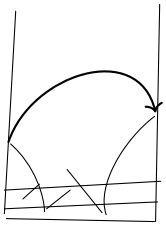


Diagram and typed description.

Original source details: GELFAND-P001341

$c_0(y)$ is obtained by integrating along const. horocycles!



Uncertain reading. The abbreviation before "horocycles" tentatively reads «пост.» ("const.").

Original source details: GELFAND-P001341

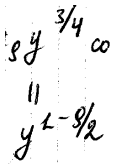
If

$$c_0(y) \sim \sum a_\lambda y^\lambda, \quad \text{then} \quad \int_0^1 y^{\lambda+s-2}\,dy = \frac{1}{\lambda-1+s}$$

$\Rightarrow$ a pole arises at $s = 1-\lambda$.

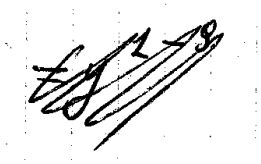
Take Φ to be the characteristic function of a region and then replicate it. We get [Circled insertion pointing to "region"] of volume $= \frac{1}{3}$ of the total

$$a_1 + \sum a_\rho \underbrace{y^{3/4}}_{\| \; y^{1-\rho/2}} \cos\left(\frac{\tau}{2}\log y - \varphi\right) \tag{47}$$



Uncertain reading. Below $y^{3/4}$ the manuscript gives $y^{1-\rho/2}$, with two vertical strokes.

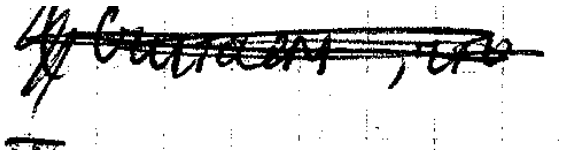
Original source details: GELFAND-P001341



Undeciphered passage. An expression at right is crossed out, possibly $+y^{1-\rho}$.

Original source details: GELFAND-P001341

The asymptotics of intersections are determined by the zeros of the ζ-function.



Undeciphered passage. The line is crossed out. The final word may read «что» ("that").

Original source details: GELFAND-P001341

(48) Theorem Let $\Phi(z)$ be Γ-invariant and $c_0(y)$ the constant term for Φ

$$\Rightarrow \quad \int_0^\infty c_0(y) y^{s-2}\,dy$$

poles at $s = 1, \rho/2$

functional eq. $s \mapsto 1 - s$

Residues [wavy underline]: at $s = 1$ this is a_1 in (47)

GELFAND-P001342

[source number: 73]

$$\frac{3}{\pi} \int_{\Gamma \backslash H} \Phi(z)\, dV = \lim_{y \to 0} c_0(y) = c_0(0) \tag{49}$$

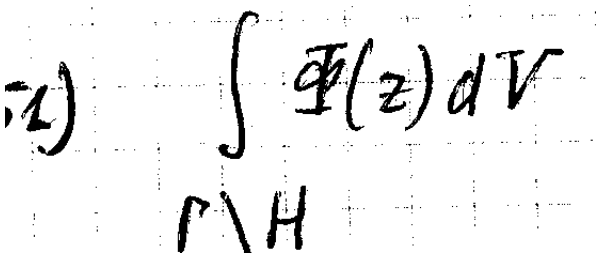
Use: instead of integrating, one can calculate $c_0(0)$, which is easy!

Application. [wavy underline] Let

$$\varphi(z) = y^k K_n(z, -\bar{z}), \quad k \geq 4 \text{ even}, \quad n \geq 1 \tag{50}$$

$$\int_{\Gamma \backslash H} \Phi(z)\, dV = c_0(0) \qquad \text{can be computed!} \tag{51}$$

(but this was known)



Uncertain reading. In (51), no coefficient is written before the integral.

Original source details: GELFAND-P001342

More interesting.

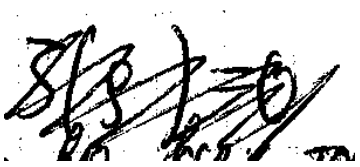
$$\begin{aligned} \int_{\Gamma \backslash H} \Phi(z) E(z,s)\, dV &= \sum_{t \in \mathbb{Z}} \int_{\Gamma \backslash H} \left(\sum_{\substack{Q \\ \operatorname{disc}(Q) = t^2 - 4n}} R_{Q,t}(z) \right) E(z,s)\, dV = \\ &= \sum_{t \in \mathbb{Z}} \underset{\substack{\uparrow \\ \text{some expression}}}{(\cdots)} \quad \zeta_{\mathbb{Q}(\sqrt{t^2-4n})}(s) \end{aligned} \tag{52}$$

$$\forall \text{ field ext. } \mathbb{Q} \subset^{K}_{|\,2} \qquad \zeta_K(s) = \sum \frac{1}{N(\,)^s} = \zeta(s) L(s)$$

$$\zeta(\rho) = 0 \quad \Rightarrow \quad \zeta_K(\rho) = 0$$

It follows that

$$\int_{\Gamma \backslash H} \Phi(z) E(z, \rho)\, dV = 0 \quad \text{provided that } \zeta(\rho) = 0! \tag{53}$$



Uncertain reading. Below (53), $\zeta(\rho) = 0$ is crossed out.

Original source details: GELFAND-P001342

Introduce the space of all such functions

$$V = \{\varphi(z) \text{ on } \Gamma\backslash H\}$$

$$\int_{\Gamma\backslash H} \psi(z)\varphi(z) = 0$$

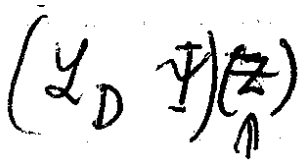


Undeciphered passage. A fragment under the last integral sign and above $\Gamma\backslash H$ is crossed out.

Original source details: GELFAND-P001342

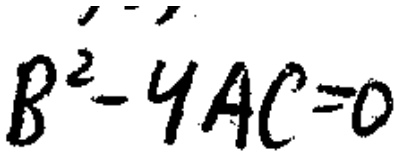
[At right after a double rule] This is a characterization of the roots ρ!!
(Spectrum of the representation in V!) (Laplace op.)
[Below and to the left] Spectrum of the Laplace op.
[Downward arrow] if it is unitary, we obtain the Riemann hypothesis.
[At left, with an arrow to (54)] The Φ form a large set
(The absence of other zeros is unproved)

$$(\mathcal{L}_D\Psi)(\underset{\Gamma\backslash H}{z}) = \sum_{\substack{A,B,C\in\mathbb{Z} \\ B^2-4AC=0}} \Psi\left(\frac{A|z|^2 + Bx + C}{y}\right) \tag{54}$$



Uncertain reading. The looped letter with subscript D has been read as $\mathcal{L}_D$.

Original source details: GELFAND-P001342



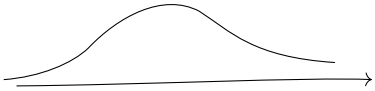
Uncertain reading. The condition under the sum is written $B^2 - 4AC = 0$.

Original source details: GELFAND-P001342

Automatically Γ-invariant



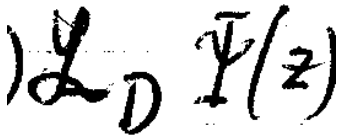
Diagram and typed description.



Original source details: GELFAND-P001342

Ψ is smooth and decaying
this describes W $\quad$ $V = W^{\perp}$

$$\int_{\Gamma\backslash H} E(z,s)\mathcal{L}_D\Psi(z)\,dV = (*)\zeta_{\mathbb{Q}(\sqrt{D})}(s). \tag{55}$$



Uncertain reading. The looped letter with subscript D has been read as $\mathcal{L}_D$.

Original source details: GELFAND-P001343

varying D, we obtain the Riemann ζ-function
one should take the standard operator.

$$\Delta = y^2\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right), \qquad \Delta(W)\subset W$$



Undeciphered passage. A clipped stroke is visible in the frame beyond the right edge of the page.

Original source details: GELFAND-P001343

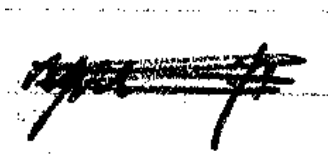
16/II-87. Gelfand Sem.

Problem. (Kazakov). How many triangulations of the sphere [crossed out] with Δ's colored in 2 colors (white and black), with fixed numbers of black and white ones and fixed boundary length (between white and black).



Uncertain reading. The surname by the problem reads "Kazakov".

Original source details: GELFAND-P001343



Undeciphered passage. An entry is crossed out after "sphere".

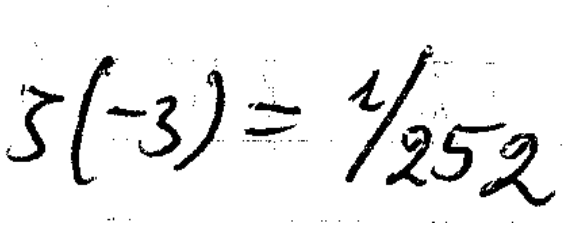
Original source details: GELFAND-P001343

Don Zagier.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \text{then } \zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90} \tag{1}$$

$$\zeta(2n) = \frac{(-1)^{n-1} 2^{2n-1} B_{2n}}{(2n)!} \pi^{2n}$$

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(-1) = -\frac{1}{12}, \quad \zeta(-2) = 0, \quad \zeta(-3) = \frac{1}{252} \tag{2}$$



Uncertain reading. In (2) the value written for $\zeta(-3)$ is $1/252$.

Original source details: GELFAND-P001343

$$\zeta(1-2g) = \chi(\mathcal{M}_g) \qquad (?) \tag{3}$$

(Harer, Zagier)

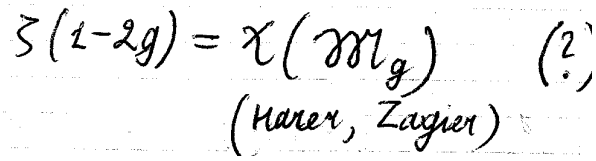
$\mathcal{M}_g$ is the moduli space of [crossed out] surfaces of genus g



Undeciphered passage. A short fragment is crossed out before the abbreviation for "surfaces".

Original source details: GELFAND-P001343

$\chi(\mathcal{M}_g)$ is its Euler char.



Uncertain reading. A question mark in parentheses appears to the right of (3).

Original source details: GELFAND-P001343

$k = 4, 6, 8, 10, \ldots$

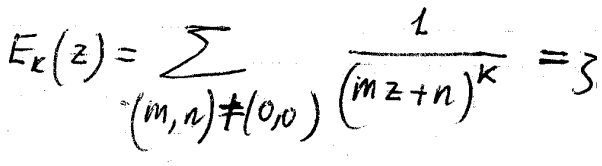
$$E_k(z) = \frac{1}{2} \sum_{(c,d)=1} \frac{1}{(cz+d)^k} \tag{4}$$

$$\begin{aligned} \zeta(k) E_k(z) &= \sum_{(m,n)\neq(0,0)} \frac{1}{(mz+n)^k} = \\ &= \zeta(+k) + \frac{(2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n, \end{aligned} \tag{5}$$



Undeciphered passage. A symbol is crossed out before k in the first $\zeta(k)$ in (5).

Original source details: GELFAND-P001343



Uncertain reading. In (5) the sum over (m, n) directly follows the equals sign.

Original source details: GELFAND-P001343

GELFAND-P001344

[source number: 74]

where

$$\sigma_{k-1}(n) = \sum_{d|n} d^{k-1}, \qquad q = e^{2\pi i z}$$

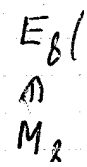
$$M_4 \ni E_4\zeta(4) = \zeta(4) + \frac{8\pi^4}{3}(q + 28q^2 + \ldots) \qquad \text{(weight 4)}$$



Uncertain reading. The formula of weight 4 reads $28q^2$.

Original source details: GELFAND-P001344

$$E_8(z)\zeta(8) = \zeta(8) + \underline{\ldots\pi^8}(q + 129q^2 + \ldots) \qquad \text{(weight 8)}$$
$$\Uparrow$$
$$M_8$$



Uncertain reading. Below E_8, M_8 is written with an upward arrow.

Original source details: GELFAND-P001344



Uncertain reading. The exponent of π has been read as 8. A similar glyph appears as the exponent of π in the formula for $\zeta(8)$ below.

Original source details: GELFAND-P001344



Undeciphered passage. A line is drawn below $\ldots\pi^8$. The space under it is blank.

Original source details: GELFAND-P001344

$$\dim M_8 = 1, \qquad \dim M_4 = 1$$

$$\Rightarrow E_4^2 = cE_8 \qquad \text{(the weights agree)}$$



Undeciphered passage. An entry to the left of the arrow to $E_4^2 = cE_8$ is crossed out.

Original source details: GELFAND-P001344

$$\zeta(4) = \pi^4/90 \quad \Rightarrow \quad \zeta(8) = \pi^8/9450$$

Thus:

From $\underset{\pi^4/90}{\zeta(4)}$ and $\underset{\pi^6/945}{\zeta(6)}$ one obtains all $\zeta(2n)$

$$E_4 E_6 = cE_{10} \quad \text{etc.}$$

Hecke L-series

Let

$$f(z) \in M_k(SL_2(\mathbb{Z}))$$

$$f(z) = \sum_{n=0}^{\infty} a_n q^n \qquad \text{mod. form of weight } k$$

$$a_0 = 0 \Rightarrow f \in S_k \qquad \begin{matrix}\text{(parabolic)} \\ \text{cusp form}\end{matrix}$$

$$L(f,s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

$$\Gamma(s) = \lambda^s \int_0^{\infty} y^{s-1} e^{-\lambda y}\, dy$$

$$\Gamma(s) \sum_{n=1}^{\infty} \frac{a_n}{(2\pi n)^s} = \int_0^{\infty} y^{s-1} \left(\sum_{n=1}^{\infty} a_n e^{-2\pi n y} \right) dy$$

Undeciphered passage. A short entry after a_n in the numerator is crossed out.



Original source details: GELFAND-P001344

$$\Gamma(s)(2\pi)^{-s} L(f,s) = \int_0^{\infty} y^{s-1} f(iy)\, dy \tag{6}$$

[At right] (if f is parabolic), $f \in S_k$

The integral on the right always converges,

$$f\left(-\frac{1}{z}\right) = z^k f(z), \qquad f\left(\frac{i}{y}\right) = (-1)^{k/2} y^k f(iy)$$

Hence

$$\underbrace{\Gamma(s)(2\pi)^{-s}L(f,s)}_{\|\ L^*(f,s)} = (-1)^{k/2}\,(s \to k-s) \tag{7}$$

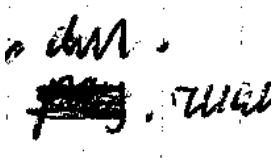
(functional eq.)

$$L^*(f,s) = (-1)^{k/2}L^*(f,k-s) \tag{8}$$

(9) Thm. (Eichler + Shimura + Manin)

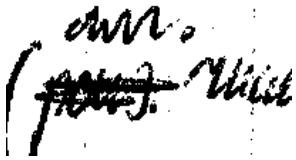
If $f(z) \in S_k(SL_2\mathbb{Z})$ is a Hecke form, then $\exists\,\omega_+,\omega_- \in \mathbb{R}$ (depending only on f) such that

$$L^*(f,n) = (\text{alg. number})\,\omega_-, \qquad n = 2,4,\ldots,k-2,$$
$$L^*(f,n) = (\text{alg. number})\,\omega_+, \qquad n = 1,3,\ldots,k-1.$$



Original source details: GELFAND-P001345

Undeciphered passage. In the first line, possibly «рац.» (“rat.”) is crossed out before “number”. “Alg.” is written above.



Original source details: GELFAND-P001345

Undeciphered passage. In the second line, possibly «рац.» (“rat.”) is crossed out before “number”. “Alg.” is written above.

ω_-, ω_+ are probably transcendental. This is proved for other groups (with complex multiplication) rather than $SL_2(\mathbb{Z})$.

$$\underset{\text{period}}{r_n(f)} = \int_0^\infty f(iy) y^n \, dy, \qquad n = 0, 1, \ldots, k-2. \tag{10}$$

Example.

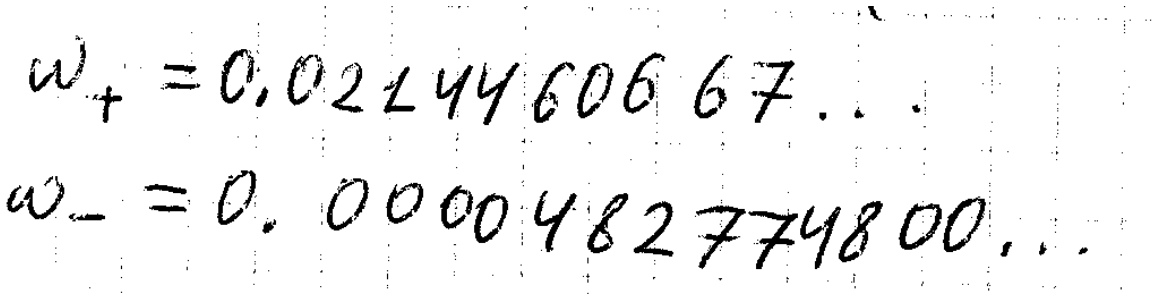
$$\Delta(z) \in S_{12}(SL_2(\mathbb{Z})) = q - 24q^2 + 252q^3 + \ldots =$$
$$= q \prod_1^\infty (1-q^n)^{24}$$

	10	9	8	7	6	
n	0	1	2	3	4	5
$r_n(\Delta)$		$\frac{384}{5}\omega_-$		$40\omega_-$		$\underbrace{32\omega_-}_{\text{def. of } \omega_-}$
	$\frac{192}{691}\omega_+$		$\frac{16}{135}\omega_+$		$\frac{8}{105}\omega_+$	

$$r_n = r_{10-n}$$

(only ratios matter)

$$\omega_+ = 0.0214460667\ldots$$
$$\omega_- = 0.0000482774800\ldots$$



Original source details: GELFAND-P001345

Uncertain reading. The decimal entries read $0.0214460667\ldots$ and $0.0000482774800\ldots$.

$$G_k(z) = \underbrace{-\frac{B_k}{2k}}_{\zeta(1-k)} E_k(z) \qquad \text{(Eisenstein series)} \tag{10'}$$



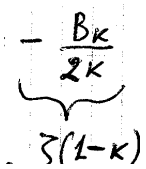
Original source details: GELFAND-P001345

Uncertain reading. The initial of $G_k(z)$ is overwritten. There may be an E underneath.



Original source details: GELFAND-P001345

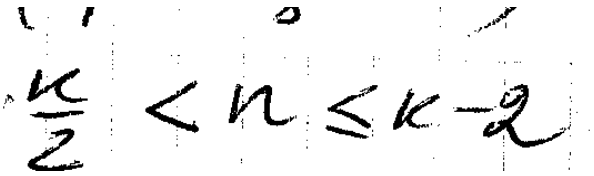
Undeciphered passage. An entry between the fraction and $E_k(z)$ is crossed out.



Uncertain reading. $\zeta(1-k)$ is written under $-B_k/(2k)$.

Original source details: GELFAND-P001345

$$\frac{k}{2} < n \leq k-2$$



Uncertain reading. The range at the right of (11) is $k/2 < n \leq k-2$.

Original source details: GELFAND-P001345

$$-\frac{4n}{B_{2n}} \left(\underset{\uparrow S_k}{f}\,,\ \underset{\uparrow M_{2n}}{G_{2n}} \underset{\uparrow M_{k-2n}}{G_{k-2n}} \right) \tag{11}$$

a. (one of them is parabolic ⇒ the inner product converges) [arrow down to the integrand] inner product:

$$\left((f,g) = \int_{\Gamma\backslash H} f(z)\overline{g(z)} y^k \frac{dx\,dy}{y^2} \right)$$

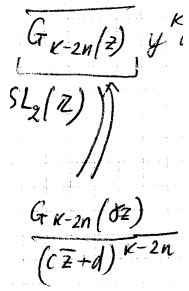
GELFAND-P001346

[source number: 75]

$$(12) \qquad (11) = \int_{\Gamma\backslash H} f(z) \sum_{\left(\begin{smallmatrix}\circ & \circ \\ c & d\end{smallmatrix}\right) \in \left(\begin{smallmatrix}1 & * \\ 0 & 1\end{smallmatrix}\right)\backslash SL_2(\mathbb{Z})} \frac{\overbrace{\overline{G_{k-2n}(z)}}}{(c\bar{z}+d)^{2n}}\, y^k\, dV$$

A double arrow leads from the fraction to the brace.

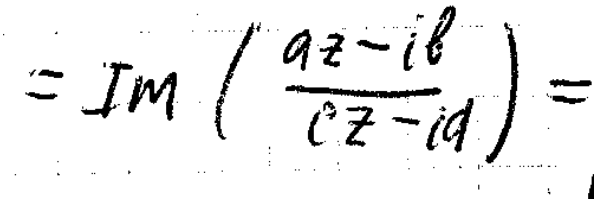
$$\frac{G_{k-2n}(\gamma z)}{(c\bar{z}+d)^{k-2n}} \qquad \left(\text{since } G(\gamma z) = (cz+d)^{k-2n} G(z)\right)$$



Uncertain reading. No overbar is visible over the numerator $G_{k-2n}(\gamma z)$.

Original source details: GELFAND-P001346

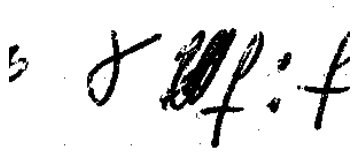
$$\operatorname{Im}(\gamma z) = \operatorname{Im}\left(\frac{az - ib}{cz - id}\right) = \frac{y}{|cz + d|^2}$$



Uncertain reading. The separate strokes in the fraction are read as $-i$.

Original source details: GELFAND-P001346

And one can insert γ into f, giving $f(\gamma z)$, by replacing the factor.



Uncertain reading. The preposition before f is overwritten, provisionally read as «в» (into).

Original source details: GELFAND-P001346

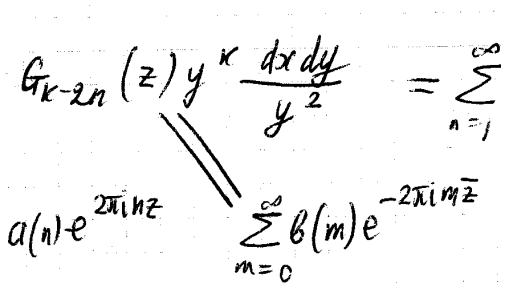
We obtain:

$$\sum_\gamma \int_{\mathcal{F}} f(\gamma z)\overline{G_{k-2n}(\gamma z)}\operatorname{Im}(\gamma z)^k\, dV =$$

$$= \sum_\gamma \int_{\gamma\mathcal{F}} f(z)\overline{G_{k-2n}(z)}(\operatorname{Im} z)^k\, dV =$$

$$= \int_{\underbrace{z\to z+1\setminus H}_{\int_0^\infty \int_0^1}} \underbrace{f(z)}_{\sum_{n=1}^\infty a(n)e^{2\pi i n z}} \underbrace{G_{k-2n}(z)}_{\sum_{m=0}^\infty b(m)e^{-2\pi i m\bar{z}}} y^k \frac{dx\,dy}{y^2}$$

$$= \sum_{n=1}^\infty a(n)\overline{b(n)} \int_0^\infty e^{-4\pi n y} y^{k-2}\, dy$$



Uncertain reading. The third integral has no overbar on $G_{k-2n}(z)$. In the series below it, $b(m)$ also has no overbar.

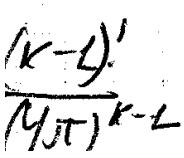
Original source details: GELFAND-P001346

Result:

$$f(z) = \sum_{n=1}^{\infty} a(n) q^n \in S_k,$$

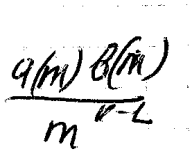
$$G(z) = \sum_{m=0}^{\infty} b(m) q^m \in M_{k-2n}, \qquad n \geq 2$$

$$-\frac{4n}{B_{2n}} \left(f, \underset{k-2n}{G} \cdot \underset{2n}{G_{2n}} \right) = \frac{(k-1)!}{(4\pi)^{k-1}} \underbrace{\sum_{m=1}^{\infty} \frac{a(m)b(m)}{m^{k-1}}}_{\text{(convolution)}} \tag{13}$$



Original source details: GELFAND-P001346

Uncertain reading. The numerator reads $(k-1)!$.



Original source details: GELFAND-P001346

Uncertain reading. No overbar is visible on $b(m)$ in (13).

$$G = G_{k-2n}(z) = (*) + \sum_{m=1}^{\infty} \sigma_{k-1-2n}(m) q^m$$

$$f = \sum_{m=1}^{\infty} a(m) q^m = \text{Hecke form}$$

[Crossed out: $= \prod_p [1 + \frac{a(p)}{p^s} +$]

$$\begin{pmatrix} a(mm') = a(m)a(m') & \text{when } (m, m') = 1 \\ a(2^a 3^b 5^c \ldots) = a(2^a)a(3^b)\ldots \end{pmatrix}$$

Then

$$(14) \qquad \sum \frac{a(m)}{m^s} = \prod_p \left[1 + \frac{a(p)}{p^s} + \frac{a(p^2)}{p^{2s}} + \ldots\right]$$

$$a(m)a(n) = \sum_{d|m,n} d^{k-1} a\left(\frac{mn}{d^2}\right)$$

$$b(p) = 1 + p^{k-1-2n}$$

$\Rightarrow$ the series is evaluated

$$\underbrace{\frac{(k-2)!}{(4\pi)^{k-1}} \sum_{m=1}^{\infty} \frac{a(m)\sigma_{k-1-2n}(m)}{m^{k-1}}}_{L(f,k-1)L(f,2n)/\zeta(2n)} = \underbrace{-\frac{4n}{B_{2n}} \left(f, \overbrace{G_{k-2n} \cdot G_{2n}}^{\alpha f + \beta f_1 + \ldots}\right)}_{\alpha \cdot (f,f)}$$

[Annotation to the expansion] and the f_j are a basis.



Original source details: GELFAND-P001347

Uncertain reading. The sign below $\alpha \cdot (f, f)$ is crossed out and its original shape cannot be read.

$$\alpha \in \overline{\mathbb{Q}}^* \quad \text{(algebraic)}$$

All coeffs. of the Eisenstein series are rational numbers.

Thus

$$L(f, 2n)/\zeta(2n) \quad \text{indep. of } n \quad \Rightarrow \quad \text{the theorem (?) is proved } \triangle$$

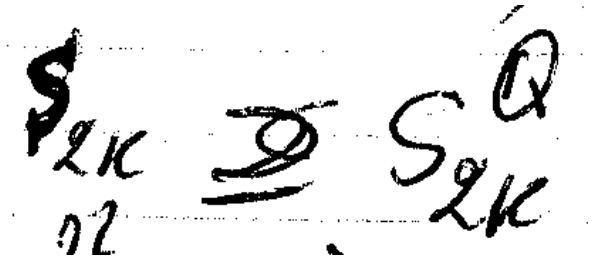


Original source details: GELFAND-P001347

Uncertain reading. The sign in parentheses reads as “?” or “o”. The following word reads “доказана” (is proved). The final mark resembles a triangle.

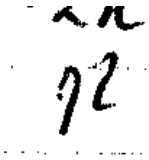
(Rankin, 1953)

$$S_{2k} \qquad S_{2k}^{\mathbb{Q}} = \left\{ f = \sum a(n) q^n,\ a(n) \in \mathbb{Q} \right\}$$



Uncertain reading. Between S_{2k} and $S_{2k}^{\mathbb{Q}}$ is a sign with overlapping strokes and a lower line.

Original source details: GELFAND-P001347



Uncertain reading. Below S_{2k} a mark resembling η precedes a raised 2.

Original source details: GELFAND-P001347

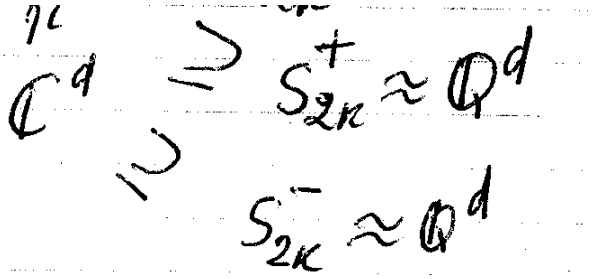


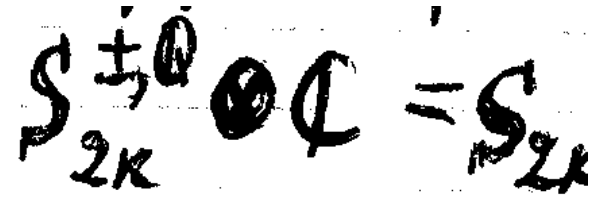
Diagram and typed description.

$$\begin{array}{ccc} & \supseteq & S_{2k}^{+} \approx \mathbb{Q}^d \\ \mathbb{C}^d & & \\ & \subseteq & S_{2k}^{-} \approx \mathbb{Q}^d \end{array}$$

Original source details: GELFAND-P001347

(e.g., 3 different rational structures in S_{2k})

$$S_{2k}^{\pm,\mathbb{Q}} \otimes \mathbb{C} = S_{2k}$$



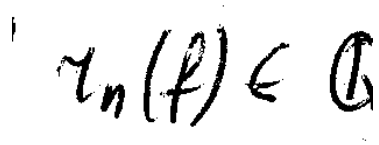
Uncertain reading. The sign between $S_{2k}^{\pm,\mathbb{Q}}$ and $\mathbb{C}$ is heavily overwritten, provisionally read as $\otimes$.

Original source details: GELFAND-P001347

GELFAND-P001348

[source number: 76]

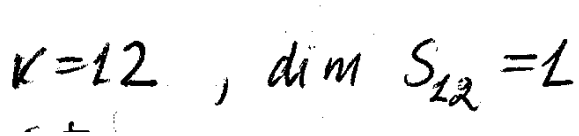
$$S^+_{2k} := \{f \in S_{2k} \mid r_n(f) \in \mathbb{Q},\ n = 0, 2, 4, \ldots, k-2\}$$
$$S^-_{2k} := \{\text{—//—} \qquad n = 1, 3, 5, \ldots\}$$



Uncertain reading. The period symbol reads r_n, as in the definition on frame GELFAND-P001345.

Original source details: GELFAND-P001348

$$k = 12, \qquad \dim S_{12} = 1$$



Uncertain reading. The example shows $k = 12$ and $\dim S_{12} = 1$ side by side.

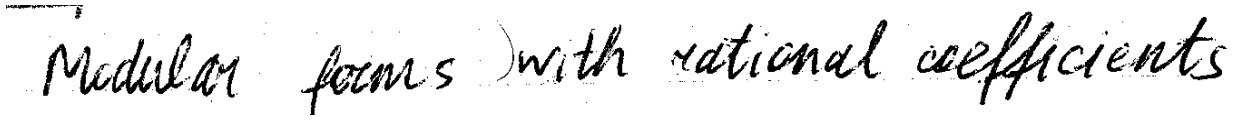
Original source details: GELFAND-P001348

$$S^+_{12} = \{f \mid r_0(f), r_2(f), r_4, r_6, r_8, r_{10} \in \mathbb{Q}\}$$

It is not obvious that the dimension is pres.!

Refs.: article by Kohnen + Zagier

Modular forms with rational coefficients



Original source details: GELFAND-P001348

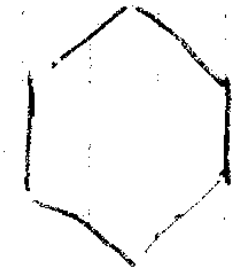
Uncertain reading. The manuscript reads “coefficients”.

(conf. Ed. Rankin — book)

GELFAND-P001349

2/III–87 V. Kazakov.

The problem of gluing surfaces of genus g from a $2n$-gon.



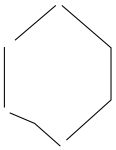
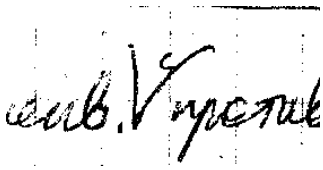


Diagram and typed description.

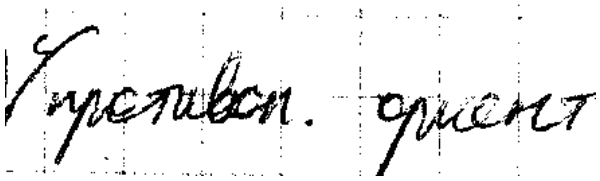
Original source details: GELFAND-P001349

$2n$-gon, gluing oppositely oriented sides (in pairs),



Original source details: GELFAND-P001349

Uncertain reading. Between “gluing” and “oppositely” is a separate mark resembling a check mark or γ.

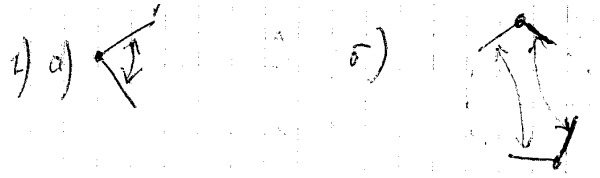


Original source details: GELFAND-P001349

Uncertain reading. The abbreviation before “oriented” is read as «противоп.» (oppositely).

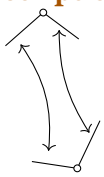
g is the genus of the resulting manifold. In how many different ways does a surface of genus g result? f_n^g is the number of such ways of gluing

(Some gluings may be forbidden. Prohibitions:



Original source details: GELFAND-P001349

Diagram and typed description.

1) a)  b) 

2) [Crossed out: OR] Only b) is forbidden. 3) Only a) is forbidden.

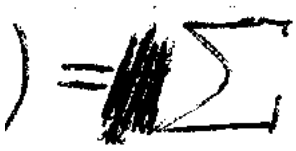
Everything reduces to the case without prohibitions.)

GELFAND-P001350

[source number: 85]

Generating fn.:

$$(1) \qquad f_n(N) = \sum_{g=0}^{\infty} N^{-2g} f_n^g$$



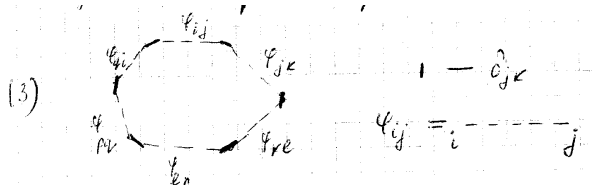
Original source details: GELFAND-P001350

Uncertain reading. A single sign before the summation sign is heavily crossed out.

Then

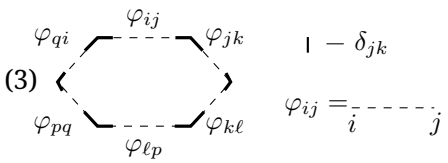
$$(2) \qquad f_n(N) = (2\pi)^{-N^2/2} \int d^{N^2}\varphi \, \frac{\mathrm{tr}}{N}(\varphi^{2n}) e^{-N \,\mathrm{tr}\, \varphi^2/2}$$
$$= \left\langle \frac{\mathrm{tr}}{N} \varphi^{2n} \right\rangle$$

where $\varphi^+ = \varphi$ is an $N \times N$ Hermitian matrix.



Original source details: GELFAND-P001350

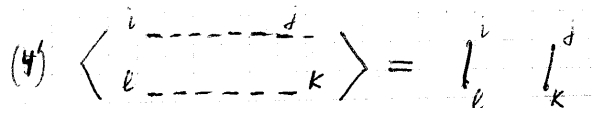
Diagram and typed description.



It is easy to check that

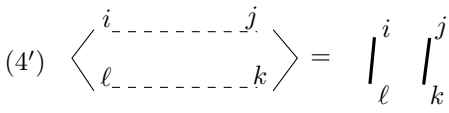
$$(4) \qquad \langle \varphi_{ij} \varphi_{k\ell} \rangle = \frac{1}{N} \delta_{i\ell} \delta_{jk}$$

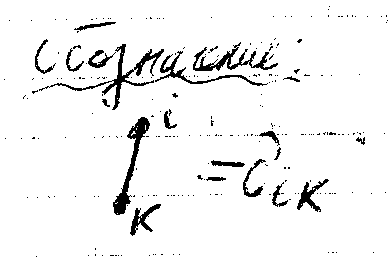
or



Original source details: GELFAND-P001350

Diagram and typed description.





Original source details: GELFAND-P001350

Notation:

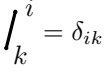
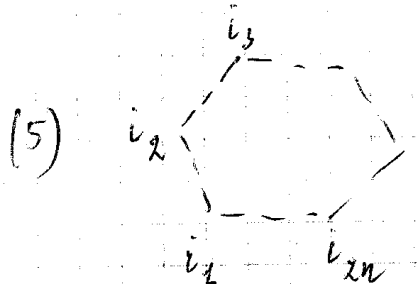
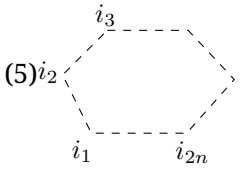


Diagram and typed description.

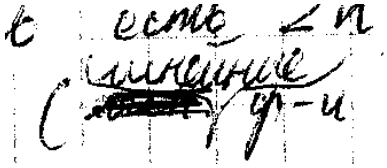


Original source details: GELFAND-P001350

Diagram and typed description.



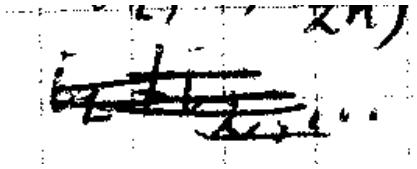
(6) Wick's theorem: Let there be $2n$ Gaussian variables $\xi_1, \xi_2, \ldots, \xi_{2n}$. Then ([Crossed-out writing] [Inserted above: linear] functions of φ_{ik})



Original source details: GELFAND-P001350

Undeciphered passage. The writing below "linear" is crossed out. Its reading is unresolved.

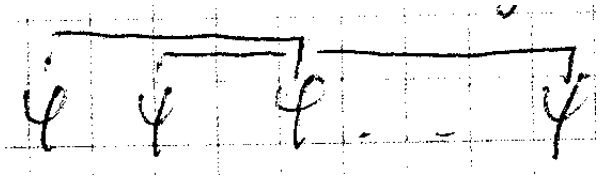
$$\langle \xi_1 \xi_2 \cdots \xi_{2n} \rangle = \sum_{P(i_1,\ldots,i_{2n})} \langle \xi_{i_1} \xi_{i_2} \rangle \langle \xi_{i_3} \xi_{i_4} \rangle \cdots$$



Original source details: GELFAND-P001350

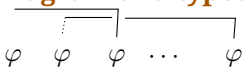
Undeciphered passage. A row of indices below the sum is crossed out. The initial index resembles i_2.

[Line from $P(i_1, \ldots, i_{2n})$] permutation [crossed out: $i_1, i_2, \ldots, i_{2n}$] $1, 2, \ldots, 2n$ The sum is over all partitions into pairs.



Original source details: GELFAND-P001350

Diagram and typed description.



(Exercise: prove?)



Original source details: GELFAND-P001350

Uncertain reading. The part «оказать» is clear. The initial stroke is faint. The likely reading is «доказать» (prove).

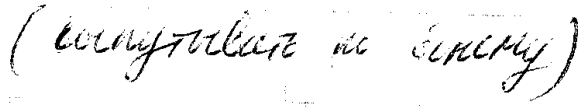
A more general theorem: let ξ be a Gaussian variable, then

$$\langle \xi f(\xi) \rangle = \langle \xi \xi \rangle \left\langle \frac{\partial f}{\partial \xi} \right\rangle \tag{7}$$

$$\langle \xi_i f(\xi) \rangle = \sum_j \langle \xi_i \xi_j \rangle \left\langle \frac{\partial f}{\partial \xi_j} \right\rangle \tag{8}$$

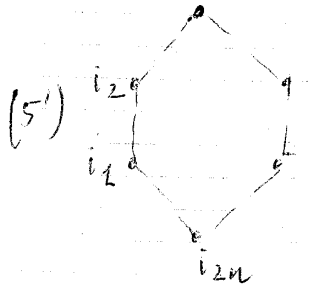
GELFAND-P001351

Using (7), (8), it is easy to prove Wick's theorem (by induction on the number of ξ's).



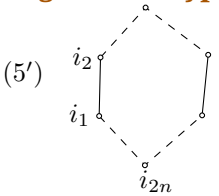
Original source details: GELFAND-P001351

Uncertain reading. The opening of the parenthesis is faint. "By induction on the number of ξ's" is provisional.



Original source details: GELFAND-P001351

Diagram and typed description.



We compute integral (2) using Wick's theorem. It is clear that

$$(9) \qquad \langle \operatorname{tr} \varphi^{2n} \rangle = \langle \varphi_{i_1 i_2} \varphi_{i_2 i_3} \cdots \varphi_{i_{2n} i_1} \rangle$$



Uncertain reading. The sign $\sum$ is crossed out between the equals sign and the bracket.

Original source details: GELFAND-P001351

Now let us apply Wick's theorem. It tells us to pair the sides! (of polygon $(5')$)
Gluing.

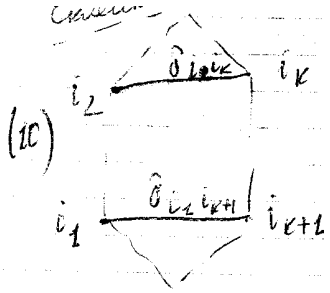
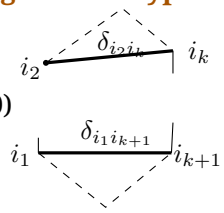
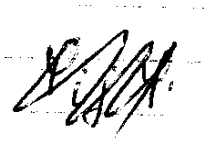


Diagram and typed description.

(10)



Original source details: GELFAND-P001351



Uncertain reading. To the right of (10), φ and another notation are crossed out. Lower strokes are visible, but the reading is unresolved.

Original source details: GELFAND-P001351

Formula (4) ensures the correct orientation in the gluing (because of Hermiticity!!)
(Choosing a matrix differently can give a nonorientable surface.)

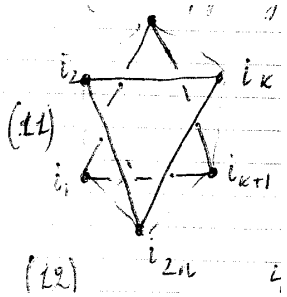
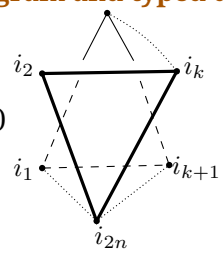


Diagram and typed description.

(11)



Original source details: GELFAND-P001351

Closed lines $\longrightarrow$ property of the Kronecker symbol, i.e. always N! For example,

$$N = \delta_{i_2 i_k} \, \delta_{i_k i_{2n}} \, \delta_{i_{2n} i_2}$$

$\forall$ closed cycles, the contribution is N

(12) Number of closed cycles = number of vertices in the glued closed figure.

(13) Contribution of each Wick pairing = (diagram)

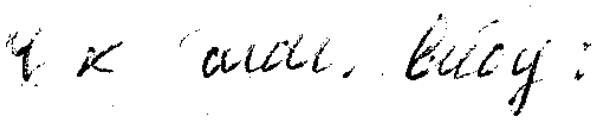
$$= N^{-(\#\text{ edges})+(\#\text{ vertices})-1} = N^{-2g}$$

(because of $\frac{N}{2} \operatorname{tr} \varphi^2$ in the exp)

GELFAND-P001352

[source number: 86]

Let us compute integral (2). We reduce it to diag. form:



Uncertain reading. The abbreviation before "form" is faint, provisionally read as "diag.".

Original source details: GELFAND-P001352

$$(14) \qquad \varphi_{ij} = \sum_k \Omega^+_{ik} \lambda_k \Omega_{kj}, \qquad \Omega \text{ is a unit. matrix}$$

$$(15) \qquad d^{N^2}\varphi = \left(\prod d\lambda_i \, [\Delta(\lambda)]^2\right) d\Omega,$$

where

$$\Delta(\lambda) = \prod_{i>j} (\lambda_i - \lambda_j)$$



Uncertain reading. A parenthesized number at left is crossed out, possibly (14) or (16).

Original source details: GELFAND-P001352

[In a box at right] (See H. Weyl: Classical groups, their invariants and representations.) (Dyson measure).

([Above the start of the list: Brezin(?)], Itzykson, Zuber, Parisi,)



Uncertain reading. The name above the start of the list is read as "Brezin". The question mark is in the manuscript.

Original source details: GELFAND-P001352

[Line down from Parisi] method of stoch. quantization quantum quantity $\longmapsto$ Markov process. The mean is computed by averaging over trajectories. [At left, arrow to (16)] works here.

$$(16) \qquad f_n(N) \sim \int \prod \left(d\lambda_i \, e^{-N\lambda_i^2/2}\right) \Delta^2(\lambda) \left(\frac{1}{N} \sum_{i=1}^N \lambda_i^{2n}\right) =$$

$$\underset{\text{symm. in } \lambda_j}{=} \int \prod \left(d\lambda_i \, e^{-N\lambda_i^2/2}\right) \Delta^2(\lambda) \lambda_1^{2n}$$

$$\sim \int \prod \left(d\lambda_i \, e^{-\lambda_i^2/2}\right) \Delta^2(\lambda) \lambda_1^{2n}$$



Uncertain reading. A separate mark is crossed out before the second integral in (16).

Original source details: GELFAND-P001352

(var. λ_i, we take out $\sqrt{N}$ in λ_i)



Uncertain reading. A separate mark is crossed out before $\sqrt{N}$.

Original source details: GELFAND-P001352

$$(17) \qquad \Delta(\lambda) = \det_{k,\ell} \|(\lambda_\ell)^k\| = \det \|H_k(\lambda_\ell)\|$$

H_k is any polynomial of degree k with leading coeff. 1. Let us choose H_k to be Hermite polynomials!! (with a normalizing factor so the leading coeff. $= 1$) (good, since H_k and H_ℓ are orthogonal)

Hence:

$$(18) \qquad f_n(N) \sim \frac{1}{C} \int \prod_{i=1}^{N} \left(d\lambda_i \, e^{-\lambda_i^2/2} \right) \lambda_1^{2n} \times \left[\sum_{\substack{p \\ \text{perm.}}} \prod_{k=1}^{N} H_k(\lambda_{p_k}) (-1)^{\operatorname{sgn} p} \right]^2 =$$



Uncertain reading. The number at left is faint, read as (18).

Original source details: GELFAND-P001352

(Parisi and Bessis devised the trick with Hermite polynomials) On squaring, every var. λ_{p_k} occurs exactly 2 times.

GELFAND-P001353

$$= \sum_{k=0}^{N-1} \frac{\int d\lambda \, e^{-\lambda^2} [H_k(\lambda)]^2 \lambda^{2k}}{\int d\lambda \, e^{-\lambda^2} (H_k(\lambda))^2}$$



Uncertain reading. The exponent on λ in the numerator is provisionally read as $2k$.

Original source details: GELFAND-P001353

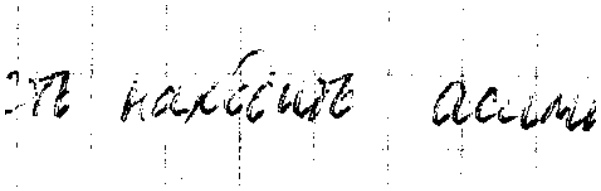
From a handbook (e.g., Abramowitz):

$$(19) \qquad \int d\lambda\, e^{i\lambda x} e^{-\lambda^2} [H_k(\lambda)]^2 = \underbrace{\sqrt{\pi}\, 2^n n!}_{\substack{\text{cancels} \\ \text{with the denominator}}} \quad e^{-x^2/4} \quad \underbrace{L_n(x^2/2)}_{\text{Laguerre poly.}}$$

In the end we get [arrow to (20)]:

$$(20) \qquad \sum_{n=0}^{\infty} \frac{(-x^2/2)^n}{(2n)!!} f_n(N) = e^{-x^2/4} \sum_{i=0}^{N-1} L_i(x^2/2)$$
$$= e^{-x^2/4} \quad \underbrace{L^1_{N-1}(x^2/2)}_{\text{assoc. Laguerre polynomials}}$$

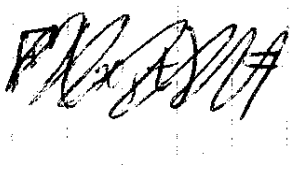
Here we are most [interested in] the asympt. as $N \to \infty$. Let us form the generating fn.:



Uncertain reading. The word before "asympt." is provisionally read as «наиболее» (most).

Original source details: GELFAND-P001353

[Crossed out before (21)] $F(x,t)$



Uncertain reading. Two horizontal strokes are visible under the crossing out after $F(x,t)$, possibly an equals sign.

Original source details: GELFAND-P001353

$$(21) \qquad f_n(t) = \sum_N t^N f_n(N)$$

The answer for it is:

$$(22) \qquad f_n(t) = \frac{(2n)!}{n!} \frac{(t+1)^n}{(1-t)^{n+2}}$$

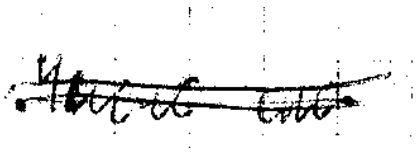
We are interested in the series expansion in t. Zakharevich: answer (after D. Zagier)

$$(23) \qquad \sum_{g=0}^{\infty} \frac{(n+1)!(n-2g)!}{(2n)!} f_g^n x^g = \left(\frac{\sqrt{x/2}}{\operatorname{th} \sqrt{x/2}} \right)^{n+1}$$

[An isolated left parenthesis in the margin] (when $n - 2g < 0$, multiply by $(2n)!$ and move it to the right-hand side) $f_g^n (n-2g)!$ is thereby defined also for $2g > n$(!).

Variant: 2 other combinatorial problems (worked on by Kazakov and Migdal).

① How to tile a $2n$-gon with k triangles so that the resulting surface has genus g.



Uncertain reading. "Number of gl." is visible beneath the crossing out.

Original source details: GELFAND-P001353

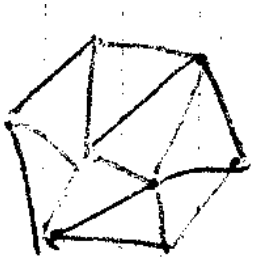
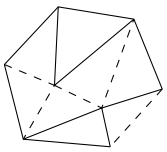
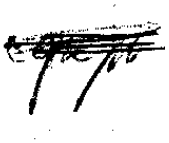


Diagram and typed description.



Original source details: GELFAND-P001353

(attaching handles is allowed!) (Simplicial complex of a given genus) $f_k^{2n}(g)$ (Bessis). (Number of Triangulations [crossed out] of a surface of genus g with k triangles, poss. $2n$-gons)



Undeciphered passage. The word between "Triangulations" and "surface" is crossed out and illegible.

Original source details: GELFAND-P001353

GELFAND-P001354

[source number: 87]

(1971–79, Itzykson et al.)

②

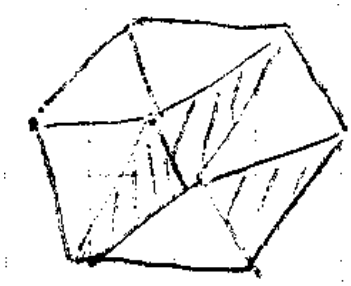
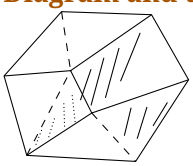
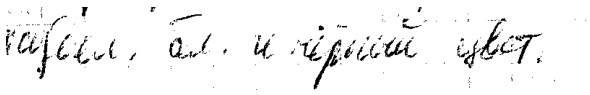


Diagram and typed description.



Original source details: GELFAND-P001354

The same, but we colour it and fix the boundary gluing, using white and black.



Uncertain reading. The faint abbreviation before "white and black" may read «раскр.» or «раздел.».

Original source details: GELFAND-P001354

GELFAND-P001355

9/III–87. Gelfand Sem. I. M. G. There is a paper which says the following. Consider

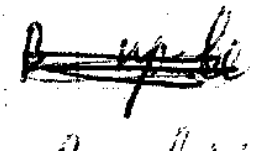
$$z \longmapsto \frac{\alpha z + \beta}{\gamma z + \delta}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{Z}, \qquad \text{in the half-plane, } \operatorname{Im} z > 0$$



Uncertain reading. A short undeciphered entry is crossed out after $\mathbb{Z}$.

Original source details: GELFAND-P001355

$$\alpha\delta - \gamma\beta > 0$$

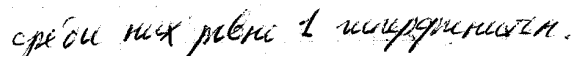


Original source details: GELFAND-P001355

Uncertain reading. A phrase is crossed out before “There exist”, possibly «в кр-ве» (abbreviation unresolved).

There exist sets of nonzero intermediate Hausd. measure, which arise in connection with the principal series.

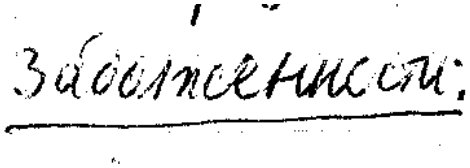
Conjecture: factors arise in reprs. of ∞-dimensional groups. In physics, factors of type $\overline{\text{III}}_\infty$ arise. Exactly 1 among them is hyperfinite.



Original source details: GELFAND-P001355

Uncertain reading. The last word in the line is read as «гиперфинитн.» (hyperfinite).

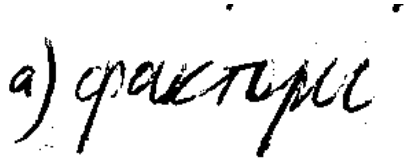
Talkativeness?:



Original source details: GELFAND-P001355

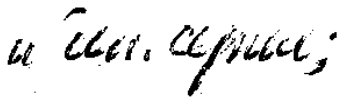
Uncertain reading. The heading is provisionally read as «Заболтенности» (talkativeness) or «Заболезненности» (ailments).

a) factors, b) attractors, c) chaos, Hausdorff measures and infin. series,



Original source details: GELFAND-P001355

Uncertain reading. The first word in the list reads “factors”.



Original source details: GELFAND-P001355

Uncertain reading. The abbreviation before “series” is provisionally read as «беск.» (infin.).

d) characteristic functions.

A. Beilinson. Basic facts about the Riemann ζ-function and its generalizations.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

1. Euler product:

$$\zeta(s) = \prod_{p} (1 - p^{-s})^{-1}$$

2. Analytic continuation: a simple pole at $s = 1$, holomorphic at the other points.
3. Functional equation. Introduce a more complete ζ-function:

$$\zeta^*(s) = \underbrace{\pi^{-s/2}\Gamma(s/2)}_{\text{Euler factor numbered } \infty} \zeta(s),$$

then $\zeta^*(s)$ is invariant under $s \leftrightarrow 1 - s$ (unlike the p-metric)

[In the left margin] Remark 1. Kirillov's. The π [crossed out] factor is the volume of an s-dimensional ball.



Original source details: GELFAND-P001355

Uncertain reading. Before the crossed-out sign the letter resembles π or Π.



Original source details: GELFAND-P001355

Undeciphered passage. A short sign is crossed out between the first letter and "factor".



Original source details: GELFAND-P001355

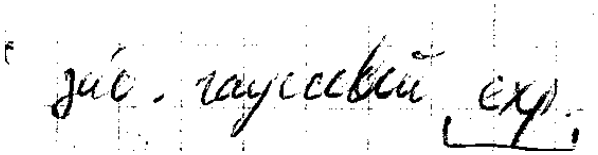
Uncertain reading. The bottom line after the dash reads "volume of an s-dimensional ball".

$$\mathbb{R}^* \qquad t^{-s} \qquad e^{-t^2}$$

Γ arises as a Mellin transform.

$$\mathbb{Q}_p^* \qquad |t|^s \qquad \text{char. fn. of} \quad \underbrace{(\mathbb{Z}_p \setminus 0)}_{\text{open subgroup}} \subset \mathbb{Q}_p^*$$

its analogue in $\mathbb{R}^*$ [abbr.] of a Gaussian exp.



Uncertain reading. After $\mathbb{R}^*$ is a faint abbreviation, possibly «дис.».

Original source details: GELFAND-P001356

4. Trivial zeros: the functional eq. makes it clear that $\zeta(s)$ has zeros (of multiplicity 1) at $s = -2k$, $k \in \mathbb{Z}$, $k > 0$. (Because of the Γ factor.)



Uncertain reading. A short entry under item 4 is heavily crossed out.

Original source details: GELFAND-P001356

5. Rational values:

$$\zeta(1-n) = -\frac{B_n}{n}$$

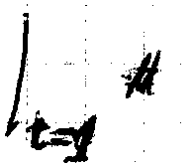
(at negative points)

$$\zeta(-n) = \sum k^n = \left(t\frac{d}{dt}\right)^n \frac{1}{1-t}\bigg|_{t=1} \qquad \#$$



Uncertain reading. A short entry before $(t\,d/dt)^n$ is heavily crossed out.

Original source details: GELFAND-P001356



Uncertain reading. To the right of the evaluation $t = 1$ is a heavy mark of intersecting strokes, read as #.

Original source details: GELFAND-P001356

(more precisely, take the value of the first regular term)

6. Regulator formulas: $\forall n$, $\exists$ a natural subgroup in $\mathbb{R}$, $\Gamma_n \subset \mathbb{R}$, such that [Crossed out] $\zeta(1-2n) = (\text{vol}\,\mathbb{R}/\Gamma_n) \cdot (\text{rational number})$.

$$\zeta'(-2n) = (\text{vol}\,\mathbb{R}/\Gamma_n) \cdot (\text{rational number})$$

$\Gamma_n =$ image of $K_{4n+1}(\mathbb{Z})$ in $\mathbb{R}$.

7. Arithmetic properties of Bernoulli numbers: p-adic properties of $\zeta(1-n)$

$$\zeta(s)(1-p^{-s})$$

$$\zeta(1-n)(1-p^{-1+n}) = \int_{\mathbb{Z}_p^*} x^{1-n}\, d\mu$$

where $d\mu$ is a natural measure on $\mathbb{Z}_p^*$ (with values in $\mathbb{Z}_p$)

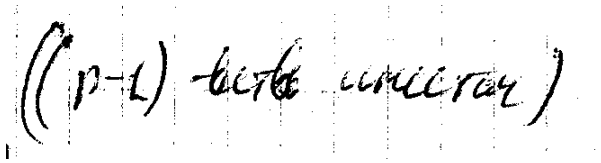
$$\mathbb{Z}_p^* = (1+p\mathbb{Z}_p) \times \sqrt[p-1]{1}$$



Uncertain reading. A mark resembling E is crossed out at left.

Original source details: GELFAND-P001356

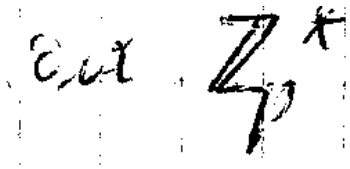
($(p-1)$th roots?)



Uncertain reading. The word after "$(p-1)$th" is faint, provisionally read as "roots".

Original source details: GELFAND-P001356

The set of characters [undeciphered] $\mathbb{Z}_p^*$ consists of $(p-1)$ components.



Uncertain reading. Before $\mathbb{Z}_p^*$ is a short word resembling «сил».

Original source details: GELFAND-P001356

8. Riemann hypothesis.

9. Connection with the distribution of prime numbers.

GELFAND-P001357

[source number: 91]

Generalizations of the Riemann ζ-function.

Artin L-functions $\quad L(\rho, s)$

$$\rho : \mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q} \underset{\text{continuous}}{\longrightarrow} GL(n, \mathbb{C})$$

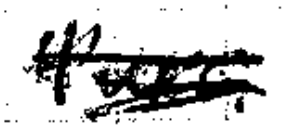


Uncertain reading. A short entry is crossed out between Gal and $\overline{\mathbb{Q}}/\mathbb{Q}$.

Original source details: GELFAND-P001357

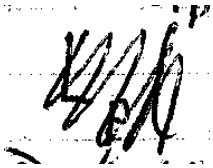
$$L(\rho, s) = \prod_p L_p(\rho, s), \qquad p \text{ is a prime number}$$

$\overline{\mathbb{Q}}_p$ is an alg. closure of the field $\mathbb{Q}_p$. Fix an embedding $\overline{\mathbb{Q}} \subset \overline{\mathbb{Q}}_p \quad \Rightarrow$



Uncertain reading. A word before $\overline{\mathbb{Q}}_p$ is crossed out.

Original source details: GELFAND-P001357



Uncertain reading. Below the start of the line a crossed-out entry resembles $\mathbb{Q}_p/\mathbb{Q}$ or $\overline{\mathbb{Q}}_p/\mathbb{Q}$.

Original source details: GELFAND-P001357

$\Rightarrow$ an embedding

$$\mathrm{Gal}\,\overline{\mathbb{Q}}_p/\mathbb{Q}_p \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q}$$

Only the conjugacy class of these maps is invariantly determined.

Inertia group:

$$I \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}_p/\mathbb{Q}_p$$

[Line from I] acts trivially on $\overline{\mathbb{F}}_p = \overline{\mathbb{Z}/p\mathbb{Z}}$

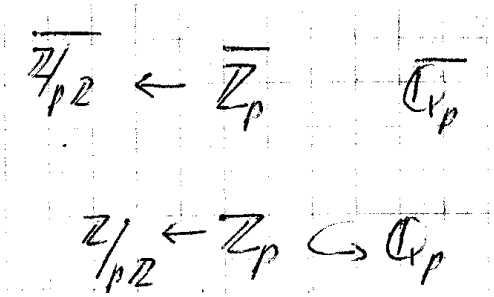
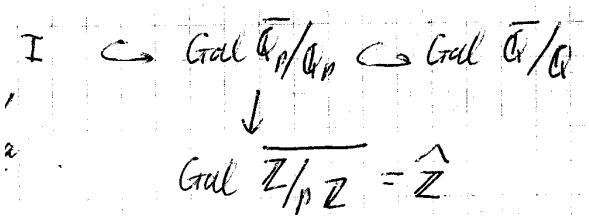


Diagram and typed description.

$\overline{\mathbb{Z}/p\mathbb{Z}} \longleftarrow \overline{\mathbb{Z}}_p \qquad \overline{\mathbb{Q}}_p$

$\mathbb{Z}/p\mathbb{Z} \longleftarrow \mathbb{Z}_p \hookrightarrow \mathbb{Q}_p$

Original source details: GELFAND-P001357



Original source details: GELFAND-P001357

Diagram and typed description.

$$I \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}_p/\mathbb{Q}_p \hookrightarrow \mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q}$$
$$\downarrow$$
$$\mathrm{Gal}\,\overline{\mathbb{Z}/p\mathbb{Z}} = \widehat{\mathbb{Z}}$$

[In the left margin, line to I] fixes $\sqrt[N]{1}$, $(N,p)=1$



Original source details: GELFAND-P001357

Diagram and typed description. Below $\widehat{\mathbb{Z}}$ before "generator" is a short vertical stroke over a slanting line. It may be an underlined 1.

generator Fr_p (Frobenius elem. $x \mapsto x^p$)

ρ continuous $\Rightarrow$ for a.e. p except a finite number, $\rho|_I$ is trivial. Unramified p: $\rho(I_p) = 1$

Refs.: Cassels, Fröhlich, "Alg. Number Theory"

$$L_p(\rho, s) = \det\left(1 - \rho(\mathrm{Fr}_p)p^{-s}\right)^{-1}$$

Example 1. Dirichlet L-functions

ρ is a 1-dimensional repr. given by a Dirichlet character

$$\chi : (\mathbb{Z}/N\mathbb{Z})^* \longrightarrow \mathbb{C}^*$$

From it one constructs a char. of the Galois group

$$\mathrm{Gal}\,\overline{\mathbb{Q}}/\mathbb{Q} \longrightarrow (\mathbb{Z}/N\mathbb{Z})^* \xrightarrow{\chi} \mathbb{C}^*$$

$$L(\chi, s) = \sum_{(n,N)=1} \frac{\chi(n)}{n^s} = \prod_{(p,N)=1} \left(1 - \chi(p)p^{-s}\right)^{-1}$$

GELFAND-P001358

Example 2. $K/\mathbb{Q} \subset \overline{\mathbb{Q}}$

[In the upper right margin:]



Original source details: GELFAND-P001358

Uncertain reading. an isolated stroke; no legible page number

$$\operatorname{Gal}\overline{\mathbb{Q}}/K \subset \operatorname{Gal}\overline{\mathbb{Q}}/\mathbb{Q}$$

Take the induced representation, and then the corresponding L-function is $\zeta_K(s)$

$$L(\operatorname{Ind}_{K/\mathbb{Q}}, s) = \zeta_K(s) = \prod_{\substack{p \text{ prime} \\ \text{ideals in } K}} (1 - N(p)^{-s})^{-1}$$
$$= \sum_{\substack{\text{integral} \\ \text{ideals } I \text{ in } K}} N(I)^{-s}$$

Properties

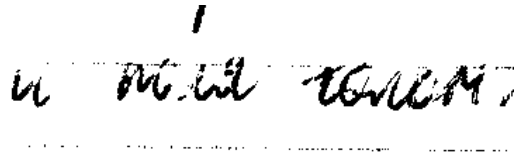
0. $L(\rho, s)$ uniquely determines ρ

! K is not recovered from $\zeta_K(s)$

$$L(\rho_1 \oplus \rho_2, s) = L(\rho_1, s)L(\rho_2, s)$$

1. Analytic continuation

L-functions of a field? almost all?[?] admit holomorphic continuation.



Uncertain reading. of a field? almost all?

Original source details: GELFAND-P001358

All admit meromorphic continuation

Conjecture. $\forall$ irreducible nontrivial ρ: $L(\rho, s)$ is holomorphic in s.

[Insertion beneath the line:] (if separate curved stroke; ρ?[?] passed through (abbreviated reading uncertain)[?] $GL(n, \mathbb{C})$ — an Artin function)



Uncertain reading. separate curved stroke; ρ?

Original source details: GELFAND-P001358



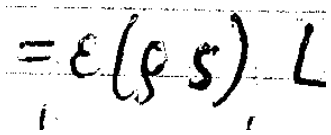
Uncertain reading. passed through (abbreviated reading uncertain)

Original source details: GELFAND-P001358

For l-adic representations, even meromorphicity is unknown

2. Functional equation. Known for Artin L-functions

$$L^*(\rho, s) = \underbrace{\varepsilon(\rho s)}_{\substack{\text{an important const is contained here} \\ \text{(Gauss sums)}}} L^*(\rho^*, 1-s)$$



Original source details: GELFAND-P001358

Uncertain reading. no comma can be made out in $\varepsilon(\rho s)$

Trivial zeros and so on.